

Eris and Harmonia

By Matt Reid

12th March 2021

Introduction and thanks

This paper marks the culmination of some 11 years of research into Jim Sanborn's Kryptos installation located on the grounds of the CIA Headquarters in Langley Virginia. Since its original dedication in 1990 public awareness of the Kryptos cipher challenge has grown year on year. The fact that only partial decryptions for three of the four ciphers have ever been publicly acknowledged have only served to increase its mystique amongst the amateur crypto community that has sprung up in its wake. Its stubbornness to yield its final secrets even with the gradual release of some 25% of its plain text over the last decade does not appear to have changed the odds stacked against any would be cryptographer in any way.

The framework and results presented herein will no doubt represent something of a paradigm shift in how both K4 and the sculpture itself as a 'whole' piece are currently viewed. As will be seen from the eventual conclusion, the artist has not only given us four encrypted messages, but also the blueprints for two physical devices, the first of which would seemingly allow us to decode K4's unique cipher scheme. Seasoned programmers will know that for many years the computer journals ran regular columns that challenged readers to construct functional programs using the most minimal amount of code possible. I would hope that after this paper has been digested, appraised and researched further that it will become apparent just how much has been squeezed into the 1735 letters that comprise Kryptos. I would also hope that the technical brilliance of the installation as a complete scheme will finally be realised and that parts of this paper will eventually find their way into educational settings for replication in STEM based teaching activities. On that basis please feel free to replicate any or all parts of this paper for use as an educational or research resource.

November the 3rd 2021 will mark the 31st anniversary of the sculptures original dedication by the then Agency Director William Webster. This in itself was also an important date as it came 31 years to the day after Dwight D Eisenhower laid the foundation stone for the Original Headquarters Building in 1959. At that same ceremony in 1959 a time capsule was buried that supposedly 'contained secrets'. Whether this was simply a joke made by Allan Dulles at the expense of the assembled press or not may never be known as the official manifest for the capsule did not appear to contain anything of any great secrecy, but what we do know is that 31 years later a sculpture was installed that really did contain secrets. This paper won't offer any background on the Kryptos story as it will be assumed that the reader is already familiar with the chronology of how the first three parts were broken. It will concentrate instead purely on the nature of the proposed cipher method and the potential meaning of the final message. This solution 'beyond the solution' will put an entirely new perspective on the sculpture and show that it is in fact far more than a redundant copper sheet once K4 is broken and reveals instead a new function that takes it into perpetuity.

This introduction would of course be incomplete without a sincere expression of gratitude to Jim Sanborn, Ed Scheidt, their patrons and any other unnamed parties behind the scenes for seeding the world with such a constructive and educational curiosity. I hope this paper in itself goes some way to paying the necessary respect to their collective artistic and technical skills.

Lastly I'd like to thank all those on the public forums, microsites and especially Larry McElhiney who moderates the Kryptos.groups.io site along with Matt Coleman over at the Kryptos Facebook Page and Elonka Dunin who took the time out to cast an eye over my initial paper back in 2010. As someone that came late to this in 2009 I'd also like to thank Elonka as well as all the other crypto enthusiasts that continue to publicly communicate so much about the sculpture and cryptography in general to the outside world. Anyone unfamiliar with Kryptos that wants additional background may also want to visit Elonka's Kryptos site at <https://elonka.com/kryptos/> before proceeding any further with this document.

Overview

This paper presents evidence showing that the Kryptos cipher text is a palimpsest written over the top of 12 cryptograms, each of which show a consistent underlying structure that can be uncovered and decrypted using the K4 cipher text as a key. A framework is put forward that shows the sculpture must first be closed to form a cylinder consisting of 32 columns and 28 rows. Each row must then be given freedom to move about the axis in the same way as a Jefferson Cylinder. Groupings of three consecutive rows are identified that give rise to 6 ‘rail positions’ on which all 12 solves are predicated. Each of these rails are designated by the same 4 letter marker, with 3 being in the horizontal plane and 3 being in the vertical. Two distinct types of symmetry operation are used to mirror any row shifts in the relation $n^* = 29 - n$, where n is the row number being shifted. These two algorithms are provided via cryptically hidden crib examples which are then used repeatedly for the remaining solves. A defined start and end position is found for each rail position based on the principle of ‘invariance under transformation’, which naturally gives rise to the identification of 8 letters for each rail configuration where there are two configurations per rail. The positioning of the rails that are marked in the vertical plane along with the information that can be gleaned from the BERLIN CLOCK plain text, shows that a transition occurs at Rail Position 3, from which point on the letters are read off vertically rather than horizontally. In each case the algorithms naturally give rise to the identification of 8 letters in each cryptogram. Once each cryptogram has been broken, all cipher and plain text is right shifted by one letter in the Kryptos alphabet. In order to read the plain text for the 8 letters in each cryptogram it is necessary to first **encrypt** them using the same Vigenere scheme used for parts K1 and K2, using the K4 cipher text as the key.

Amendments to the previous paper

At the time of publishing this is believed to be an original piece of work that draws upon and extends the findings already put into the public domain by the author’s previous paper, ‘Proposed Outline Method For Decryption of K4’, published to the Kryptos Group forum in May 2020. This date was of course prior to the latest ‘EAST’ clue gradually emerging via group member Sukhwant Singh’s posts that occurred later in August of that year. The new clue highlighted a discrepancy in the previous work whereby the retrieval of ‘NORTH EAST’ in the correct positions was achievable in isolation, but could not be used to successfully recover ‘EAST NORTH EAST’.

I’m indebted to James Churchill a member of the Kryptos Group’s sister site on Facebook for an observation around repeating letter strings that he shared on December 15th 2020. This intriguing pattern in the K2 cipher text represented the break that was needed in order to explain why the new ‘EAST’ clue could not be accommodated within the original framework despite the general premise being sound. Most importantly, it also highlighted a viable basis for how the previous framework could be amended in order to achieve a full solve, the results of which are now discussed in detail.

Results

The modified method put forward in this paper generates a coherent plain text for the remaining K4 cipher as well as all verified plain text letters for K4 in their correct positions. As ever, this is a large document and the author makes no apologies for that as a full set of cipher and plain text that has been correctly transformed by the appropriate right shift in the Kryptos alphabet has been provided should anyone wish to test or refine any of the results for themselves. This paper may represent something of an initial pass at the final plain text, but the general basis of the method is generated from first principles and is believed to be sound. The complete K4 string derived in this paper is 96 letters in length with a 97th letter (R) that remains unmapped, giving an overall string of.

KRYPTO SITE AM EAST / ROOND EAST NORTH EAST B STREET LEADS TO IT / A
CRONOMITER OF BERLIN CLOCKS BRILLIANSE / MORNING STAR(R)

The forward slashes represent my interpretation of how the message should be split, but is informed by a conscious bias brought about by an awareness of the sculptures other underlying properties that are discussed as part of the conclusion. This would appear at first glance to be a reasonably cryptic message replete with as many spelling mistakes as K1-3 put together, but there are aspects of it that give reason for confidence that portions of it at the very least (beyond the known plain text) are indeed correct. The attributes of this message that are deemed to be of interest are discussed in detail later, but before we proceed to the method it is perhaps worth taking the 1000 foot view and looking at two subtle features that follow from the nature of the general method used.

The first is that the proposed solution is predicated on the existence of 6 'Rail Positions'. Each of these Rail Positions consist of 3 consecutive rows of 32 letters, which sum in total to 96 letters. As we have a single letter remaining in K4 once we are done it can readily be seen that the plain text eventually creates a result that fits perfectly into the exact same dimensions as one of these standard Rail Positions.

The second and perhaps most subtle feature of this method is what at first glance may appear to be just a numerical coincidence, but equally may also be the intentional signature of the master puzzle makers that have put this entire scheme together.

It will be seen shortly in the discussion on method that we are selecting 8 characters for each cryptogram over rows that are 32 letters in length. Those that have researched the CIA's Langley campus for any additional art, inscriptions or seals looking for potential clues will know that etched into the wall of the original building's main lobby is a biblical verse which also characterizes the intelligence mission in a free society. It reads:

"And ye shall know the truth and the truth shall make you free."

John VIII-XXXII

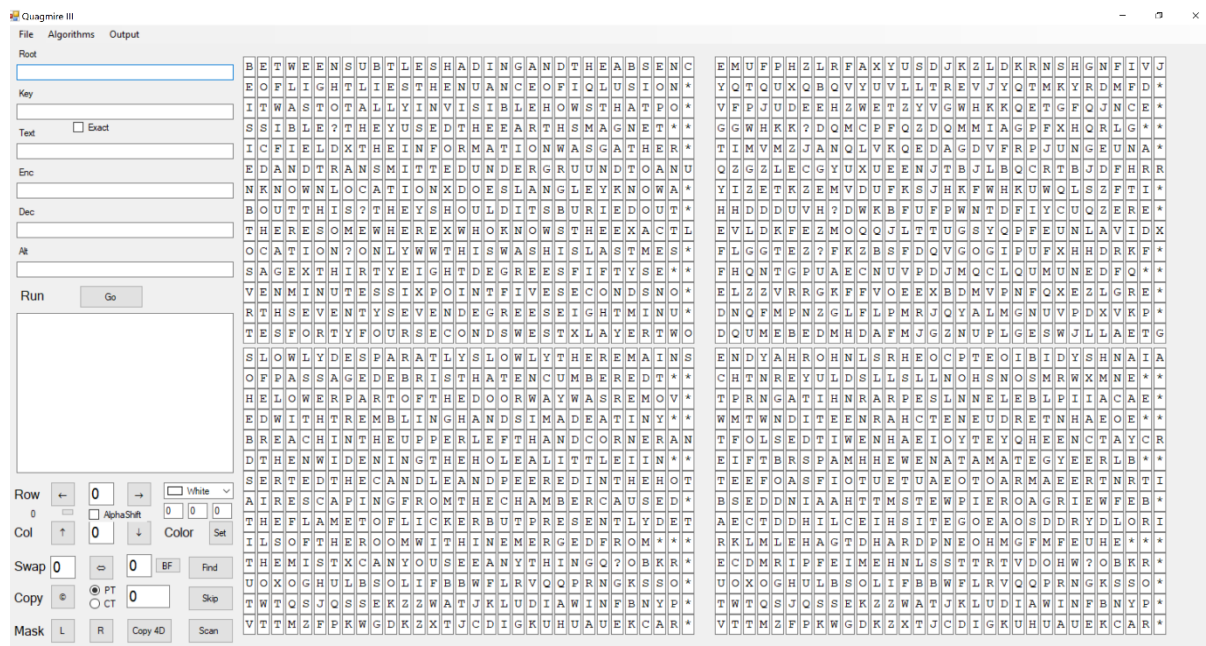
Prerequisites

In order to successfully replicate the results described in this paper, a software tool that is capable of applying the following transforms to the Kryptos cipher and plain Text is recommended, but this is purely for expedience. The results presented could equally be derived in long hand fashion simply using ‘pencil and paper’, the Kryptos cipher text and its associated tableau. The recommended transform capabilities for any software tool are:

- A method of showing the plaintext and cipher text side by side that allows transforms to be applied to both sides simultaneously. This leads to a display with four panels visible on screen as shown below.
- The shifting of individual rows to the right or left over a field of 32 characters, whereby a right shift will make the current 32nd character the 1st character and shift all subsequent character positions right by 1 position.
- A method of reciprocating any transform (or its opposite) in row n with a corresponding transform in row $(29-n)$. *This is a symmetrical transform that will be explained later.*
- A method of transforming the cipher and plain text in bulk by right shifting through the KRYPTOS alphabet by one letter at a time. (This operation can be done by hand by simply utilising a single row from the decryption tableau as a guide).

Screenshots used in this paper make use of bespoke software written by the author in order to achieve the functional requirements outlined above, but written tables are provided at the end of the paper in order to allow access to the same outputs as the software.

As can be seen here, the software tool gives a simultaneous view of both the plaintext and the cipher text. Any positional shifts applied to a given row are made in both views. Row 23 extends over 33 characters, but this is compensated for by the software, despite not showing in the matrix view below.



Evolution

Whilst this paper presents an end to end method for the decryption of K4, it should be born in mind that this paper is just one in a long line of previous attempts, each of which have fallen short of a complete solution. Anyone wanting to follow the evolution of the ideas from the outset can track them through the Kryptos.Groups.Io forum by reviewing the author's previous papers in the following order. All are available in the personal folders section under 'Matz K4 Decryption Candidate' and 'Matz Machine Paradigm'.

In order of appearance they are:

File	Date	Title	Notes
KryptosK4.pdf	October 2010	Kryptos K4. Evidence for a 44 Character Plain Text	Something of a naïve first paper on Kryptos, but with some interesting insights that are still relevant to this latest paper.
Valkyrie.pdf	October 2013	Valkyrie Evidence of a palimpsest in K1 and K2 discoverable via steganographic transforms of the Kryptos cipher and plain texts	A catalogue of interesting phenomena, but most importantly the first paper to document the steganographic scheme for the coordinates that was then taken further in the following two papers.
The Machine.pdf	June 2017	The Machine	A detailed breakdown of the various clues gathered to date and how they formed a link between the sculpture text, its physical layout and the Antikythera Mechanism.
The Machine r1.pdf	September 2017	The Machine	This was an edited version of the previous paper that removed any conjecture about K4 and added an additional section on using the tableau panels to compute the saros cycle.
Proposed Outline Decryption Method.pdf	May 2020	Proposed Outline Method For Decryption Of Kryptos K4	The first draft of the framework described in this paper.

Part 1

Method

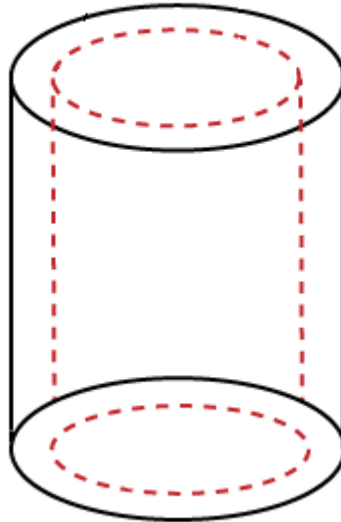
Creating a physical rendition of the cipher machine

Before discussing the decryption method in detail, we need to first construct a physical device that we will use for all steps of the K4 decryption process. The design of this device is not arbitrary, but is instead an informed design choice based on Jim Sanborn's Antipodes sculpture that was created in 1992 and is now on display at the Hirshhorn Museum in Washington DC.



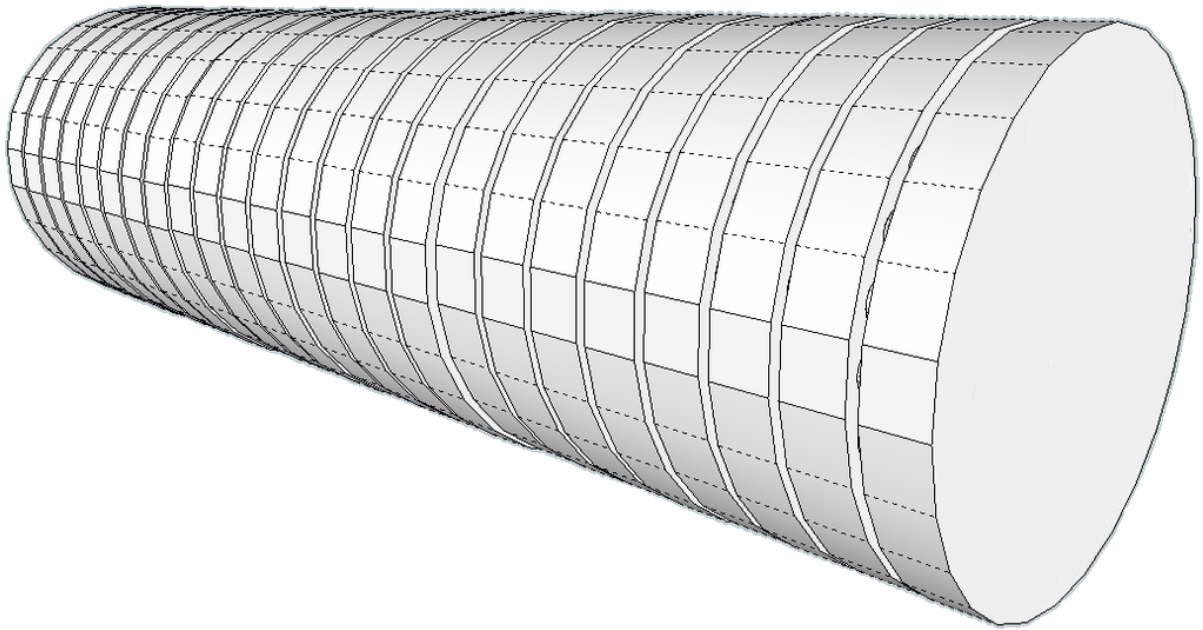
The design of Antipodes design as seen above would suggest that if we wrap the copper sheets like a scroll then it is possible to create two cylinders, one within the other. If we remove the extraneous letters T and L from both the cipher and tableau panels respectively then we can see that by putting the text into a regular matrix arrangement the cipher text has dimensions of 32 columns by 28 rows and similarly the tableau text has dimensions of 31 columns by 28 rows.

With these dimensions in mind we can see that it is possible to close both the cipher text and tableau panels into two closed cylinders, one of which will fit neatly within the other.



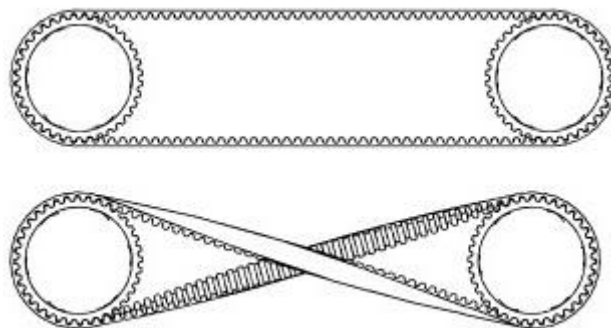
To complete the necessary construction of our cipher cylinder we need to add some additional kinetic elements. Firstly we will need to allow the inner cylinder to rotate freely about the axis. This means that the inner tableau cylinder will always be matched against 31 letters of the outer 32 letter cipher cylinder meaning that rotation about the central axis will cause different letters from the inner cylinder to be juxtaposed against different letters of the outer cipher text cylinder each time it moves by 1 letter position. The functionality that this feature provides will be discussed shortly.

The outer cipher text cylinder needs to have its 28 individual rows of 32 letters broken apart and given the freedom to move freely about the central axis in that same way as a Jefferson Cylinder. I've taken this rendition of a Jefferson Cylinder from the web and reworked it slightly, so it is deficient in 4 of the 28 rows, but shows the general concept of how we will need the rows to be able to move.



The final feature that we will need to add is reasonably unique and I would keen to hear from any readers of this paper as to whether they can propose a more elegant solution to this design requirement than the one I outline below.

As there are 28 individual rows there will naturally be an axis of symmetry between rows 14 and 15. Our final design requirement is that for any rotation of a particular row we will require the ability to shift the corresponding reciprocal row (about the axis of symmetry) either symmetrically or counter-symmetrically about the central axis. We will revert shortly to an idealised theoretical version of this setup whereby we only require one cylinder, but from a mechanical perspective its possible to achieve this functionality by creating a second replica of the cipher text cylinder and inverting it so that row 1 is at the bottom rather the top. Attaching a belt between rows of the two cylinders then gives symmetrical movement about the axis for the reciprocal rows. Attaching the belt in a twisted or figure of eight configuration would allow for counter symmetrical rotation instead.



A necessary part of the decryption will require us to be walked through two crib examples, both of which make use of the plaintext, so our final step will be to create a second version of this device that replaces the cipher text with the known plain text for parts 1 to 3. To complete rows 25 to 28 on this cylinder, we shall simply fill in with the standard K4 cipher text. Intriguingly, the idea of two cylinders existing one as a counterpart to the other has already been alluded in other works by Jim Sanborn such as the 2001 ‘Lux’ installation in Fort Meyers.



Flattening the device to a 2D representation

For the purposes of this paper the text shown will have to be in a flat 2D format, so any row shifts to the left or right will result in the characters at the edges wrapping round as they would do on the physical cylinder. In case it is not already obvious this also means that spaces are treated as physical characters that move with the respective rows and are not constant white space that pad row ends. An example is given here for the rotation of right shift rotation of row 2 for the cypher text cylinder.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

Row alignments after right shift of 1 letter on row 2. Note the movement of the space as well as the letters.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J
	Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

Rail Positions

After some exhaustive analysis to determine their uniqueness, it became apparent that the cipher text cylinder has six defined ‘rail position’ markers that are each defined by a distinct pattern of the repeating letters, R, E, V and F.

A ROT13 operation on these same letters gives ERIS. As Eris is also the name of the Greek goddess of chaos, strife and discord it was thought fitting that she should be entrusted as the guardian of K4’s concealment. Her counterpart in mythology is Harmonia, who symbolises union and marriage. The solution described herein seeks to gradually bring together a number of seemingly disparate clues into a single unified solution hence the title of this paper. .

The first two of these six markers are immediately apparent in the cipher text without the need for any row shifts and can be found in the following locations in rows 1-14.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G		
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A	
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I	
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E	
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F	
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q		
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E	
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P	
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G	

The presence of a third marker is then discovered via a left shift of 13 letters in row 1. As will be seen later in the paper this left shift of 13 characters is absolutely fundamental to making the initial break in K4.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G		
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A	
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I	
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E	
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F	
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q		
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E	
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P	
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

It can be seen that these three markers are in the horizontal plane and are all contained within the first 14 rows or ‘top left panel’. Three further markers can be identified that exist in the vertical plane and are more generally spread about both panels, whereby the letters RE extend upwards across rows and are capped with the letters F and V, each with a separation of 6 letters within the cylinder scheme.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G		
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A	
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I	
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E	
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F	
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q		
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E	
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P	
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G	
E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A
C	H	T	N	R	E	Y	U	L	D	S	L	L	S	L	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E		
T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E	
W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E		
T	F	O	L	S	E	D	T	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	C	R
E	I	F	T	B	R	S	P	A	M	H	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	E	R	L	B		
T	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	R	T	N	R	T	I
B	S	E	D	D	N	I	A	A	H	T	T	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B	
A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R	I
R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	P	N	E	O	H	M	G	F	M	F	E	U	H	E			
E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R	
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	

Creating a normalised picture for the vertical markers by shifting the rows appropriately gives a clearer picture of how they are supposed to look.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E		V	F
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G		
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A	
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I	
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E	
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F	
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q		
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E	
D	X	V	K	P		D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G	
E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A
C	H	T	N	R	E	Y	U	L	D	S	L	L	S	L	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E		
T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E	
W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E		
T	F	O	L	S	E	D	T	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	C	R
E	I	F	T	B	R	S	P	A	M	H	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	E	R	L	B		
T	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	R	T	N	R	T	I
B	S	E	D	D	N	I	A	A	H	T	T	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B	
A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R	I
P	N	E	O	H	M	G	F	M	F	E	U	H	E			R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	
S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R		E	C	D	M	R	I	P	F	E	I	M	E	H	N	L
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	

The known plain text for K4 along with the decryption method presented in this paper allows us to discern the order in which these rails are to be used, which are as follows.

Rail Position 1

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

Rail Position 2

Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I	
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E	
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X

Rail Position 3

F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q		
E	L	Z	Z	V	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E		
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P	

Rail Position 4

R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	P	N	E	O	H	M	G	F	M	F	E	U	H	E			
E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R	
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	

Rail Position 5

D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P	
D	Q		U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G
E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A

Rail Position 6

V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G		
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A	

The most important takeaway from the observation of these six defined rail positions is that they represent our first piece of evidence that the Kryptos cipher text is in fact a palimpsest written over the top of 12 discrete cryptograms.

Rail Positions 5 & 6

Rail positions 5 and 6 are both slightly anomalous for two very different, but related reasons. The overall solution will eventually be seen to be dependent on the use of two types of symmetry operation that occur in conjunction with the reciprocal rows for each of these rails. With regards Rail Position 5 this is impossible as the rows do not sit evenly across the axis of symmetry, which may account for why Jim Sanborn felt compelled to give BERLIN as the first clue as this would flush out any would be solves that may have stalled at this particular step. The method used on Rail Position 5 still uses the same general principles as the rest of the puzzle, but dispenses with the symmetry operations used by the other rails.

With regards Rail Position 6 there is an obvious issue with ‘too much’ symmetry being in play as it mirrors the positions for Rail Position 4 that have themselves already used the underlying symmetry relationships for those rows. If the puzzle maker did not take action to address this issue then the general solutions to Rail Position 4 could also be used to instantly break the alignment settings for Rail Position 6.

As a result of this it is conjectured that the puzzle maker has simply dropped one row down (or under) in order to compose Rail Position 6 of rows 4, 5 and 6 as shown here.

Rail Position 6 (Alternate. Repositioned in order to break symmetry with Rail Position 4)

G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G		
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A	
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R

A consequence of this downward row shift is the presence of the letters QED and END, which are a fitting marker to imply that this alternative row configuration may represent the final intended rail of the K4 solution. If F and V(B) had their proper six letter separation in that row then the plain text GRUUND would sit between them. It can be seen that GRUND is spelt out in the cipher text above.

Symmetry Operations

In order to understand the method used we will first need to explain two distinct types of symmetry operation that can be carried out on the cylinder. This has already been explained in the context of the physical model in 3D, but we shall revisit the topic again here so that it is explicit within the context of this 2D paper rendition.

The rows for the rail positions defined above (with the exception of rail position 5) have symmetrical counterparts defined by the relationship:

$n^* = 29 - n$, where n is the row number of any rail row being transformed by a shift operation. If we take rail position 1 as an example then this formula tells us that we would also have to perform a corresponding operation in row 28, by virtue of the relation $28 = 29 - 1$.

We can perform either a symmetrical or a counter-symmetrical shift operation on the reciprocal row. The reciprocal rows for rail position 1 are actually the full 3 rows of K4, so we can illustrate the effect of a symmetrical right shift by one letter on row 1 as follows.

Before symmetrical right shift of 1 letter in row 1

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	

After symmetrical right shift of 1 letter in row 1

J	E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	

A **counter-symmetrical** operation on row 1 would lead to a **left shift** in row 28 giving this result instead

J	E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	
T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R		V

Conventions used and glossary of terms

It would appear that we are dealing with a unique cipher type and as a result we will have to define some conventions and also introduce some new terms in order to best describe the decryption process concisely. The drafting of these terms will be seen to pay dividends later by allowing us to reduce the K4 decryption to a simple two-step process that is repeated for all 12 cryptograms.

C(n). The cryptogram number. This paper puts forward the idea that K4 is made up of 12 discrete cryptograms, which shall be labelled C(1) to C(12).

KRS(n) In the discussion on the nature of the physical cipher device it was noted that the inner cylinder on which the tableau is written can rotate by one letter position at a time. In order to emulate the effect of the changing juxtaposition of tableau letters against cipher text or plain text letters we shall simulate this by right shifting by one letter value each time using the Kryptos alphabet.

KRYPTOSABCDEFGHIJLMNQUVWXZ

To perform a right shift operation on any arbitrary piece of text using the Kryptos alphabet we simply need to look up the current position for any letter of that text in the alphabet above and move one position to the right. The most simple example to use would of course be the word KRYPTOS itself. It can be seen that a right shift of one letter through the Kryptos alphabet will result in the letters RYPTOSA being returned. Another subsequent right shift operation would result in the letters YPTOSAB being our next output. Every time we break a cryptogram for a given rail configuration a global ‘Kryptos Right Shift’ is applied to all of the plain and cipher text on both of our cylinders. There is no **KRS** applied for **C(1)**, so its initial setting is $n=0$, but all subsequent right shifts performed are **KRS(n+1)** every time we have extracted a cryptogram and are about to move to either the next configuration or the next rail position.

RP(n). The rail position number 1-6. Each rail position will have two configurations.

SA(CS/SS). The two configurations for each rail position correspond to the algorithm that is necessary to apply in each instance. Odd numbered cryptograms will always have the counter-Symmetric algorithm applied (CS) and even numbered cryptograms will always make use of the simple-symmetric algorithm (SS).

For each of the two rail position configurations we shall define three alignment settings, which are the **IRP**, **NSP** and **TSP**.

IRP is the ‘initial row position’ and is useful to document each time, so that anyone wishing to verify these results or do more research can simply cut and paste the text, which will already have had the correct amount of right shifting applied.

NSP is the ‘normalised symmetry position’ and is the row alignment in which 3 letter mirroring *within* the row can be seen as well as the characters that will remain ‘invariant under transformation’ when we move the middle rail to the ‘transformed symmetry position’.

TSP is the ‘transformed symmetry position’ in which the middle row of the rail will have its invariant letters shifted to their second possible configuration other than the **NSP**.

IUT 2:2[i/j] These are the 4 letters that remain ‘invariant under transformation’ from the **NSP** to the **TSP**. Letters in the top or bottom row will be defined as i and the letters in the middle row as j.

IUT 3:4[i/j] These are the 3 letters that initially occur in a contiguous block that is either horizontal, split or vertical in the **NSP** that will remain invariant and be present in the **TSP** position as 3 out of 4 letters of a 2:2 block of 4 letters.

IUT H:V[i/j] The i/j parameters are for letters that will be read in the horizontal plane for the **NSP** and in the vertical plane for the **TSP**.

RML[ijk]. These are the 3 letters that are mirrored within the row for any given **NSP**.

SML[ijk]. These are the letters that are mirrored between the rail position rows and their reciprocal row positions when the counter-symmetrical algorithm has been run to arrive at the **TSP**.

It is appreciated that these terms and conventions may initially seem complex before viewing the plain text derivations, but it will allow for all cryptograms to be broken on a repeatable basis using a common template that will make for easy comparison between cases.

R(n) For any given rail position we shall use this to denote a specific rail, namely 1, 2 or 3.

RS/LS(n) The amount of left or right shift that has been applied to a given rail between transformations. For all of the 12 cryptograms it is only ever rail 2 that is shifted left or right in order to transition from the **NSP** to the **TSP**.

LBA. We shall refer to a ‘letter block alignment’ as two letters that sit directly over each other between rows.

A brief note on Vigenère encryption

The final part of the decryption process appears to cause some confusion even amongst those familiar with the cryptographic techniques involved. The principle of symmetrical encryption means that just like the rest of the framework that has utilised symmetry to some degree, the final method of decoding is no different. When using the Vigenère scheme, the use of encryption is purely the reversal of the process that was used to decrypt parts one and two. The puzzle makers have chosen to only reveal the final plain text when the mirror image of the original decryption process is undertaken. An example is given here for OBKRUOX, the first 7 letters of K4.

The key required to **decrypt** OBKRUOX to give a plain text KRYPTOS is OCYTZDT.

The plain text required to be **encrypted** with a key of OBKRUOX to give a cipher text of KRYPTOS is UNYYCKB.

The puzzle makers have opted to use this second method instead. As the process is symmetrical the level of cryptographic security is identical. This means that the end result that is read as plain text is ‘technically’ classified as cipher text, but the output of this process for each 8 letter cryptogram is clearly intended to be read as the plain text.

General Method

The discovery of the general method and how it was derived from the two crib examples that accompany cryptograms 1 and 2 is discussed in a low level of detail in part 3, but for those not looking to replicate results and that purely want to see the end results then part 2 will give something of a pictorial overview of the complete scheme. The underlying principle is reasonably simple, but is made obscure by the application of two additional layers of obfuscation.

The basis on which the entire scheme is predicated is that of ‘invariance under transformation’ or IUT.

There are three types of IUT scheme that are used repeatedly for each cryptogram. *Note. The third type of IUT only starts to appear from C(7) onwards.*

The first type of IUT is what we have just defined in our glossary of terms as the **IUT 2:2** type. This is perhaps the easiest type to explain and we shall use C(1) as an example of how this works.

We can see that by shifting the first row of the cipher text left by 13 letters that we arrive at the following alignment scheme for the first two rows of the cipher text.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	

If we look at the letters U and N in the above alignment then we can clearly see the 4 ‘letter block alignment’ LBA that forms UN/YY in rows 1 and 2 respectively.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

Applying a left shift of 10 letters to the bottom row results in the following new relationship between the rows. The Y that was previously the left most Y in the 4 letter block retains its colouring to show its new position after the row shift.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		Y	Q	T	Q	U	X	Q	B	Q	V
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

It is not only the invariant letter blocks that are important, but also the start and end position for the row alignments. The initial left shift of 13 letters in row 1 put us into what we have specified as being the ‘normalised symmetry position’ or NSP.

The NSP position is one in which there will be a defined symmetry characteristic of three letters occurring in the vertical plane twice in one rail position. In the case of the first cryptogram this occurs in the corresponding plain text for rows 1-3 and can be shown here. Remember that to get into the NSP alignment all we have done is to left shift row 1 by 13 letters.

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N	
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	

When we apply a shift to the middle rail to take up the second possible configuration that is not the NSP then we say that we have shifted to the ‘Transformed Symmetry Position’ TSP.

The entire solve only requires that we find the required NSP for each rail position and shift the middle rail in order to arrive at the TSP. The 8 letters that we use to construct each cryptogram result will always be read from the TSP. The number of row configurations that allow for these symmetrical relationships are much higher than one might suspect, but as we shall see the degrees of freedom that allow us to define the NSP are limited by a number of other criteria. The first of these criteria is that the NSP has to allow for the presence of the IUT 3:4 type, which is the presence of 3 letters in the NSP that must be present in the TSP and will give us 3 out of 4 letters in an eventual set of 2 LBA's in the 2:2 configuration.

An example is given here for the C(1) NSP.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

The same letters are then found in their 2:2 configuration in the TSP.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		Y	Q	T	Q	U	X	Q	B	Q	V
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

The format of the IUT 3:4 type can be one of 3 types. Horizontal, Split or Vertical. Examples of those configurations are given here. The DK/BQ configuration given in the previous example is obviously an example of a split configuration. In this diagram the Horizontal configuration is shown in orange, the Split in blue and the Vertical in green.

The final IUT type is IUT H:V and occurs when 2 LBA's can be read off horizontally in the NSP and then vertically in the TSP. An example is given here for C(9), which encapsulates the majority of the BERLIN CLOCK section of K4. *As the cryptograms are read 8 letters at a time, the eventual plaintext for C(9) is ERLI NCLO.*

The NSP position for C(9) is as follows and as can be seen from the coloured blocks the invariant letters are IL and MN when read horizontally.

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M	I	Q	K	X	U	A	R	P	E	Z	U
I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W

Upon shifting the middle rail to the TSP position we can see that the invariance has shifted to the vertical plane.

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
I	Q	K	X	U	A	R	P	E	Z	U	N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M
I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W

From cryptogram C(7) onwards all LBA's are read vertically rather than horizontally. This tallies with the change in the rail marker orientations for rail positions 4 onwards, but interestingly a gradual transition is noted in cryptograms C(5) and C(6) whereby the last four letters are read in a hybrid diagonal manner for C(5) and in the vertical plane for the final 4 letters of C(6).

There are some additional features of interest which will be loosely described here as Symmetry Mirror Letters, but these are artifacts that occur consistently in the TSP. For all odd cryptograms up to C(5) there is a mirrored set of 3 diagonal letters that would appear to be additional confirmation that the row settings are correct and that we haven't wandered off track due to the available degrees of freedom. An example of SML shall be given here for C(1) in which the rail position rows (in rows 1-3) show a mirroring relationship in their reciprocal rows (rows 26-28).

PT

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		E	O	F	L	I	G	H	T	L	I
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	

*PT

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
A	W	I	N	F	B	N	Y	P		T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I
G	K	U	H	U	A	U	E	K	C	A	R		V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

There are two distinct algorithms that make use of the IUT scheme. The counter symmetric algorithm used for odd numbered cryptograms is very straight forward and should be self-explanatory to some degree even before we delve into a lower level of detail in the next section, but the simple-symmetric algorithm used for the even numbered cryptograms is a different beast altogether. This algorithm dictates that we have to solve using the same IUT scheme used for the odd numbered cryptograms, but from the perspective of the reciprocal rows. The invariant letters selected from the Rail Position in focus do not used the IUT2:2 scheme, but the IUT 3:4 scheme instead, which makes solving the even numbered cryptograms much harder than for the odds in both respects. For the purposes of this first pass it may seem confusing at first as to how these even numbered cryptograms are broken, but the method derivation in part 2 should start to clear that up the missing detail before all 12 cryptograms are re-examined again at a much lower level of detail in part 3. This will also give readers the materials they need to take their own research further with a complete set of correctly transformed grids for each cryptogram.

For the purposes of parsing the overview of the results given in part 2 the only thing that needs to be borne in mind is that we will be alternating between the Counter-Symmetric and Simple-Symmetric for C(1-8) and C(11-12). How these alternations in scheme are mapped to the K4 key is given here.

E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R

Orange corresponds to the CS algorithm for the odds and green to the SS algorithm for the evens. The blue region specifies cryptograms C(9-10) which have no symmetry relationships. The final R is the K4 letter that doesn't map.

A distinct colour scheme has been devised to differentiate between the IUT 2:2 ,IUT 3:4, RML and SML as they transition from the NSP to TSP positions. Remember that this transform only ever requires the movement of the middle rail. If anything is still not clear after this high level overview then please look for more detail against the particular cryptogram in the next section. For these initial results, screenshots will be taken directly from the authors bespoke software that will show the plain and cipher text side by side with the appropriate level of Kryptos right shifting applied. As a final reminder before the first pass results are shown, all plain and cipher text is shifted right by 1 letter value in the Kryptos alphabet every time we break a cryptogram.

The colour convention used will be.

	RML Row Mirrored Letters
	SML Symmetry Mirrored Letters
	IUT 2:2 Blocks of 2 across rows that remain invariant
	IUT 2:2 Blocks of 2 that are the dynamic component of the invariant scheme
	IUT 3:4 Blocks of 4 letters that contain 3 of the final 4 in the TSP

All results will only require the display of two distinct row alignments, which are the NSP and TSP. It will be seen that it is only the position of the middle row that is changed in each case. The current rail position and it's reciprocal rows are additionally delimited within the grey boundary. The use of the previously defined terms and conventions should now become as apparent as we move through on a case by case basis.

Remember also that LBA's are consistently read off in the vertical plane from C(7) onwards.

Part 2

Cryptogram Solves

C(1 – 12)

First Pass

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E	
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N	*	
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	*	
S	S	I	B	L	E	?	T	H	E	Y	U	S	E	D	T	H	E	E	A	R	T	H	S	M	A	G	N	E	T	*	*	
I	C	F	I	E	L	D	X	T	H	E	I	N	F	O	R	M	A	T	I	O	N	W	A	S	G	A	T	H	E	R	*	
E	D	A	N	D	T	R	A	N	S	M	I	T	T	E	D	U	N	D	E	R	G	R	U	N	D	T	O	A	N	U		
N	K	N	O	W	N	L	O	C	A	T	I	O	N	X	D	O	E	S	L	A	N	G	L	E	Y	K	N	O	W	A	*	
B	O	U	T	H	I	S	?	T	H	E	Y	S	H	O	U	L	D	I	T	S	B	U	R	I	E	D	O	U	T	A	*	
T	H	E	R	E	S	O	M	E	W	H	E	R	E	X	W	H	O	K	N	O	W	S	T	H	E	E	X	A	C	T	L	
O	C	A	T	I	O	N	?	O	N	L	Y	W	H	I	S	W	A	S	H	I	S	L	A	S	T	M	E	S	*			
S	A	G	E	X	T	H	I	R	T	Y	E	I	G	H	T	D	E	G	R	E	E	S	F	I	F	T	Y	S	E	*	*	
V	E	N	M	I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	*	
R	T	H	S	E	V	E	N	T	Y	S	E	V	E	N	D	E	G	R	E	E	S	E	I	G	H	T	M	I	N	U	*	
T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	E	T	X	L	A	Y	E	R	T	W	O			
S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S	
O	F	P	A	S	S	A	G	E	D	E	B	R	I	S	T	H	A	T	E	N	C	U	M	B	E	R	E	D	T	*	*	
H	E	L	O	W	E	R	P	A	R	T	O	F	T	H	E	D	O	O	R	W	A	S	R	E	M	O	V	*				
E	D	W	I	T	H	T	R	E	M	B	L	I	N	G	H	A	N	D	S	I	M	A	D	E	A	T	I	N	Y	*	*	
B	R	E	A	C	H	I	N	T	H	E	U	P	P	E	R	L	E	F	T	H	A	N	D	C	O	R	N	E	R	A	N	
D	T	H	E	N	W	I	D	E	N	I	N	G	T	H	E	H	O	L	E	A	L	I	T	T	L	E	I	I	N	*	*	
S	E	R	T	E	D	T	H	E	C	A	N	D	L	E	A	N	D	P	E	E	R	E	D	I	N	T	H	E	H	O	T	
A	I	R	E	S	C	A	P	I	N	G	F	R	O	M	T	H	E	C	H	A	M	B	E	R	C	A	U	S	E	D	*	
T	H	E	F	L	A	M	E	T	O	F	L	I	C	K	E	R	B	U	T	P	R	E	S	E	N	T	L	Y	D	E	T	
I	L	S	O	F	T	H	E	R	O	O	M	W	I	T	H	I	N	E	M	E	R	G	E	D	F	R	O	M	*	*	*	
T	H	E	M	I	S	T	C	A	N	O	Y																					

C(1) KRS(0) SA(CS) SML[ANL] IUT2:2[UN/YY] IUT3 :4[DK/BQ] TSP

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E		U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y		
E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	S	U	S	I	O	N	*	E	O	F	L	I	G	H	T	L	I		Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	*	Y	Q	T	Q	U	X	Q	H	Q	V	
I	T	W	A	S	T	O	T	A	L	L	I	N	V	I	S	I	B	L	E	H		H	O	W	S	T	H	A	T	P	O	*		V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	*	
S	S	I	B	L	E	?	T	H	E	Y		U	S	E	D	T	H	E	E	A	R	T	H	S	M	A	G	N	E	T	*	*		G	G	W	H	K	K	?	D	O	M	C	P	F	Q	Z	D	O	M	M	I	A	G	P	F	X	H	O	R	L	G	*	*	
I	C	F	I	E	L	D	X	T	H	E	I	N	F	O	R	M	A	T	I	O	N	W	A	S	G	A	T	H	E	R	*			T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	E	G	U	A	*		
E	K	A	N	D	T	R	A	N	S	M	I	T	T	E	D	U	N	D	E	R	G	R	O	U	N	D	T	O	A	N	U			Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	N	J	T	B	J	L	B	H	Q	C	R	T	B	J	D	F	H	R	*	
N	K	N	O	W	N	L	O	C	A	T	I	O	N	X	D	O	E	S	L	A	N	G	L	E	Y	K	N	O	W	A	*			Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	Q	U	L	S	Z	F	T	I	*			
B	O	U	T	H	I	S	?	T	H	E	Y	S	H	O	U	L	D	I	T	S	B	U	R	I	E	D	O	U	T	*			H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	*			
T	H	E	R	E	S	O	M	E	W	H	E	R	E	X	W	H	O	K	N	O	W	S	T	H	E	E	X	A	C	T	L			E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X	
O	C	A	T	I	O	N	?	O	N	L	Y	W	W	T	H	I	S	W	A	S	H	I	S	L	A	S	T	M	E	S	*			F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F	*	
S	A	G	E	X	T	H	I	R	T	Y	E	I	G	H	T	D	E	G	R	E	E	S	F	I	F	T	Y	S	E	*			F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q	*	*		
R	V	N	M	I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	*			E	L	Z	Z	V	M	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	G	F	Q	X	E	Z	L	G	R	*
T	H	S	E	V	E	N	T	Y	S	E	V	E	N	D	E	G	R	E	E	S	E	I	G	H	T	M	I	N	U	*			D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	N	G	U	V	P	D	X	V	K	P	*		
T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	W	E	S	T	X	L	A	Y	E	R	T	W	O			D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	S	W	J	L	A	E	T	G			
S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S			E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A	
O	F	P	A	S	S	A	G	E	D	E	B	R	I	S	T	H	A	T	E	N	C	U	M	B	E	R	E	D	T	*	*			C	H	T	N	R	E	Y	U	L	D	S	L	S	L	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E	*	*		
H	E	L	O	W	E	R	P	A	R	T	O	F	T	H	E	D	O	O	R	W	A	Y	W	A	S	R	E	M	O	V	*			T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E	*	
E	D	W	I	T	H	T	R	E	M	B	L	I	N	G	H	A	N	D	S	I	M	A	D	E	A	T	I	N	Y	*	*			W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E	*	*	
B	R	E	A	C	H	I	N	T	H	E	U	P	P	E	R	L	E	F	T	H	A	N	D	C	O	R	N	E	R	A	N			T	F	O	L	S	E	D	T	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	C	R	
D	T	H	E	N	W	I	D	E	N	I	N	G	T	H	E	H	O	L	E	A	L	I	T	T	L	E	I	I	N	*			E	I	F	T	B	R	S	P	A	M	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	E	R	L	B	*	*			
S	E	R	T	E	D	T	H	E	C	A	N	D	L	E	A	N	D	P	E	E	R	E	D	I	N	T	H	E	H	O	T			T	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	R	T	N	R	T	I	*
A	I	R	E	S	C	A	P	I	N	G	F	R	O	M	T	H	E	C	H	A	M	B	E	R	C	A	U	S	E	D	*			B	S	E	D	D	N	I	A	A	H	T	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B	*		
T	H	E	F	L	A	M	E	T	O	F	L	I	C	K	E	R	B	U	T	P	R	E	S	E	N	T	L	Y	D	E	T			A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R	I	
I	L	S	O	F	T	H	E	R	O	O	M	W	I	T	H	I	N	E	M	E	R	G	E	D	F	R	O	M	*	*	*			R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	P	N	E	O	H	M	G	F	M	F	E	U	H	E	*	*	*	
T	H	E	M	I	S	T	X	C	A	N	Y	O	U	S	E	A	N	Y	T	H	I	N	G	Q	?	O	B	K	R	*			E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	R	T	V	D	O	H	W	?	O	B	K	R	*			
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	P	R	N	G	K	S	S	O	*			U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	P	R	N	G	K	S	S	O	*			
A	W	I	N	F	B	N	Y	P	*	T	W	T	Q	S	J	O	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I			A	W	I	N	F	B	N	Y	P	*	T	W	T	Q	S	J	O	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	
G	K	U	H	U	A	U	E	K	C	A	R	*	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I			G	K	U	H	U	A	U	E	K	C	A	R	*	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	

23

C(2) RP1 KRS(1) SA(SS) SML[HUN] IUT2:2[AS/MV] IUT3 :4[WL/FN] IUT3 :4[NG/RR] NSP

QDCFOXFFQAVCOMFAIBEJQHBOEIOIFBCAF	WLFNVGTIKMYGBZPVAELRKKMERYYQAIHQGJ
Q*FSGMJHIOMJFAOIFQVBQDFSGJUMVJAS	E*PUOUVZUCUWVWMMOYFWLPUONRPFYENG
ISXAOIBOTS*JOXBASOSOBMMFJQWJAJCMF	UFOHGULODF*WGTLVEFFIKXFOKFPWHXIRR
AAJCMF?OIFPVAFEOIFFBYOIANBHQFO**	HXXIRR?EUNDTGUEUNNJBHTGTZIUYMHH**
JDGJFMEZOIFJQGSYNBOJSQXBAHBOIFY*	OJNWNKLBQUMWRUFEBHEWGYTLVQHFBVQB*
FEBQEOYBQANJOOFQEFYHYVVQEOSBQV	UKHKMFDPVZVFFQLOCLMCUDYOCLEGIYY
QRQSXQMSDBOJSQZESFAMBQHMFPRQSXB*	PJKFORKKFNWEVGRALIRGXIRVXUMAKGOJ*
CSVOOIJAJ?OIFPAISVMEJOACVYJFESVO*	IIIEEEVWI?EXRCGVGTXQOEGJPDVUKFYF*
OIFYFASNFXIFYFZXISRQSAOIFFZBDOM	FWMERGFKNSSUULMOOVHAPUTGFGVQMBWJEZ
SDBOJSQ?SQMPXXOIJAXBAIJAMBAAONFA*	GMMHOFK?GRKCAGEUWHSHTJTVGZIIIEYRG*
ABHFZIOIJYOPFJHIOEFHYFFAGJGOPAF**	GIUQOHTVBFDQVWTELNUDMUVNVQFEGU**
WFQNJQVOFAAJZTSJQOGJWFADFDSQEAQS*	FMKKWYHYHRRGGWSFFZCENWTQGUZFKMHYF*
YOIAFWFQOPAFWFQEFHYFFAFJHIONJQV*	EQUGNTQKHMGMNTNYLUPBMNHQVWTEZWRRT*
OFAGSYOPGGSVYAFADSQEAAXFAOZMBPFFYOXS	EUVNFCFENIEBGNLHKQVMTMHFAAXLMMBFOH
AMSXMPPEFATBYBOMPAMSXMPPOIFYFNBQJQA	FQEPBBIYSIQMAYIIFSDDTOFSJJCJEPAIQBJB
SGTBAAABHFEFCYJAOIBOFQDVCIFYFEO**	DIOQYFPVMEAMMAMMQSIAQSANYXZNQF**
IFMSXFYTYBYOSGOIFESSYXBPXBAYFNBSW*	OTYQHBBOJIIQYBYTFAMQQFMFCMTJJBDDBF*
FEXJOIOYFNCMJQHIBQEAJNBEBFBOJQFP**	XNOXQEQJOFFQYBIDOFQFVEYFOQIBFSF**
CYFBDIJQOIFVTTFYMFQGOIBQEDSYQFYBQ	OGSMFAFEQJXQFQIBFJSPOFPUIFFQDOBPDY
EOIFQXJEFQJQHIOIFISMFBMJOOMFJJQ**	FJGOCYATBNIIIFXFBQOBNNBOFHFFYQMC**
AFYOFEOIFDBQEMFBOETFFFYFEJQOIFISO	OJFGSBBAGJSOVFOVBFSSOSBYNBBFFYOQYOJ
BJYFADBTJQHGYSNOIFDIBNCFYDBVAFE*	CAFEEQJBBIOONAOFXJTJFYSBHYJFXGFC*
OIFGMBNBFOSGMJDRFYCVOTYFAFQOMPEFO	BFDQEEIJMDJFIJAJOFHSSFBSSAEEYPEMSYJ
JMASGOIFYSSNXJOIJQFNFYHFEGYSN**	YRMNMFIHBOEIBYETQFSINHGNGFVIF***
OIFNJAOZDBQPSVAFFBQPOIJQHUT?SCRY*	FDENYJTGJFNFIQMAAOYOOWESIX?SCRY*
UUTYQHRAAS*VSZSHIVMCASMJGCCXGMYW	UUTYQHRAAS*VSZSHIVMCASMJGCCXGMYW
T*OXOUALUAAFRKKXBOLRMVEJJBXJQGCQP	T*OXOUALUAAFRKKXBOLRMVEJJBXJQGCQP
Y*WOONKGTTRXHERKZOLDEJHRVIVBVFRDB	Y*WOONKGTTRXHERKZOLDEJHRVIVBVFRDB

R(2)LS(12) to the TSP gives.

C(2) RP1 KRS(1) SA(SS) IUT2:2[AS/MV] IUT3 :4[LW/FR] IUT3 :4[PR/RG] TSP

QDCFOXFFQAVCOMFAIBEJQHBOEIOIFBCAF	WLFNVGTIKMYGBZPVAELRKKMERYYQAIHQGJ
FAOIFQVBQDFSGJUMVJASQ*FSGMJHIOMJ	PVWMMMOYFWLPUONRPFYENG*PUOUVZUCUW
ISXAOIBOTS*JOXBASOSOBMMFJQWJAJCMF	UFOHGULODF*WGTLVEFFIKXFOKFPWHXIRR
AAJCMF?OIFPVAFEOIFFBYOIANBHQFO**	HXXIRR?EUNDTGUEUNNJBHTGTZIUYMHH**
JDGJFMEZOIFJQGSYNBOJSQXBAHBOIFY*	OJNWNKLBQUMWRUFEBHEWGYTLVQHFBVQB*
FEBQEOYBQANJOOFQEFYHYVVQEOSBQV	UKHKMFDPVZVFFQLOCLMCUDYOCLEGIYY
QRQSXQMSDBOJSQZESFAMBQHMFPRQSXB*	PJKFORKKFNWEVGRALIRGXIRVXUMAKGOJ*
CSVOOIJAJ?OIFPAISVMEJOACVYJFESVO*	IIIEEEVWI?EXRCGVGTXQOEGJPDVUKFYF*
OIFYFASNFXIFYFZXISRQSAOIFFZBDOM	FWMERGFKNSSUULMOOVHAPUTGFGVQMBWJEZ
SDBOJSQ?SQMPXXOIJAXBAIJAMBAAONFA*	GMMHOFK?GRKCAGEUWHSHTJTVGZIIIEYRG*
ABHFZIOIJYOPFJHIOEFHYFFAGJGOPAF**	GIUQOHTVBFDQVWTELNUDMUVNVQFEGU**
WFQNJQVOFAAJZTSJQOGJWFADFDSQEAQS*	FMKKWYHYHRRGGWSFFZCENWTQGUZFKMHYF*
YOIAFWFQOPAFWFQEFHYFFAFJHIONJQV*	EQUGNTQKHMGMNTNYLUPBMNHQVWTEZWRRT*
OFAGSYOPGGSVYAFADSQEAAXFAOZMBPFFYOXS	EUVNFCFENIEBGNLHKQVMTMHFAAXLMMBFOH
AMSXMPPEFATBYBOMPAMSXMPPOIFYFNBQJQA	FQEPBBIYSIQMAYIIFSDDTOFSJJCJEPAIQBJB
SGTBAAABHFEFCYJAOIBOFQDVCIFYFEO**	DIOQYFPVMEAMMAMMQSIAQSANYXZNQF**
IFMSXFYTYBYOSGOIFESSYXBPXBAYFNBSW*	OTYQHBBOJIIQYBYTFAMQQFMFCMTJJBDDBF*
FEXJOIOYFNCMJQHIBQEAJNBEBFBOJQFP**	XNOXQEQJOFFQYBIDOFQFVEYFOQIBFSF**
CYFBDIJQOIFVTTFYMFQGOIBQEDSYQFYBQ	OGSMFAFEQJXQFQIBFJSPOFPUIFFQDOBPDY
EOIFQXJEFQJQHIOIFISMFBMJOOMFJJQ**	FJGOCYATBNIIIFXFBQOBNNBOFHFFYQMC**
AFYOFEOIFDBQEMFBOETFFFYFEJQOIFISO	OJFGSBBAGJSOVFOVBFSSOSBYNBBFFYOQYOJ
BJYFADBTJQHGYSNOIFDIBNCFYDBVAFE*	CAFEEQJBBIOONAOFXJTJFYSBHYJFXGFC*
OIFGMBNBFOSGMJDRFYCVOTYFAFQOMPEFO	BFDQEEIJMDJFIJAJOFHSSFBSSAEEYPEMSYJ
JMASGOIFYSSNXJOIJQFNFYHFEGYSN**	YRMNMFIHBOEIBYETQFSINHGNGFVIF***
OIFNJAOZDBQPSVAFFBQPOIJQHUT?SCRY*	FDENYJTGJFNFIQMAAOYOOWESIX?SCRY*
UUTYQHRAAS*VSZSHIVMCASMJGCCXGMYW	UUTYQHRAAS*VSZSHIVMCASMJGCCXGMYW
RKKXBOLRMVEJJBXJQGCQP	RKKXBOLRMVEJJBXJQGCQP
Y*WOONKGTTRXHERKZOLDEJHRVIVBVFRDB	Y*WOONKGTTRXHERKZOLDEJHRVIVBVFRDB

C(2) KRS(1) SA(SS)	K4 Key	Encryption
LWFR PRRG	GHUL BSOL	TEAM EAST

C(3) RP2 KRS(2) SA(CS) RML[NFW] IUT2:2[GZ/OJ] IUT3 :4[FW/FN] NSP

D	G	S	Z	G	G	U	B	W	D	S	N	G	B	J	C	F	L	U	I	C	U	F	S	J	G	C	D	B	G	U	E		
G	A	H	N	L	I	J	S	N	L	G	B	S	J	G	U	W	C	U	E	G	A	H	L	V	N	W	B	L	A	U	*		
L	S	Z	C	B	S	A	S	C	N	N	T	L	U	X	L	B	L	D	N	G	J	A	Z	B	S	J	C	S	O	A	*		
B	B	L	D	N	G	?	S	J	G	T	W	B	G	F	S	J	G	C	P	S	J	B	Q	C	I	U	G	S	*	*			
L	E	H	L	G	N	F	K	S	J	G	L	U	H	A	P	Q	C	S	L	A	U	Z	C	B	I	C	S	J	G	P	*		
G	F	C	U	F	S	P	C	U	B	Q	L	S	S	G	F	W	U	F	G	P	I	P	W	U	F	S	A	C	U	W			
U	Y	U	A	Z	U	N	A	E	C	S	L	A	U	K	F	A	G	B	N	C	U	I	N	G	T	Y	U	A	Z	C	*		
S	J	G	T	B	J	A	W	N	F	L	S	B	D	W	P	L	G	F	A	W	S	*	D	A	W	S	S	J	L	B	?		
S	J	G	G	K	C	E	S	N	S	J	G	P	G	B	A	Q	G	Z	J	G	P	G	K	Z	J	A	Y	U	A	Z	B		
A	E	C	S	L	A	U	?	A	U	N	T	Z	Z	S	J	L	B	Z	C	B	J	L	B	N	C	B	S	Q	G	B	*		
B	C	I	G	K	S	J	L	P	S	T	G	L	I	J	S	F	G	I	P	G	G	B	H	L	H	S	T	B	G	*	*		
X	G	U	Q	L	U	W	S	G	B	B	L	K	O	A	L	U	S	H	L	X	G	B	G	E	A	U	F	B	U	A	*		
P	S	J	B	G	X	G	U	S	T	B	G	X	G	U	F	G	I	P	G	G	B	G	L	I	J	S	Q	L	U	W	*		
S	G	B	H	A	P	S	T	H	A	W	P	B	G	E	A	U	F	B	Z	G	B	S	K	N	C	T	G	P	S	Z	A		
B	N	A	Z	N	T	F	G	B	O	C	P	C	S	N	T	B	N	A	Z	N	T	S	J	G	P	G	Q	C	L	U	B		
A	H	O	C	B	B	C	I	G	F	G	D	P	L	S	J	C	S	G	U	E	W	Q	D	G	P	G	F	S	*	*			
J	G	N	A	Z	G	P	O	C	P	S	A	H	S	J	G	F	A	A	P	Z	C	T	Z	C	B	P	G	Q	A	X	*		
G	F	Z	L	S	J	S	P	G	Q	D	N	L	U	I	J	C	U	F	B	L	Q	C	F	G	C	S	L	U	T	*	*		
D	P	G	C	E	J	L	U	S	J	G	W	O	O	G	P	N	G	H	S	J	C	U	F	E	A	P	U	G	P	C	U		
F	S	J	G	U	Z	L	F	G	U	L	I	U	I	S	J	G	J	A	N	G	C	N	L	S	S	N	G	L	L	U	*	*	
B	G	P	S	G	F	S	J	G	E	C	U	F	N	G	C	U	F	O	G	G	P	F	L	U	S	J	G	J	A	S	*		
C	L	P	G	B	E	C	O	L	U	I	H	P	A	Q	S	J	G	E	J	C	Q	D	G	P	E	C	W	B	G	F	*		
S	J	G	H	N	C	Q	G	S	A	H	N	L	E	Y	G	P	D	W	S	O	P	G	B	G	U	S	N	T	F	G	S		
L	N	B	A	H	S	J	G	P	A	A	Q	Z	L	S	J	L	U	G	Q	G	P	I	G	F	H	P	A	Q	*	*	*		
S	J	G	Q	L	B	S	K	E	C	U	T	A	W	B	G	G	C	U	T	S	J	L	U	I	V	?	A	D	Y	P	*		
W	A	K	A	I	J	W	N	D	B	A	N	L	H	D	D	Z	H	N	P	X	V	O	P	U	I	Y	B	B	A	*			
S	Z	S	V	B	M	V	B	B	G	Y	R	R	Z	C	S	M	Y	N	W	F	L	C	Z	L	U	H	D	U	T	O	*		
X	S	S	Q	R	H	O	Y	Z	I	F	Y	R	K	S	M	E	F	L	I	Y	W	J	W	C	W	G	Y	E	C	P	*		
G	Q	W	H	O	J	R	N	P	H	C	R	T	W	B	F	M	Y	R	N	F	Y	P	U	B	J	I	U	H	L	X	M		
T	V	S	V	W	K	V	D	V	X	T	W	X	N	N	S	P	G	X	M	T	V	S	Q	Y	T	P	F	Q	H	F	*		
X	H	O	M	W	F	G	G	J	R	Z	G	S	R	T	X	I	Z	J	Y	V	G	S	I	H	V	M	U	E	G	*			
I	I	Z	J	Y	?	F	V	Q	E	O	H	V	R	F	V	Q	Q	L	C	I	O	H	K	J	V	P	N	I	*	*			
S	L	Q	X	Q	R	M	C	U	V	N	X	Y	V	G	F	C	I	F	X	H	P	O	M	W	U	I	G	W	U	C	*		
V	R	I	R	N	G	E	I	T	W	K	W	G	G	U	M	S	D	M	N	D	V	E	P	S	D	M	F	H	J	P			
T	L	R	G	S	Y	R	G	Q	X	F	W	H	Y	B	M	J	Y	H	Z	J	Y	W	Z	V	N	B	R	H	S	L	*		
F	Z	Y	D	H	W	H	Z	U	S	F	H	L	T	E	W	V	R	G	P	G	*	J	F	F	F	W	X	J	?				
G	W	U	N	C	X	L	F	K	G	X	N	F	Y	H	G	R	O	A	V	V	M	N	S	S	W	I	B	T	V	O	H		
H	N	I	I	S	G	R	?	H	Y	R	D	B	H	F	V	X	I	A	I	L	O	W	H	K	J	J	F	P	Y	H	*		
H	J	V	U	S	I	O	W	C	G	E	U	W	X	O	F	M	Q	V	E	N	V	W	Q	W	U	G	F	H	V	*	*		
G	N	R	R	X	P	I	Y	H	H	X	A	G	G	K	D	F	Q	X	O	U	H	V	K	G	R	N	I	P	G	*			
F	U	V	H	Q	O	U	R	I	N	H	N	O	Q	P	M	V	T	C	N	Q	I	U	W	X	O	F	K	X	Y	O	*		
F	V	W	Q	G	D	G	F	Q	J	F	C	H	Q	M	I	R	U	W	O	N	I	G	B	Z	M	N	N	C	G	S	I		
G	U	F	T	C	J	P	A	J	U	N	B	P	J	G	A	E	O	S	G	A	L	D	L	F	T	B	J	U	C	L			
E	J	S	U	P	G	T	W	N	F	B	N	B	N	U	A	J	B	U	A	B	O	P	Z	K	Q	U	G	*	*				
S	O	P	U	I	C	S	L	J	U	P	C	P	O	G	B	N	U	U	G	N	G	D	N	O	L	L	C	E	C	G	*		
Z	Q	S	Z	U	F	L	S	G	G	U	P	C	J	E	S	G	U	G	W	F	P	G	S	U	J	C	G	A	G	*	*		
S	H	A	N	B	G	F	S	L	Z	G	U	J	C	G	L	A	T	S	G	T	V	J	G	G	U	E	S	C	T	E	P		
G	L	H	S	D	P	B	O	C	Q	J	J	G	Z	G	U	C	S	C	Q	C	S	G	I	T	G	G	P	N	D	*	*		
S	G	G	H	A	C	B	H	L	A	S	W	G	S	W	C	G	A	S	C	P	Q	C	G	G	S	P	U	P	S	L			
D	B	G	F	F	U	L	C	C	J	S	S	Q	B	S	G	Z	O	L	G	P	A	C	I	P	L	G	Z	H	G	D	*		
C	G	E	S	F	F	J	L	N	E	G	L	J	B	L	S	G	I	A	G	C	A	B	F	F	P	T	F	N	A	P	L		
P	Y	N	Q	N	G	J	C	I	S	F	J	C	P	F	O	U	G	A	J	Q	I	H	Q	H	G	W	J	G	*	*	*		
G	E	F	Q	P	L	O	H	G	L	Q	G	J	U	N	B	B	S	S	P	S	X	F	A	J	Z	?	A	D	Y	P	*		
W	A	K	A	I	J	W	N	D	B	A	N	L	H	D	D	Z	H	N	P	X	V	O	P	U	I	Y	B	B	A	*			
S	Z	S	V	B	M	V	B	B	G	Y	R	R	Z	C	S	M	Y	N	W	F	L	C	Z	L	U	H	D	U	T	O	*		
X	S	S	Q	R	H	O	Y	Z	I	F	Y	R	K	S	M	E	F	L	I	Y	W	J	W	C	W	G	Y	E	C	P	*		

R(2)LS(4) to the TSP gives.

C(3) RP2 KRS(2) SA(CS) SML[GHG] IUT2:2[GZ/OJ] IUT3 :4[EW/NF] NSP

D	G	S	Z	G	G	U	B	W	D	S	N	G	B	J	C	F	L	U	I	C	U	F	S	J	G	C	D	B	G	U	E	
G	A	H	N	L	I	J	S	N	L	G	B	S	J	G	U	W	C	U	E	G	A	H	L	V	N	W	B	L	A	U	*	
L	S	Z	C	B	S	A	S	C	N	N	T	L	U	X	L	B	L	D	N	G	J	A	Z	B	S	J	C	S	O	A	*	
B	B	L	D	N	G	?	S	J	G	T	W	B	G	F	S	J	G	C	P	S	J	B	Q	C	I	U	G	S	*	*		
L	E	H	L	G	N	F	K	S	J	G	L	U	H	A	P	Q	C	S	L	A	U	Z	C	B	I	C	S	J	G	P	*	
G	F	C	U	F	S	P	C	U	B	Q	L	S	S	G	F	W	U	F	G	P	I	P	W	U	F	S	A	C	U	W		
U	Y	U	A	Z	U	N	A	E	C	S	L	A	U	K	F	A	G	B	N	C	U	I	N	G	T	Y	U	A	Z	C	*	
B	J	A	W	N	F	L	S	B	D	W	P	L	G	F	A	W	S	*	D	A	W	S	S	J	L	B	?	S	J	G	T	
S	J	G	G	K	C	E	S	N	S	J	G	P	G	B	A	Q	G	Z	J	G	P	G	K	Z	J	A	Y	U	A	Z	B	
A	E	C	S	L	A	U	?	A	U	N	T	Z	Z	S	J	L	B	Z	C	B	J	L	B	N	C	B	S	Q	G	B	*	
B	C	I	G	K	S	J	L	P	S	T	G	L	I	J	S	F	G	I	P	G	G	B	H	L	H	S	T	B	G	*	*	
X	G	U	Q	L	U	W	S	G	B	B	L	K	O	A	L	U	S	H	L	X	G	B	G	E	A	U	F	B	U	A	*	
P	S	J	B	G	X	G	U	S	T	B	G	X	G	U	F	G	I	P	G	G	B	G	L	I	J	S	Q	L	U	W	*	
S	G	B	H	A	P	S	T	H	A	W	P	B	G	E	A	U	F	B	Z	G	B	S	K	N	C	T	G	P	S	Z	A	
B	N	A	Z	N	T	F	G	B	O	C	P	C	S	N	T	B	N	A	Z	N	T	S	J	G	P	G	Q	C	L	U	B	
A	H	O	C	B	B	C	I	G	F	G	D	P	L	S	J	C	S	G	U	E	W	Q	D	G	P	G	F	S	*	*		
J	G	N	A	Z	G	P	O	C	P	S	A	H	S	J	G	F	A	A	P	Z	C	T	Z	C	B	P	G	Q	A	X	*	
G	F	P	Z	L	S	J	S	P	G	Q	D	N	L	U	I	J	C	U	F	B	L	Q	C	F	G	C	S	L	U	T	*	
D	P	G	C	E	J	L	S	J	G	W	O	O	G	P	N	G	H	S	J	C	U	F	E	A	P	U	G	P	C	U		
U	L	U	I	S	J	G	J	A	N	C	N	L	S	S	N	G	L	L	U	*	*	F	S	J	G	U	Z	L	F	G		
G	G	P	G	F	L	U	S	J	G	J	A	S	B	G	P	S	G	F	S	J	G	E	C	U	F	N	G	C	U	F	O	
C	L	P	G	B	E	C	O	L	U	I	H	P	A	Q	S	J	G	E	J	C	Q	D	G	E	C	W	B	G	F	*		
S	J	G	H	N	C	Q	G	S	A	H	N	L	E	Y	G	P	D	W	S	O	P	G	B	G	U	S	N	T	F	G	S	
L	N	B	A	H	S	J	G	P	A	A	Q	Z	L	S	J	L	U	G	Q	G	P	I	G	F	H	P	A	Q	*	*	*	
S	J	G	Q	L	B	S	K	E	C	U	T	A	W	B	G	G	C	U	T	S	J	L	U	I	V	?	A	D	Y	P	*	
W	A	K	A	I	J	W	N	D	B	A	N	L	H	D	D	Z	H	N	P	X	V	O	P	U	I	Y	B	B	A	*		
S	Z	S	V	M	V	B	B	G	Y	R	R	Z	C	S	M	Y	N	W	F	L	C	Z	L	U	H	D	T	O	*			
X	S	S	Q	R	H	O	Y	Z	I	F	Y	R	K	S	M	E	F	L	I	Y	W	J	W	C	W	G	Y	E	C	P	*	
G	Q	W	H	O	J	R	N	P	H	C	K	T	W	B	F	M	Y	R	N	F	Y	P	U	B	J	I	U	H	L	X	M	
T	V	S	V	W	K	V	D	V	X	T	W	X	N	S	P	G	X	M	T	V	S	Q	Y	T	P	F	Q	H	F	*		
X	H	O	M	W	F	G	G	J	R	Z	G	S	R	T	X	I	Z	J	Y	V	G	S	I	H	V	M	U	E	G	*		
I	I	Z	J	Y	?	F	V	Q	E	O	H	V	R	F	V	Q	Q	L	C	I	O	H	K	J	V	P	N	I	*	*		
S	L	Q	X	Q	R	M	C	U	V	N	X	Y	V	G	F	C	I	F	X	H	P	O	M	W	U	I	G	W	*			
V	R	I	R	N	G	E	I	T	W	K	W	G	G	U	M	S	D	M	N	D	V	E	P	S	D	M	F	H	J	P		
T	L	R	G	S	Y	R	G	Q	X	F	W	H	Y	B	M	J	Y	H	Z	J	Y	W	Z	V	N	B	R	H	S	L	*	
H	W	H	G	Z	U	S	F	H	L	T	E	W	V	R	G	P	G	J	J	F	F	W	X	J	?	F	Z	Y	D			
G	W	U	N	C	X	L	F	K	C	X	N	F	Y	H	G	R	Q	A	V	V	M	N	S	S	W	I	B	T	O	H		
H	N	I	I	S	G	R	?	H	Y	R	D	B	H	F	V	X	I	A	I	L	O	W	H	K	J	J	F	P	Y	H	*	
H	J	V	U	S	I	O	W	C	G	E	U	W	X	O	F	M	Q	V	E	N	V	W	Q	W	U	G	F	H	V	*	*	
G	N	R	R	X	P	P	I	Y	H	X	A	G	K	D	F	Q	X	O	U	H	Y	K	R	G	N	I	P	G	*			
F	U	V	H	Q	O	U	R	I	N	H	N	O	Q	P	M	V	T	C	N	O	I	U	W	X	O	F	K	X	Y	O		
F	V	W	Q	D	G	F	Q	J	F	C	H	Q	M	I	R	U	W	O	N	I	G	B	Z	M	N	N	C	G	S	I		
G	U	F	T	C	J	P	A	J	U	N	B	P	J	G	A	E	O	S	G	A	L	D	L	F	T	B	J	U	C	L	C	
E	J	S	U	P	G	T	W	N	F	B	N	N	B	N	U	A	J	B	U	A	B	Q	P	Z	K	Q	U	G	*	*		
S	O	P	U	I	C	S	L	J	U	P	C	P	O	G	B	N	U	U	G	N	G	D	N	O	L	L	C	E	C	G	*	
Z	Q	S	Z	U	F	L	S	G	G	U	P	C	J	E	S	G	U	F	F	G	P	S	J	U	C	E	C	A	G	*	*	
S	H	A	N	B	G	F	S	L	Z	G	U	J	C	G	L	T	S	G	T	V	J	G	U	J	E	S	C	T	E	P	C	
Q	J	J	G	Z	G	U	C	S	C	Q	C	S	G	I	T	G	P	N	D	*	*	G	L	H	S	D	P	B	O	C		
A	C	P	Q	C	G	G	P	S	U	P	S	L	S	G	G	H	A	C	B	H	L	A	S	W	G	S	W	C	G	A	S	
D	B	G	F	F	U	L	C	C	J	S	S	Q	B	S	G	Z	O	L	G	P	A	C	I	P	L	G	Z	H	D	*		
C	G	E	S	F	F	J	L	N	E	G	L	J	B	L	S	G	I	A	G	C	A	B	F	F	F	T	F	N	A	P	L	
P	Y	N	Q	N	G	J	C	I	S	F	J	C	P	F	O	U	G	A	J	Q	I	H	Q	H	G	W	J	G	*	*	*	
G	E	F	Q	P	L	O	H	G	L	Q	G	J	U	N	B	B	S	S	P	S	X	F	A	J	Z	?	A	D	Y	P	*	
W	A	K	A	I	J	W	N	D	B	A	N	L	H	D	D	Z	H	N	P	X	V	F	V	O	P	U	I	Y	B	B	A	*
S	Z	S	V	M	V	B	B	G	Y	R	R	Z	C	S	M	Y	N	W	F	L	C	Z	L	U	H	D	T	O	*			
X	S	S	Q	R	H	O	Y	Z	I	F	Y	R	K	S	M	E	F	L	I	Y	W	J	W	C	W	G	Y	E	C	P	*	

E	H	A	K	H	H	V	C	X	E	A	Q	H	C	L	D	G	M	V	J	D	V	G	A	L	H	D	E	C	H	V	F		H	U	X	I	S	L	Y	Q	T	I	D	R	O	X	C	G	N	P	Y	Q	G	P	T	V	C	L	J	V	I	M	Z	N	
H	B	I	Q	M	J	L	A	Q	M	H	C	A	L	H	V	X	D	V	F	H	B	I	M	W	Q	X	C	M	B	V	*		O	W	A	W	X	R	W	E	W	Z	O	X	Z	Q	Q	A	N	T	H	Z	N	O	W	A	U	P	O	T	G	U	I	G	*
M	A	K	D	C	A	B	A	D	Q	Q	O	M	V	Z	M	C	M	E	Q	H	L	B	K	C	A	L	D	A	S	B	*		Z	I	S	N	X	G	H	H	L	Y	K	H	A	Y	O	Z	J	K	L	P	P	W	H	A	J	I	W	N	V	F	H	*	
C	C	M	E	Q	H	?	A	L	H	O	X	C	H	G	A	L	H	D	T	A	L	C	U	D	J	V	H	A	*	*		J	J	K	L	P	P	?	G	W	U	F	S	I	W	Y	G	W	U	U	M	D	J	S	I	R	L	W	T	Q	J	*	*		
M	F	I	M	H	Q	G	R	A	L	H	M	V	I	B	T	U	D	A	M	B	V	K	D	C	J	D	A	L	H	T	*		A	M	U	Z	U	Y	N	D	V	W	Q	Z	P	W	H	G	D	J	G	Z	I	T	S	N	X	V	J	H	X	V	*		
H	G	D	V	G	A	T	D	V	C	U	M	A	H	A	X	U	V	G	H	T	J	T	X	X	V	G	A	B	D	V	X		W	Y	J	Y	Q	H	F	J	O	X	R	X	H	H	N	A	E	N	E	W	F	T	A	E	N	G	I	L	T				
A	M	B	V	R	G	B	H	C	Q	D	V	J	Q	H	O	P	V	B	K	D	*	V	P	V	B	K	V	Q	B	F	D		G	X	I	P	C	N	L	F	I	O	K	P	X	K	W	Q	C	Y	I	A	M	*	O	M	Y	H	A	B	Y	H	U	Z	
A	C	E	X	T	M	H	G	B	X	A	*	E	B	X	A	A	L	M	C	?	A	L	H	O	C	L	B	X	Q	G	M		G	I	M	O	F	X	W	Y	I	T	H	*	L	L	G	G	X	Z	L	?	G	K	P	E	I	X	I	S	K	V	A		
A	L	H	T	H	C	B	U	H	K	L	H	T	H	R	K	L	B	P	V	B	K	C	A	L	H	H	R	D	F	A	Q		H	Z	Q	G	P	I	H	Y	U	B	W	W	N	Q	A	A	X	J	C	O	W	S	I	H	X	V	Q	D	Z	M	G	R	
B	F	D	A	M	B	V	?	B	V	Q	O	K	K	A	L	M	C	K	D	C	L	M	C	Q	D	C	A	U	H	C	*		I	Q	J	J	A	H	Y	?	I	P	Y	E	C	I	G	W	Z	J	B	J	M	S	X	I	R	L	L	G	T	P	I	*	
C	D	J	H	R	A	L	M	T	A	O	H	M	J	L	A	G	H	J	T	H	H	C	I	M	I	A	O	C	H	*	*		I	L	W	V	A	J	S	X	D	H	F	V	X	Z	S	G	N	U	W	F	Q	W	X	U	X	V	H	G	I	W	*	*	
Z	H	V	U	M	V	X	A	H	C	O	M	R	S	B	M	V	A	I	M	Z	H	C	H	F	B	V	G	C	V	B	*		H	Q	Y	Y	Z	T	J	P	I	Z	B	H	H	R	E	G	U	Z	S	V	I	W	R	H	Y	Q	J	T	H	*			
T	A	L	C	H	Z	H	V	A	O	C	R	Z																																																					

C(4) RP2 KRS(3) SA(SS) RML[HHT] IUT2:2[DH/BA] IUT3 :4[AP/XI] TSP

E	H	A	K	H	H	V	C	X	E	A	Q	H	C	L	D	G	M	V	J	D	V	G	A	L	H	D	E	C	H	V	F		H	U	X	I	S	L	Y	Q	T	I	D	R	O	X	C	G	N	P	Y	Q	G	P	T	V	C	L	J	V	I	M	Z	N	
H	B	I	Q	M	J	L	A	Q	M	H	C	A	L	H	V	X	D	V	F	H	B	I	M	W	Q	X	C	M	B	V	*		O	W	A	W	X	R	W	E	W	Z	O	X	Z	Q	Q	A	T	H	Z	N	O	W	A	U	P	O	T	G	U	I	G	*	
M	A	K	D	C	A	B	A	D	Q	Q	O	M	V	Z	M	C	M	E	Q	H	L	B	K	C	A	L	D	A	S	B	*		Z	I	S	N	X	G	H	H	L	Y	K	H	A	Y	O	Z	J	K	L	P	P	F	W	H	A	J	I	W	N	V	F	H	*
C	C	M	E	Q	H	?	A	L	H	O	X	C	H	G	A	L	H	H	D	T	A	L	C	U	D	J	V	H	A	*		J	J	K	L	P	P	?	G	W	U	F	S	I	W	Y	G	W	U	U	M	D	J	S	I	R	L	W	T	Q	J	*	*		
M	F	I	M	H	Q	G	R	A	L	H	M	V	I	B	T	U	D	A	M	B	V	K	D	C	J	D	A	L	H	T	*		A	M	U	Z	Y	Q	H	D	V	W	Q	Z	P	H	W	G	D	J	G	Z	I	S	N	X	V	J	H	X	V	D	*		
H	G	D	V	G	A	T	D	V	C	M	A	A	H	G	X	V	G	H	T	J	T	X	X	V	G	A	B	D	V	X		W	Y	J	Y	Q	F	H	J	O	X	R	X	H	V	N	A	E	N	Q	E	W	F	T	A	E	N	G	I	L	T	*			
A	M	B	V	R	G	B	H	C	Q	D	V	J	Q	H	O	P	V	B	K	D	*	V	P	V	B	K	V	Q	B	F	D		G	X	I	P	C	N	L	P	I	K	L	P	X	K	W	Q	C	Y	I	A	M	*	O	M	Y	H	A	P	Y	H	U	Z	*
X	Q	G	M	A	C	E	X	T	M	H	G	B	X	A	*	E	B	X	A	A	L	M	C	?	A	L	H	O	C	L	B		S	K	V	A	G	I	M	O	F	X	W	Y	H	T	H	*	L	L	G	G	X	Z	L	?	G	K	P	E	I	X	I	*	
A	L	H	T	H	C	B	U	H	K	L	H	T	H	R	K	L	B	P	V	B	K	C	A	L	H	H	R	D	F	A	Q		H	Z	Q	G	P	I	H	Y	U	B	W	W	N	Q	A	A	X	J	C	O	W	S	I	H	X	V	Q	D	Z	N	G	R	*
B	F	D	A	M	B	V	?	B	V	Q	O	K	K	A	L	M	C	K	D	C	L	M	C	Q	D	C	A	U	H	C	*		I	Q	J	J	A	H	Y	?	I	P	Y	E	C	I	G	W	Z	J	B	J	M	S	X	I	R	L	L	G	T	P	I	*	
C	D	J	H	R	A	L	M	T	A	O	H	M	J	L	A	G	H	J	T	H	H	C	I	M	I	A	O	A	O	C	H	*		I	L	W	V	A	J	S	X	D	H	F	V	X	Z	S	G	N	U	W	F	Q	W	X	U	X	V	H	G	I	W	*	*
Z	H	V	U	M	V	X	A	H	C	M	R	S	B	M	V	A	I	M	Z	H	C	H	F	B	V	G	C	V	B	*		H	Q	Y	Z	T	T	J	P	I	I	Z	B	H	H	R	E	G	U	Z	S	V	I	W	R	H	Y	Q	J	T	H	*			
T	A	L	C	H	Z	H	V	A	O	C	H	Z	H	V	G	J	T	H	C	H	M	J	L	A	U	M	V	X	*		G	W	V	I	U	S	H	V	Y	J	Q	I	Q	S	T	N	W	O	D	U	J	V	X	Z	S	G	R	Z	S	*	*				
A	H	C	I	B	T	A	O	I	B	X	T	C	H	F	B	V	G	C	K	H	C	A	R	Q	D	O	H	T	A	K	B		G	W	X	U	H	E	V	G	U	L	G	D	I	U	N	J	Y	V	X	S	Q	J	H	C	K	N	Q	Q	D	H	A	J	*
C	Q	B	K	Q	O	G	H	C	S	D	T	D	A	Q	O	C	Q	B	K	Q	O	A	L	H	T	H	U	D	M	V	C		H	V	G	O	D	L	T	B	L	V	Q	C	T	L	H	B	F	S	A	H	B	M	E	M	G	O	C	L	V	D	M	D	*
B	I	S	D	C	C	D	J	H	G	H	E	T	M	C	A	L	D	A	H	V	F	X	U	E	H	T	H	G	A	*	*		F	L	A	V	T	H	O	X	Q	G	C	Q	C	Q	V	B	L	C	V	B	C	U	T	K	R	U	V	H	*	*			
L	H	Q	B	K	H	T	S	D	T	A	B	I	A	L	H	G	B	B	T	K	D	O	K	D	C	T	H	U	B	Z	*		A	S	T	V	J	D	A	M	L	V	T	D	T	S	H	C	Q	V	V	H	Q	H	E	Q	S	M	M	D	F	D	H	*	*
H	G	K	M	A	L	T	H	U	E	Q	M	V	J	L	D	V	G	C	M	U	D	G	H	D	A	M	V	O	*	*		K	U	A	K	F	G	M	A	H	H	V	T	D	L	F	A	H	V	H	X	G	T	H	A	V	L	D	H	B	H	*	*		
E	T	H	D	F	L	M	V	A	L	H	X	S	S	H	T	O	H	I	A	L	D	V	G	F	B	T	V	H	T	D	V		A	I	B	Q	C	H	G	A	M	K	H	V	L	D	H	M	B	O	A	H	O	W	L	H	V	F	A	D	O	F	*	*	
G	A	L	H	V	K	M	G	H	V	M	V	J	A	L	H	L	B	Q	H	D	M	A	A	Q	H	M	M	V	*		H	M	I	A	E	T	C	S	D	U	L	L	H	K	H	V	D	A	D	U	D	A	H	J	O	H	T	Q	E	*	*				
V	G	S	H	H	T	H	G	M	V	A	L	H	L	B	A	C	T	A	H	G	A	L	H	F	D	V	G	Q	H	D		H	B	A	B	D	T	U	D	H	T	A	V	T	A	M	A	H	H	I	B	D	C	I	M	B	A	X	H	A	D	*			
J	I	T	B	U	A	L	H	F	L	D	U	E	H	T	F	D	X	C	H	G	*	D	M	T	H	C	F	D	S	M	V		A	A	U	C	A	H	K	S	M	H	T	B	D	J	T	M	H	K	I	H	E	*	E	C	H	G	G	V	M	D	D	L	*
A	L	H	I	Q	D	U	H	A	B	I	Q	M	F	P	H	T	E	X	A	S	T	H	C	H	V	A	Q	O	G	H	A		D	H	F	A	G	G	L	M	Q	F	H	M	L	C	M	A	H	J	B	H	D	B	C	G	G	T	O	G	Q	B	T	M	*
M	Q	C	B	I	A	L	H	T	B	B	U	K	M	A	L	M	V	H	U	H	T	J	H	G	I	T	B	U	*	*	*		T	P	Q	U	Q	H	L	D	J	A	G	L	D	T	G	S	V	H	B	L	U	J	I	U	I	H	X	L	H	*	*	*	
A	L	H	U	M	C	A	R	F	D	V	O	B	X	C	H	H	D	V	O	A	L	M	V	J	W	?	B	E	P	T	*		H	F	G	U	T	M	S	I	H	M	U	H	L	V	Q	C	C	A	A	T	A	Z	G	B	L	K	?	B	E	P	T	*	
X	A	B	R	B	J	L	X	Q	E	C	B	Q	M	I	E	K	I	Q	T	Z	W	S	T	V	J	P	C	B	*		X	A	B	R	B	J	L	X	Q	E	C	B	Q	M	I	E	E	K	I	Q	T	Z	W	S	T	V	J	P	C	B	*				
A	A	K	A	W	C	N	W	C	H	P	Y	Y	K	D	A	N	F	Q	X	G	M	D	K	M	V	I	E	O	S	*		A	A	K	A	W	C	N	W	C	H	P	Y	Y	K	D	A	N	F	Q	X	G	M	D	K	M	V	I	E	O	S	*			
Z	A	A	U	Y	I	S	P	K	J	G	P	Y	R	A	N	F	G	M	J	P	X	L	X	D	X	H	P	F	D	T	*		Z	A	A	U	Y	I	S	P	K	J	G	P	Y	R	A	N	F	G	M	J	P	X	L	X	D	X	H	P	F	D	T	*	

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C(5) RP3 KRS(4) SA(CS) RML[QOW] IUT2:2[XK/YP] IUT3 :4[LO/AW] NSP

F	I	B	R	I	I	W	D	Z	F	B	U	I	D	M	E	H	N	W	L	E	W	H	B	M	I	E	F	D	I	W	G	
I	C	J	U	N	L	M	B	U	N	I	D	B	M	I	W	Z	E	W	G	I	C	J	N	X	U	Z	D	N	C	W	*	
N	B	R	E	D	B	C	B	E	U	S	N	W	K	N	D	N	F	U	I	M	C	R	D	B	M	E	B	A	C	*		
D	D	N	F	U	I	?	B	M	I	S	Z	D	I	H	B	M	I	E	O	B	M	D	V	E	L	W	I	B	*	*		
N	G	J	N	I	U	H	Y	B	M	I	N	W	J	C	O	V	E	B	N	C	W	R	E	D	L	E	B	M	I	O	*	
I	H	E	W	H	B	O	E	W	D	V	N	B	B	I	H	Z	W	H	I	O	L	O	Z	Z	W	H	B	C	E	W	Z	
W	T	W	C	R	W	U	C	G	E	B	N	C	W	Y	H	C	I	D	U	E	W	L	U	I	S	T	W	C	R	E	*	
F	C	Z	B	B	M	N	D	?	B	M	I	S	D	M	C	Z	U	H	N	B	D	F	Z	O	N	I	H	C	Z	B	*	
B	M	I	O	I	D	C	V	I	R	M	I	O	I	Y	R	M	C	T	W	C	R	D	B	M	I	I	Y	E	G	B	U	
C	G	E	B	N	C	W	?	C	W	U	S	R	R	B	M	N	D	R	E	D	M	N	D	U	E	D	B	V	I	D	*	
D	E	L	I	Y	B	M	N	O	B	S	I	N	L	M	B	H	I	L	O	I	D	J	N	J	B	S	D	I	*	*		
A	C	N	W	B	J	N	K	I	D	I	G	C	W	H	D	W	C	*	K	I	W	V	N	W	Z	B	I	D	D	N	Y	
I	N	L	M	B	V	N	W	Z	*	O	B	M	D	I	K	I	W	B	S	D	I	K	I	W	H	I	L	O	I	D		
B	I	D	J	C	O	B	S	J	C	Z	O	D	I	G	C	W	H	D	R	I	D	B	Y	U	E	S	I	O	B	R	C	
D	U	C	R	U	S	H	I	D	A	E	O	E	B	U	S	D	U	C	R	U	S	B	M	I	O	I	V	E	N	W	D	
I	F	O	N	D	B	M	E	B	I	W	G	Z	V	F	I	O	I	H	B	*	*	C	J	A	E	D	D	E	L	I	H	
O	R	E	S	R	E	D	O	I	V	C	K	*	M	I	U	C	R	I	O	A	E	O	B	C	J	B	M	I	H	C	C	
I	H	R	N	B	M	B	O	I	V	F	U	N	W	L	M	E	W	H	D	N	V	E	H	I	E	B	N	W	S	*	*	
F	O	I	E	G	M	N	W	B	M	I	Z	A	A	I	O	U	I	J	B	M	E	W	H	G	C	O	W	I	O	E	W	
H	B	M	I	W	R	N	H	I	W	N	L	B	M	I	M	C	U	I	E	U	N	B	B	U	I	N	N	W	*	*		
D	I	O	I	H	B	M	I	G	E	W	H	U	I	E	W	H	A	I	I	O	I	H	N	W	B	M	I	M	C	B		
E	N	O	I	D	G	E	A	N	W	L	J	O	C	V	B	M	I	G	M	E	V	F	I	O	G	E	Z	D	I	H	*	
B	M	I	J	U	E	V	I	B	C	J	U	N	G	T	I	O	F	Z	B	A	O	I	D	I	W	B	U	S	H	I	B	
N	U	D	C	J	B	M	I	O	C	C	V	R	N	B	M	N	W	I	V	I	O	L	I	H	J	O	C	V	*	*	*	
B	M	I	V	N	D	B	Y	G	E	W	S	C	Z	D	I	E	W	S	B	M	N	W	L	X	?	C	F	T	O	*		
Z	C	Y	C	L	M	Z	U	F	D	C	U	N	J	F	F	R	J	U	O	K	X	X	A	O	W	L	T	D	D	C	*	
B	R	B	X	D	Q	X	D	D	I	T	P	P	R	E	B	Q	T	U	Z	H	N	E	R	N	W	J	F	W	S	A	*	
K	B	B	V	P	J	A	T	R	L	H	T	P	Y	B	Q	G	H	N	L	T	Z	M	Z	E	Z	I	T	G	E	O	*	
I	V	Z	J	A	M	P	U	O	J	E	Y	S	Z	D	H	Q	T	P	U	H	T	O	W	D	M	L	W	J	N	K	R	Q
S	X	B	X	Z	Y	X	F	X	K	S	Z	K	U	U	B	O	I	K	Q	S	X	B	V	T	S	O	H	V	J	H	*	
K	J	A	Q	Z	H	I	I	M	P	R	I	B	P	S	K	L	R	M	T	T	X	I	B	L	J	X	Q	W	G	I	*	
L	L	R	M	T	T	?	H	X	V	G	A	J	X	P	H	X	V	N	E	L	A	J	Y	M	X	O	U	L	*	*		
B	N	V	K	V	P	Q	E	W	X	U	K	T	X	I	H	E	L	H	K	J	O	A	Q	Z	W	L	I	Z	W	E	*	
X	P	L	P	U	I	G	L	S	Z	Y	Z	I	I	W	Q	B	F	Q	U	F	X	G	O	B	F	Q	H	J	M	O		
S	N	P	I	B	T	P	I	V	K	H	Z	J	T	D	Q	M	T	J	R	M	T	Z	R	X	U	D	P	J	B	N	*	
M	M	H	H	H	Z	K	M	?	H	R	T	F	J	Z	J	A	R	W	B	H	J	N	S	G	Z	X	P	I	O	I	*	
I	K	U	H	T	J	I	P	V	C	X	X	Q	U	B	B	Z	L	D	S	X	A	J	I	Z	W	U	E	K	N	H	Y	
J	U	L	L	B	I	P	?	J	T	P	F	D	J	H	X	K	L	C	L	N	A	Z	J	Y	M	M	H	O	T	J	*	
J	M	X	W	B	L	A	Z	E	I	G	W	Z	X	A	H	Q	V	X	G	U	X	Z	V	Z	W	I	H	J	X	*	*	
I	I	Y	F	H	V	K	A	W	J	X	Y	I	P	U	L	O	I	*	I	U	P	K	O	L	T	J	J	K	C			
W	Z	K	A	H	Y	K	T	A	*	H	W	X	J	V	A	W	P	L	U	J	U	A	V	O	Q	X	S	E	U	V	L	
H	X	Z	V	I	F	I	H	V	M	H	E	J	V	Q	L	P	W	Z	A	U	L	I	D	R	Q	U	E	I	B	L		
I	W	H	S	E	M	O	C	M	W	U	D	O	M	I	C	G	A	B	I	C	N	F	N	H	S	D	M	W	E	N	E	
D	U	U	D	U	U	W	C	M	D	W	C	D	V	O	R	Y	V	V	W	I	*	*	G	M	B	W	O	I	S	Z	U	H
I	U	I	F	U	A	N	N	E	G	E	I	*	B	A	O	W	L	E	B	N	M	W	O	E	O	A	I	D	U	W	W	
R	V	B	R	W	H	N	B	I	I	W	O	E	M	G	B	I	W	I	Z	H	O	I	B	W	M	E	I	C	I	*	*	
B	J	C	U	D	I	H	B	N	R	I	W	M	E	I	N	C	S	B	I	S	X	M	I	I	W	G	B	E	S	G	O	
I	N	J	B	F	O	D	A	E	V	M	M	I	R	I	W	E	B	E	V	E	B	I	L	S	I	I	O	U	F	*	*	
B	I	I	J	C	E	D	J	N	C	B	Z	I	B	Z	E	I	C	B	C	E	O	V	E	I	I	O	B	W	O	B	N	
F	D	I	H	H	W	N	E	E	M	B	B	V	D	B	I	R	A	N	I	O	C	E	L	O	N	I	R	J	I	F	*	
E	I	G	B	H	H	M	N	U	G	I	N	M	D	N	B	I	L	C	I	E	C	D	H	H	O	S	H	U	C	O	N	
O	T	U	V	U	I	M	E	L	B	H	M	E	O	H	A	W	I	C	M	V	L	J	V	J	I	Z	M	I	*	*	*	
I	G	H	V	O	N	A	J	I	N	V	I	M	W	U	D	D	B	B	O	B	K	H	C	M	R	?	C	F	T	O	*	
Z	C	Y	C	L	M	Z	U	F	D	C	U	N	J	F	F	R	J	U	O	K	X	X	A	O	W	L	T	D	D	C	*	
B	R	B	X	D	Q	X	D	D	I	T	P	P	R	E	B	Q	T	U	Z	H	N	E	R	N	W	J	F	W	S	A	*	
K	B	B	V	P	J	A	T	R	L	H	T	P	Y	B	Q	G	H	N	L	T	Z	M	Z	E	Z	I	T	G	E	O	*	

R(2)LS(9) to the TSP gives.

C(5) RP3 KRS(4) SA(CS) SML[IIW] IUT2:2[XK/YP] IUT3 :4[AW/LO] TSP

F	I	B	R	I	I	W	D	Z	F	B	U	I	D	M	E	H	N	W	L	E	W	H	B	M	I	E	F	D	I	W	G		
I	C	J	U	N	L	M	B	U	N	I	D	B	M	I	W	Z	E	W	G	I	C	J	N	X	U	Z	D	N	C	W	*		
N	B	R	E	D	B	C	B	E	U	S	N	W	K	N	D	N	F	U	I	M	C	R	D	B	M	E	B	A	C	*			
D	D	N	F	U	I	?	B	M	I	S	Z	D	I	H	B	M	I	E	O	B	M	D	V	E	L	W	I	B	*	*			
N	G	J	N	I	U	H	Y	B	M	I	N	W	J	C	O	V	E	B	N	C	W	R	E	D	L	E	B	M	I	O	*		
I	H	E	W	H	B	O	E	W	D	V	N	B	B	I	H	Z	W	H	I	O	L	O	Z	Z	W	H	B	C	E	W	Z		
W	T	W	C	R	W	U	C	G	E	B	N	C	W	Y	H	C	I	D	U	E	W	L	U	I	S	T	W	C	R	E	*		
F	C	Z	B	B	M	N	D	?	B	M	I	S	D	M	C	Z	U	H	N	B	D	F	Z	O	N	I	H	C	Z	B	*		
B	M	I	O	I	D	C	V	I	R	M	I	O	I	Y	R	M	C	T	W	C	R	D	B	M	I	I	Y	E	G	B	U		
C	G	E	B	N	C	W	?	C	W	U	S	R	R	B	M	N	D	R	E	D	M	N	D	U	E	D	B	V	I	D	*		
D	E	L	I	Y	B	M	N	O	B	S	I	N	L	M	B	H	I	L	O	I	D	J	N	J	B	S	D	I	*	*			
D	I	G	C	W	H	D	W	C	*	K	I	W	V	N	W	Z	B	I	D	D	N	Y	A	C	N	W	B	J	N	K	I		
I	N	L	M	B	V	N	W	Z	*	O	B	M	D	I	K	I	W	B	S	D	I	K	I	W	H	I	L	O	I	D			
B	I	D	J	C	O	B	S	J	C	Z	O	D	I	G	C	W	H	D	R	I	D	B	Y	U	E	S	I	O	B	R	C		
D	U	C	R	U	S	H	I	D	A	E	O	E	B	U	S	D	U	C	R	U	S	B	M	I	O	I	V	E	N	W	D		
I	F	O	N	D	B	M	E	B	I	W	G	Z	V	F	I	O	I	H	B	*	*	C	J	A	E	D	D	E	L	I	H		
B	C	J	B	M	I	H	C	C	O	R	E	S	R	E	D	O	I	V	C	K	*	M	I	U	C	R	I	O	A	E	O		
I	H	R	N	B	M	B	O	I	V	F	U	N	W	L	M	E	W	H	D	N	V	E	H	I	E	B	N	W	S	*	*		
F	O	I	E	G	M	N	W	B	M	I	Z	A	A	I	O	I	J	B	M	E	W	H	G	C	O	W	I	O	E	W			
H	B	M	I	W	R	N	H	I	W	N	W	L	M	I	M	I	C	I	E	U	N	B	B	U	I	N	N	W	*	*			
D	I	O	B	I	H	B	M	I	E	G	W	H	U	I	E	W	H	A	I	O	I	H	N	W	B	M	I	M	C	B			
E	N	O	I	D	G	E	A	N	W	L	J	O	C	V	B	M	I	G	M	E	V	F	I	O	G	E	Z	D	I	H	*		
B	M	I	J	U	E	V	I	B	C	J	U	N	G	T	I	O	F	Z	B	A	O	I	D	I	W	B	U	S	H	I	B		
N	U	D	C	J	B	M	I	O	C	C	V	R	N	B	M	N	W	I	V	I	O	L	I	H	J	O	C	V	*	*	*		
B	M	I	V	N	D	B	Y	G	E	W	S	C	Z	D	I	E	I	E	W	S	B	M	X	N	W	L	X	?	C	F	T	O	*
Z	C	Y	C	L	M	Z	U	F	D	C	U	N	J	F	E	R	J	U	O	K	X	A	O	W	L	T	D	D	C	*			
B	R	B	X	D	O	X	D	D	I	T	P	P	R	E	B	Q	T	U	Z	H	N	E	R	N	W	J	F	W	S	A	*		
K	B	B	V	P	J	A	T	R	L	H	T	P	Y	B	Q	G	H	N	L	T	Z	M	Z	E	Z	I	T	G	E	O	*		
I	V	Z	J	A	M	P	U	O	J	E	Y	S	Z	D	H	Q	T	P	U	H	T	O	W	D	M	L	W	J	N	K	*		
S	X	B	X	Z	Y	X	F	X	K	S	Z	K	U	B	O	I	K	Q	S	X	B	V	T	S	O	H	V	J	H	*			
K	J	A	Q	Z	H	I	I	M	P	R	I	B	P	S	K	L	R	M	T	T	X	I	B	L	J	X	Q	W	G	I	*		
L	L	R	M	T	T	?	H	X	V	G	A	J	X	P	H	X	V	V	N	E	L	A	J	Y	M	X	O	U	L	*	*		
B	N	V	K	V	P	G	E	W	X	U	K	T	X	I	H	E	L	H	K	J	O	A	Q	Z	W	L	I	Z	W	E	*		
X	P	L	P	U	I	G	L	S	Z	Y	Z	I	I	W	Q	B	F	Q	U	F	X	G	O	B	F	Q	H	J	M	O	*		
S	N	P	I	B	T	F	I	V	K	H	Z	J	T	D	Q	M	T	J	R	M	T	Z	R	X	U	D	P	J	B	N	*		
M	M	H	H	H	Z	K	M	?	H	R	T	F	J	Z	J	A	R	W	B	H	J	N	S	G	Z	X	P	I	O	I	*		
I	K	U	H	T	J	I	P	V	C	X	X	Q	U	B	B	Z	L	D	S	X	A	J	I	Z	W	U	E	K	N	H	Y	*	
J	U	L	L	B	I	P	?	J	T	P	F	D	J	H	X	K	L	C	L	N	A	Z	J	Y	M	M	H	O	T	J	*		
J	M	X	W	B	L	A	Z	E	I	G	W	Z	K	A	H	Q	L	V	X	G	U	X	Z	V	Z	W	I	H	J	X	*	*	
J	X	Y	I	P	U	L	O	I	*	I	U	P	K	O	E	Q	T	J	J	K	C	I	I	Y	F	H	V	A	K	W			
W	Z	K	A	H	Y	K	T	A	*	H	W	X	J	V	A	W	P	L	U	J	U	A	V	O	A	O	X	S	E	U	V	I	
H	X	Z	A	Y	I	F	I	H	V	M	H	E	J	V	Q	L	P	W	Z	A	U	L	I	D	R	Q	U	E	I	B	L		
I	W	H	S	E	M	O	C	M	W	U	D	O	M	I	C	G	A	B	I	C	N	F	N	H	S	D	M	W	E	N	E		
D	U	U	D	U	U	C	M	D	W	C	D	V	O	R	Y	V	W	I	*	*	G	M	B	W	O	I	S	Z	U	H			
O	E	O	A	I	D	U	W	W	I	U	I	F	U	A	N	N	E	G	E	I	*	B	A	O	W	L	E	B	N	M	W		
R	V	B	R	W	H	N	B	I	I	W	O	E	M	G	B	I	W	I	Z	H	O	I	B	W	M	E	I	C	I	*	*	*	
B	J	C	U	D	I	H	B	N	R	I	W	M	E	I	N	C	S	B	I	S	X	M	I	I	W	G	B	E	S	G			
I	N	J	B	F	O	D	A	E	V	M	M	I	R	I	W	E	B	E	V	E	B	I	L	S	I	O	I	O	F	*	*		
B	I	I	J	C	E	D	J	N	C	B	B	I	B	Z	E	I	C	B	C	E	O	V	E	I	O	B	W	O	B	N			
F	D	I	H	H	W	N	E	E	M	B	B	V	D	B	I	R	A	N	I	O	C	E	L	O	N	I	R	J	I	F	*		
E	I	G	B	H	H	M	N	U	G	I	N	M	D	N	B	I	L	C	I	E	C	D	H	H	O	S	H	U	C	O	N		
O	T	U	V	U	I	M	E	L	B	H	M	E	O	H	A	I	C	M	V	L	J	V	J	I	Z	M	I	*	*	*			
I	G	H	V	O	N	A	J	I	N	V	I	M	W	U	D	B	B	O	B	K	H	C	M	R	?	C	F	T	O	*			
Z	C	Y	C	L	M	Z	U	F	D	C	U	N	J	F	F	R	J	U	O	K	X	A	O	W	L	T	D	D	C	*			
B	R	B	X	D	O	X	D	D	I	T	P	P	R	E	B	Q	T	U	Z	H	N	E	R	N	W	J	F	W	S	A	*		
K	B	B	V	P	J	A	T	R	L	H	T	P	Y	B	Q	G	H	N	L	T	Z	M	Z	E	Z	I	T	G	E	O	*		

C(6) RP3 KRS(5) SA(SS) RML[VVY] IUT2:2[VV/VB] IUT3 :4[HW/OR] IUT3:4[ZS/OS] TSP

GJCYJXJXERKGCVCVJENFIQXMFIXICNJJFGEJXHX	JWKLBNTVSLFPAPAKEIUOTVIOXENMXLQRU
JDLVQMNQVQJECNJKKFXHJDLQZVKEQDX*	AZCZKRPZGZRAKRVVCSJRUAZCWOASIWLI*
QCYFECDCFCVVAQXKQEQGVJNDYECNFCBD*	RLBURIJJNTYJCTARMYNOOZJCMZUXHJ*
EEQGVJ?CNJAKEJICNJJFSCNEWFMXJC**	MMYNNOO?IZWHBLZTIZZWQFMBLPNZSVM**
QHLQJVIPCNJQXLDSWFCQDXYFEMFCNJS*	CQWRWTUFXZVROZJIFMIRLSBUXMXJKXF*
JIFXICSFEXEQCCJIKXIJSMSKKXICDFXK	ZTMTVJHMAKPKJXUCGUVGZHSBGUILNSS
XOXDYXVDHFCQDIXPIDJEVFXMVJAOXDYF*	AQTJJCOTJWRIKLOEUNOLYNOKYZVETLCQ*
GDKCCNQE?CNJAEENDKVIQCEGKSQJIDKC*	NNIIIIKRN?IYOGKLKLBXYXCILQAHKZTJSJ*
CNJSJEDWJYNJSJYPYNDODXDEECNJJPFHCV	JRVIOLOLJTWDDZUVCCKMEAZBLJXVFRQIP
DHFCQDX?DXVAYYCNQEYFENQEVFECWJE*	LVMMCJT?LOTGELIZRMDMQBKLFPNNISOL*
QLCAEJ**EFMJPCNQSCAJQMNCIJMSJJJEL	KXJILZ**LNZXCMBKFFJHXKRBUIUWZHVZKW
PBDQXCLQRJEJHDXIEXD*RXJWQXKCEEQ	DJJJPGIWRBXLZPJTVMSJ*JVTTTRSSMO
SCNEJRJXCAEJRJXIJMSJJJEJQMNCWQXK*	IXZLWBXTMVLVBWSUZAFFVWMMKRBIPROB*
CJELDSICALDKSEJHDXIEYJECPVFAJSCYD	IZKJWJGJIWNIFLWUMTXKBVMJEYUUVFJCM
EVDYVAIIEJBFSFCVAEVDYVACNJSJWFQXE	JXIAFNSDNXVESNJJDHBCJJDQGGQIAENXFFQF
DLBFEEEFMJJIGSQECNFCJXHKWGSJIC**	HNCXSJAKVIEVVEVVDNEXDEWSYPWXXJ**
LCNJJIDDSYFAYFESJWDR*NJVVDYJSBFS	SBJEVSXJXJGVGBQQFHJ*CBXSXMFQCNXSF
JFCQXAX**JIYQCNCSJWGVQXMFNFXIEQWFI	XNFJJDJ**YWCYXIQCJJXSFNHCJXJKISJC
GSJFHNQXCNJKBBSJSVJLCNFXIHDSXJSFX	CLDVEJICQYJXNFJQDACJAZZNJXJHCFAHS
ICNJJXYQIJXQXMCNNDVJFVQCCVJQQX**	CJLJLDFELQDCKJCKFJDCDFSFJJSXCSQ
EJSCJICNJJHFXIIVJFJIBJJSJIXQCNJND	GEJIIIXQFFNCCEWCJYBQJSDFMSSQJYLJG*
FQSJEHFBQXMLSDWCNJJHNFJGJSJHFKREJI*	FJHCCIINQVHJQNEQCJMDJFDEIISAIIVDSQ
CNJJLVFWJCDLVQHOJSGKCBSJEJXCVVAIJC	SOVWVJNFMFCINFISIBXJDNWMLWLJKNJ***
QVEDLNCNJSDDWYQCNQXJWJSMJILSDW***	JHIWSQBLJQWJNXVEECCSCRIDNY?DGOSS*
CNJJWQECPHFXADKEJJJFXACNQXMXZ?DGOSS*	KDPDMNKVGEDVQLGGYLVSRZZBSXMOEED*
KDPDMNKVGEDVQLGGYLVSRZZBSXMOEED*	CYCZEUZEJOTTYFCUOVKIQFYQXLGXAB*
CYCZEUZEJOTTYFCUOVKIQFYQXLGXAB*	RCJWTLBOYMIOTTPCUHIQMOKNKFKJOHFS*
RCJWTLBOYMIOTTPCUHIQMOKNKFKJOHFS*	

R(2)RS(3) to the TSP gives.

C(6) RP3 KRS(5) SA(SS) RML[JWJ] IUT2:2[VV/VB] IUT3 :4[HW/OR] IUT3:4[ZS/OS] TSP

GJCYJXJXERKGCVCVJENFIQXMFIXICNJJFGEJXHX	JWKLBNTVSLFPAPAKEIUOTVIOXENMXLQRU
JDLVQMNQVQJECNJKKFXHJDLQZVKEQDX*	AZCZKRPZGZRAKRVVCSJRUAZCWOASIWLI*
QCYFECDCFCVVAQXKQEQGVJNDYECNFCBD*	RLBURIJJNTYJCTARMYNOOZJCMZUXHJ*
EEQGVJ?CNJAKEJICNJJFSCNEWFMXJC**	MMYNNOO?IZWHBLZTIZZWQFMBLPNZSVM**
QHLQJVIPCNJQXLDSWFCQDXYFEMFCNJS*	CQWRWTUFXZVROZJIFMIRLSBUXMXJKXF*
JIFXICSFEXEQCCJIKXIJSMSKKXICDFXK	ZTMTVJHMAKPKJXUCGUVGZHSBGUILNSS
XOXDYXVDHFCQDIXPIDJEVFXMVJAOXDYF*	AQTJJCOTJWRIKLOEUNOLYNOKYZVETLCQ*
GDKCCNQE?CNJAEENDKVIQCEGKSQJIDKC*	NNIIIIKRN?IYOGKLKLBXYXCILQAHKZTJSJ*
CNJSJEDWJYNJSJYPYNDODXDEECNJJPFHCV	JRVIOLOLJTWDDZUVCCKMEAZBLJXVFRQIP
DHFCQDX?DXVAYYCNQEYFENQEVFECWJE*	LVMMCJT?LOTGELIZRMDMQBKLFPNNISOL*
QLCAEJ**EFMJPCNQSCAJQMNCIJMSJJJEL	KXJILZ**LNZXCMBKFFJHXKRBUIUWZHVZKW
EEQPBQXCLQRJEJHDXIEXD*RXJWQXKCEEQ	LRLDJPJGIWRBXLZPJTVMSJ*JVTTTRSSMO
SCNEJRJXCAEJRJXIJMSJJJEJQMNCWQXK*	IXZLWBXTMVLVBWSUZAFFVWMMKRBIPROB*
CJELDSICALDKSEJHDXIEYJECPVFAJSCYD	IZKJWJGJIWNIFLWUMTXKBVMJEYUUVFJCM
EVDYVAIIEJBFSFCVAEVDYVACNJSJWFQXE	JXIAFNSDNXVESNJJDHBCJJDQGGQIAENXFFQF
DLBFEEEFMJJIGSQECNFCJXHKWGSJIC**	HNCXSJAKVIEVVEVVDNEXDEWSYPWXXJ**
SCDLNCNJJIDDSYFAYFESJWDR*NJVVDYJSBFS	XSFSBJSJEVXXJXJGVGBQQFHJ*CBXSXMFQCN
JFCQXAX**JIYQCNCSJWGVQXMFNFXIEQWFI	XNFJJDJ**YWCYXIQCJJXSFNHCJXJKISJC
GSJFHNQXCNJKBBSJSVJLCNFXIHDSXJSFX	CLDVEJICQYJXNFJQDACJAZZNJXJHCFAHS
ICNJJXYQIJXQXMCNNDVJFVQCCVJQQX**	CJLJLDFELQDCKJCKFJDCDFSFJJSXCSQ
EJSCJICNJJHFXIIVJFJIBJJSJIXQCNJND	GEJIIIXQFFNCCEWCJYBQJSDFMSSQJYLJG*
FQSJEHFBQXMLSDWCNJJHNFJGJSJHFKREJI*	FJHCCIINQVHJQNEQCJMDJFDEIISAIIVDSQ
CNJJLVFWJCDLVQHOJSGKCBSJEJXCVVAIJC	SOVWVJNFMFCINFISIBXJDNWMLWLJKNJ***
QVEDLNCNJSDDWYQCNQXJWJSMJILSDW***	JHIWSQBLJQWJNXVEECCSCRIDNY?DGOSS*
CNJJWQECPHFXADKEJJJFXACNQXMXZ?DGOSS*	KDPDMNKVGEDVQLGGYLVSRZZBSXMOEED*
KDPDMNKVGEDVQLGGYLVSRZZBSXMOEED*	CYCZEUZEJOTTYFCUOVKIQFYQXLGXAB*
CYCZEUZEJOTTYFCUOVKIQFYQXLGXAB*	RCJWTLBOYMIOTTPCUHIQMOKNKFKJOHFS*
RCJWTLBOYMIOTTPCUHIQMOKNKFKJOHFS*	

C(6) KRS(5) SA(SS)	K4 Key	Encryption
JQSS EKZZ	HWOR ZSOS	TLEA DSTO

C(7) RP4 KRS(6) SA(CS) SML[LJN] IUT H:V[RH/XU] IUT3 :4[MK/WN] NSP

H	L	D	P	L	L	Z	F	R	H	D	W	L	F	Q	G	J	U	Z	N	G	Z	J	D	Q	L	G	H	F	L	Z	I		L	X	R	M	C	Q	O	W	A	M	G	T	B	R	F	J	V	S	O	W	J	S	A	Z	F	Q	N	Z	M	U	Y	V
L	E	M	W	U	N	Q	D	W	L	F	D	Q	L	Z	R	G	Z	I	L	E	M	U	K	W	R	F	U	E	Z	*		B	K	D	K	R	T	K	H	K	Y	B	R	Y	W	W	D	A	L	Y	V	B	K	D	X	S	B	A	J	X	M	J	*	
W	L	Q	E	P	F	D	Q	G	D	C	E	*	U	D	P	G	F	D	E	D	G	W	B	U	Z	Y	U	F	U	H		S	S	K	L	D	N	M	K	V	Z	I	L	*	Y	M	C	V	R	J	L	L	Q	O	P	I	D	O	B	Y	N	P	Q	
Q	F	X	G	N	Z	L	D	*	*	F	F	U	H	W	L	?	D	Q	L	B	R	F	L	J	D	Q	L	L	G	A	D		C	M	T	Q	K	A	W	N	*	*	N	N	P	Q	S	S	?	J	K	X	I	C	M	K	O	J	K	X	X	U	G	N
U	I	M	U	L	W	J	T	D	Q	L	U	Z	M	E	A	X	G	D	U	E	Z	P	G	F	N	G	D	Q	L	A	*		D	U	X	Y	X	O	V	G	Z	K	W	Y	S	K	L	J	G	N	J	Y	M	A	C	V	R	Z	N	L	R	Z	G	*
L	J	G	Z	J	D	A	G	Z	F	X	U	D	D	L	J	R	Z	J	L	A	N	A	R	R	Z	J	D	E	G	Z	R		K	O	N	O	W	L	I	N	B	R	T	R	L	L	Z	V	D	H	V	W	H	K	I	A	D	H	V	J	M	Q	A	A
Z	S	Z	E	P	Z	W	E	I	G	D	U	E	Z	T	J	E	L	F	W	G	Z	N	W	L	B	S	Z	E	P	G	*		B	U	O	L	D	S	O	L	X	Y	J	R	M	S	F	V	Q	S	M	P	Q	S	R	P	K	W	F	O	M	D	U	*
H	E	R	D	D	Q	U	F	?	D	Q	L	B	F	Q	E	R	W	J	U	D	F	H	R	A	U	L	J	E	R	D	*		Q	Q	J	J	J	R	Y	Q	?	J	P	S	H	M	R	M	C	P	Z	D	J	M	U	B	I	R	K	O	L	A	L	*
D	Q	L	A	L	F	E	X	L	P	Q	L	A	L	T	P	Q	E	S	Z	E	P	F	D	Q	L	L	T	G	I	D	W		L	Y	W	J	S	M	L	O	X	E	K	K	V	W	D	D	R	N	F	B	K	C	M	L	R	Z	W	G	Y	U	J	T
E	I	G	D	U	E	Z	?	E	Z	W	B	P	P	D	Q	U	F	P	G	F	Q	U	F	W	G	F	D	X	L	F	*		M	W	N	N	D	L	O	?	M	S	O	H	F	M	J	K	Y	N	E	N	U	C	R	M	T	Q	Q	J	A	S	M	*
F	G	N	L	T	D	Q	U	A	D	B	L	U	N	Q	D	J	L	N	A	L	L	F	M	U	M	D	B	F	L	*		M	Q	K	Z	D	N	C	R	G	L	I	Z	R	Y	C	J	V	X	K	I	W	K	R	X	R	Z	L	J	M	K	*		
Y	L	Z	X	U	Z	R	D	L	F	F	U	T	C	E	U	Z	D	M	U	Y	L	F	L	I	E	Z	J	F	Z	E	*		L	W	O	O	Y	A	A	N	S	M	M	Y	E	L	L	T	H	J	X	Y	C	Z	M	K	T	L	O	W	N	A	L	*
A	D	Q	F	L	Y	L	Z	D	B	F	L	Y	L	Z	J	L	N	A	L	L	F	L	U	N	Q	D	X	U	Z	R	*		J	Z	K	M	X	C	Z	O	N	W	M	W	C	X	A	V	K	B	G	W	X	N	Z	R	Y	C	J	T	Y	S	C	*
D	L	F	M	E	A	D	B	M	E	R	A	F	L	I	E	Z	J	F	P	L	F	D	T	W	G	B	L	A	D	P	E		J	K	R	X	L	H	L	J	X	Q	J	G	M	X	V	N	O	Z	R	C	W	N	L	F	P	V	W	W	G	L	D	N
F	W	E	P	W	B	J	L	F	C	G	A	G	D	W	B	F	W	E	P	W	B	D	Q	L	A	L	X	G	U	Z	F		L	Z	J	B	G	Q	A	E	Q	Z	W	F	A	Q	L	E	I	C	D	L	E	U	H	U	J	B	F	Q	Z	G	U	G
E	M	C	G	F	F	G	N	L	J	L	H	A	U	F	D	Q	G	D	L	Z	I	R	X	H	L	A	L	J	D	*		I	Q	D	Z	A	L	B	R	W	J	F	W	F	W	Z	E	Q	F	Z	E	F	X	A	P	T	X	Z	L	*	*			
Q	L	W	E	P	L	A	C	G	A	D	E	M	D	Q	L	J	E	E	A	P	G	B	P	G	F	A	L	X	E	Y	*		D	C	A	Z	N	G	D	U	Q	Z	A	G	A	C	L	F	W	Z	Z	L	W	L	H	W	C	U	U	G	I	G	L	*
L	J	P	U	D	Q	D	A	L	X	H	W	U	Z	N	Q	G	Z	J	F	U	X	G	J	L	G	D	U	Z	B	*		P	X	D	P	Z	J	U	D	L	L	Z	A	G	Q	I	D	L	Z	L	R	J	A	L	D	Z	Q	G	L	E	B	I	*	
H	A	L	G	I	Q	U	Z	D	Q	L	R	C	C	L	A	W	L	M	D	Q	G	Z	J	I	E	A	Z	L	A	G	Z		D	M	E	W	F	L	J	D	U	P	L	Z	Q	G	L	U	E	B	D	L	B	K	Q	L	L	Z	I	D	G	B	I	A
J	D	Q	L	Z	P	U	J	L	Z	U	Z	N	D	Q	L	Q	E	W	L	G	W	U	D	D	W	L	U	Z	*		L	L	M	D	H	A	F	C	G	X	Q	Q	L	P	L	Z	G	D	G	X	G	D	L	N	B	L	L	A	W	H	*			
F	L	A	D	L	J	D	Q	L	I	G	Z	J	W	L	G	Z	J	C	L	L	A	L	J	U	Z	D	Q	L	Q	E	D		D	L	L	M	E	G	F	M	U	E	D	R	L	D	R	G	L	E	D	E	G	A	X	G	L	L	A	D	Z	A	D	U
G	U	A	L	F	I	G	C	U	Z	N	M	A	E	X	D	Q	L	I	Q	G	X	H	L	A	I	G	R	F	L	J	*		H	F	L	J	J	Z	U	G	G	Q	D	D	X	F	D	L	P	C	U	L	A	E	G	N	A	U	L	P	M	L	H	*
D	Q	L	M	W	G	X	L	D	E	M	W	U	I	S	L	A	H	R	D	C	A	L	F	L	Z	D	W	B	J	L	D		G	L	I	D	J	J	Q	U	W	I	L	U	Q	F	U	D	L	N	E	L	G	E	F	J	A	B	J	W	E	A	U	
U	W	F	E	M	D	Q	L	A	E	E	X	P	U	D	Q	U	Z	L	X	L	A	N	L	J	M	A	E	X	*	*		A	S	W	X	W	L	O	G	N	D	J	Q	G	A	J	C	Z	L	E	Q	X	N	M	X	M	L	R	Q	L	*	*		
Z	B	E	R	F	L	L	G	Z	B	D	Q	U	Z	N	K	?	E	H	S	A	*	D	Q	L	X	U	F	D	T	I	G		X	L	Q	Z	W	F	F	D	D	A	D	Y	J	E	Q	P	?	E	H	S	A	*	L	I	J	X	A	U	C	M	L	U
M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E	*	R	E	T	E	N	Q	R	W	H	F	E	W	U		M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E	*	R	E	T	E	N	Q	R	W	H	F	E	W	U
D	P	D	K	F	V	K	F	F	L	S	O	O	P	G	D	V	S	W	R	J	U	G	P	U	Z	M	H	Z	B	C	*		D	P	D	K	F	V	K	F	F	L	S	O	O	P	G	D	V	S	W	R	J	U	G	P	U	Z	M	H	Z	B	C	*
Y	D	D	X	O	M	C	S	P	N	J	S	O	T	D	V	I	J	U	N	S	R	Q	R	G	R	L	S	I	G	A	*		Y	D	D	X	O	M	C	S	P	N	J	S	O	T	D	V	I	J	U	N	S	R	Q	R	G	R	L	S	I	G	A	*

R(2)RS(8) to the TSP gives.

C(7) RP4 KRS(6) SA(CS) SML[EEA] RML[LXY] IUT H:V[XR/UH] IUT3 :4[MW/PN] TSP

H	L	D	P	L	L	Z	F	R	H	D	W	L	F	Q	G	J	U	Z	N	G	Z	J	D	Q	L	G	H	F	L	Z	I		L	X	R	M	C	Q	O	W	A	M	G	T	B	R	F	J	V	S	O	W	J	S	A	Z	F	Q	N	Z	M	U	Y	V		
L	E	M	W	U	N	Q	D	W	L	F	D	Q	L	Z	R	G	Z	I	L	E	M	U	K	W	R	F	U	E	Z	*		B	K	D	K	R	T	K	H	K	Y	B	R	Y	W	W	D	A	L	Y	V	B	K	D	X	S	B	A	J	X	M	J	*			
W	L	Q	E	P	F	D	Q	G	D	C	E	*	U	D	P	G	F	D	E	D	G	W	B	U	Z	Y	U	F	U	H		S	S	K	L	D	N	M	K	V	Z	I	L	*	Y	M	C	V	R	J	L	L	Q	O	P	I	D	O	B	Y	N	P	Q			
*	*	F	F	U	H	W	L	?	D	Q	L	B	R	F	L	J	D	Q	L	L	G	A	D	Q	F	X	G	N	Z	L	D		*	*	N	N	P	Q	S	S	?	J	K	X	I	C	M	K	O	J	K	X	X	U	G	N	C	M	T	Q	K	A	W	N		
U	I	M	U	L	W	J	T	D	Q	L	U	Z	M	E	A	X	G	D	U	E	Z	P	G	F	N	G	D	Q	L	A	*		D	U	X	Y	X	O	V	G	Z	K	W	Y	S	K	L	J	G	N	J	Y	M	A	C	V	R	Z	N	L	R	Z	G	*		
L	J	G	Z	J	D	A	G	Z	F	X	U	D	D	L	J	R	Z	J	L	A	N	A	R	R	Z	J	D	E	G	Z	R		K	O	N	O	W	L	I	N	B	R	T	R	L	L	Z	V	D	H	V	W	H	K	I	A	D	H	V	J	M	Q	A	A		
Z	S	Z	E	P	Z	W	E	I	G	D	U	E	Z	T	J	E	L	F	W	G	Z	N	W	L	B	S	Z	E	P	G	*		B	U	O	L	D	S	O	L	X	Y	J	R	M	S	F	V	Q	S	M	P	Q	S	R	P	K	W	F	O	M	D	U	*		
H	E	R	D	D	Q	U	F	?	D	Q	L	B	F	Q	E	R	W	J	U	D	F	H	R	A	U	L	J	E	R	D	*		Q	Q	J	J	J	R	Y	Q	?	J	P	S	H	M	R	M	C	P	Z	D	J	M	U	B	I	R	K	O	L	A	L	*		
D	Q	L	A	L	F	E	X	L	P	Q	L	A	L	T	P	Q	E	S	Z	E	P	F	D	Q	L	L	T	G	I	D	W		L	Y	W	J	S	M	L	O	X	E	K	K	V	W	D	D	R	N	F	B	K	C	M	L	R	Z	W	G	Y	U	J	T		
E	I	G	D	U	E	Z	?	E	Z	W	B	P	P	D	Q	U	F	P	G	F	Q	U	F	W	G	F	D	X	L	F	*		M	W	N	D	L	O	?	M	S	O	H	F	M	J	K	Y	N	E	N	U	C	R	M	T	Q	Q	J	A	S	M	*			
F	G	N	L	T	D	Q	U	A	D	B	L	U	N	Q	D	J	L	N	A	L	F	M	U	M	D	B	F	L	*	*		M	Q	K	Z	D	N	C	R	G	L	I	Z	R	Y	C	J	V	X	K	I	W	K	R	X	K	T	L	J	M	K	*	*			
Y	L	Z	X	U	Z	R	D	L	F	F	U	T	C	E	U	Z	D	M	U	Y	L	F	L	I	E	Z	J	F	Z	E	*		L	W	O	O	Y	A	C	A	N	S	M	M	Y	E	L	L	T	H	J	X	Y	C	X	Z	M	K	T	L	O	N	A	L	*	*
A	D	Q	F	L	Y	L	Z	D	B	F	L	Y	L	Z	J	L	N	A	L	F	L	U	N	Q	D	X	U	Z	R	*		J	Z	K	M	X	C	Z	O	N	W	M	G	C	X	A	V	K	B	G	W	X	N	Z	R	Y	C	J	T	Y	S	C	*			
D	L	F	M	E	A	D	B	M	E	R	A	F	L	I	E	Z	J	F	P	L	F	D	T	W	G	B	L	A	D	P	E		J	K	R	X	L	H	L	J	X	Q	J	G	M	X	V	N	O	Z	R	C	W	N	L	F	P	Y	V	W	G	L	D	N		
F	W	E	P	W	B	J	L	F	C	G	A	G	D	W	B	F	W	E	P	W	B	D	Q	L	A	L	X	G	U	Z	F		L	Z	J	B	G	A	E	Q	Z	W	F	A	Q	L	E	I	C	D	L	E	U	H	U	J	B	F	Q	Z	G	U	G			
E	M	C	G	F	F	G	N	L	J	L	H	A	U	F	D	Q	G	D	L	Z	I	R	X	H	L	A	L	J	D	*	*		I	Q	D	Z	A	L	B	R	W	J	F	W	F	W	W	Z	E	Q	F	Z	E	F	X	A	P	T	X	Z	L	*	*			
Q	L	W	E	P	L	A	C	G	A	D	E	M	D	Q	L	J	E	E	A	P	G	B	P	G	F	A	L	X	E	Y	*		D	C	A	Z	N	G	D	U	Q	Z	A	G	A	C	L	F	W	Z	Z	L	W	L	H	W	C	U	G	I	G	L	*			
L	J	P	U	D	Q	D	A	L	X	H	W	U	Z	N	Q	G	Z	J	F	X	G	J	L	G	D	U	Z	B	*	*		P	X	D	P	Z	J	U	D	L	L	Z	A	G	Q	I	D	L	Z	L	R	J	A	L	D	Z	Q	G	L	E	L	*	*			
H	A	L	G	I	Q	U	Z	D	Q	L	R	C	L	A	W	L	M	D	Q	G	Z	J	I	E	A	Z	L	A	G	Z		D	M	E	W	F	L	J	D	U	P	L	Z	Q	G	L	E	B	D	L	B	K	Q	L	Z	I	D	G	B	I	A	*				
J	D	Q	L	Z	P	U	J	L	Z	U	Z	N	D	Q	L	E	W	L	G	W	U	D	D	W	L	U	Z	*	*		L	U	M	D	H	A	F	C	G	X	Q	L	Q	L	F	L	Z	G	D	G	X	G	D	N	B	L	L	A	W	H	*	*				
F	L	A	D	L	J	D	Q	L	I	G	Z	J	W	L	G	Z	J	C	L	L	A	L	J	U	Z	D	Q	L	Q	E	D		D	L	L	M	E	G	F	M	U	E	D	R	L	D	R	G	L	E	D	E	G	A	X	G	L	L	A	D	A	Z	*	*		
G	U	A	L	F	I	G	C	U	Z	N	M	A	E	X	D	Q	L	I	Q	G	X	H	L	A	I	G	R	F	L	J	*		H	F	L	J	J	Z	U	G	G	Q	D	X	F	D	L	P	C	U	L	A	E	G	N	A	U	L	P	M	L	H	*			
D	Q	L	M	W	G	X	L	D	E	M	W	U	I	S	L	A	H	R	D	C	A	L	F	L	Z	D	W	B	J	L	D		G	L	I	D	J	J	Q	U	W	I	L	U	Q	F	U	D	L	N	E	L	G	E	F	J	A	B	J	W	E	A	U			
U	W	F	E	M	D	Q	L	A	E	E	X	P	U	D	Q	U	Z	L	X	L	A	N	L	J	M	A	E	X	*	*	*		A	S	W	X	W	L	Q	G	N	D	J	Q	G	A	J	C	Z	D	L	E	Q	X	N	M	X	M	L	R	Q	L	*	*	*	
L	X	U	F	M	D	T	I	G	Z	B	E	R	F	L	G	Z	B	D	Q	U	Z	N	K	?	E	H	S	A	*	F	D		J	X	A	U	C	M	L	U	X	L	Q	Z	W	F	F	D	A	D	Y	J	E	Q	P	?	E	H	S	A	*	L	I			
M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E	*	R	E	T	E	N	Q	R	W	H	F	E	W			M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E	*	R	E	T	E	N	Q	R	W	H	F	E	W			
D	P	D	K	F	V	K	F	L	S	O	O	P	G	D	V	S	W	R	J	U	G	P	U	Z	M	H	Z	B	C	*		D	P	D	K	F	V	K	F	L	S	O	O	P	G	D	V	S	W	R	J	U	G	P	U	Z	M	H	Z	B	C	*				
Y	D	D	X	O	M	C	S	P	N	J	S	O	T	D	V	I	J	U	N	S	R	Q	R	G	R	L	S	I	G	A	*		Y	D	D	X	O	M	C	S	P	N	J	S	O	T	D	V	I	J	U	N	S	R	Q	R	G	R	L	S	I	G	A	*		

C(8) RP4 KRS(7) SA(SS) RML[LZM] IUT2:2[BF/BG] IUT3 :4[ITN][ABV] NSP

IMETMMKGYIEXMGUHLVKQHRLLEUMHIGMKJ	MZYNDUSXBNHOCYGLWASXLABKGUQKNVPW
MFNXVQUEXVMGEUMKXHKJMFNVRRXYGVFK*	CRERYORIRPCYPXXEBMPWCRESZACBLZNL*
TGEUHEDEF*VETHGEFEHXXCVKPVGVIXMUF	EQNRWKJM*PNDWYLMUMUSTMESCPQTUAARM
IXM?EUMCYGMLEUMMHBEUGZHQKME**GGV	UAA?LRZJDNRSLRZZVHQDNOURBXQ**QQT
VJNVMXLOEUMVKNFBBZHEVEFKTHGQHEUMB*	EVZPZSWHKKRXPARMHQLPNBDWYKQMYKH*
MLHKKLEBHKGZVEEMLYKLMBOBYKLEFHKY	RSQSXMJQCYOYMMKWEIWXIRJBEIWLNUBB
KAKFTKXKFJHEVFKOLFMMGXHKQXMCARFTH*	CVSMEASMZPLYNAGWUANTUAYTRXGNSNEV*
IFYEEUVG?EUMCGUFYXLVLEGYIBVMFLFY*	UULLLYPU?LTAINYNNDTKELNVCJYRSMBM*
EUMBMGFZMTUMBMOUFUFAKFTGEUMMOHJEX	MPXLANMSZFRRWXEEYQGCARDNMYKXHPVLO
FJHEVFK?FKXCTTEUVGTHGUVGXHGEZMG*	NXQQEMS?NASIGNLRPQFQVDYNOULBAN*
GHQMOEUVBECMVQUELMQBMMGNVNECGM**	NURKEQDYHMKYPPDLWZRJXRYZYKMLNR**
PMKZVKYEMGGVODFVKENVPMMGJFKLGKF*	MXSSPBBQANNPFMMOILZPDKNROMSXQBM*
BEUGMPMKECGMPMKLMQBMMGMVQUEZVKY*	LKRNZDKSQXNXDZBWRCHXZQKYPDLOPAD*
EMGNFBECECNFYBGMJFKLGTMGEOXHCMBET*	LRYZMIMLZULHNZQWQSKYDXQMGTWXXHMEQ
GXFTXCLMGDHBHEXCGXFTXCEUMBMZHVKG	MKLCHUBFUKXGBUMFJDEMFVIVLVCUGKHVH
FNDHGGHQMMLIBVGEUEHEMKJYZIMBMLE**	JUEKBMCYXLGXGXKXKFUGKFGZBTOZKMM**
UMXFTMBDBHEFNEUMLFFBTHCTHGBMZFP*	EDBKQHEVUKBBDMGXKKMMXIMXDVHJHJM*
MLTVEUEBMMZIXVKQUHKLGVZHLMHVEVK**	TZETKLVEMMKBUJEMKMYLBMEKUHMFMM**
IBMHJUVKEUMYDDMBXMMNEUHKLJFBKMBHK	ENFXGMLEVTMKUHMVFCMCRUMMKJEHCJB
LEUMKTVLMKVKQEUUMUFXXMHXVEEXMVVK**	MVNEIBGDHZZUUMTMKHEHZZHEMQCMMBXI**
GMBEMLEUMJHKLXMHKKLDMMBMLVKEUMUFE	EMMNFFHGNVFEYMEYHMFEBHBMMEKBEV
HVBMGJHDVKQNBFEZEUUMJUHZIMBJHYGML*	IGMLLKVHHUEEZGEMTDVMBFHBQBMVTNMI*
EUMNXHZMEFNXVJAMBIIYEDBGMKEXCLME	HMJELLUVXJMVUGVEMQFMHFGLLBCLXFBV
VXGFNEUMBFFZTVEUVKMMZMBQMLNBFZ***	BAXZXMUHQELUHBBLDKMFUZQNZNMUYUM***
ZVGEQJHCKFYGMHMKCEUVKQR?FIAB*EUM	ZBVDNMVZMUKXGGEEBEPFUT?FIAB*MJL
DBKQAGGF*YFOSFQUYXIGFXVNIITNXBPRR	DBKQAGGF*YFQFQUYXIGFXVNIITNXBPRR
ETERGWRGGMASSSTHEWAXYLVHTVKNIKCD*	ETERGWRGGMASSSTHEWAXYLVHTVKNIKCD*
PEEZSNDATQLASOEWJLVQAYUYHYMAJHJB*	PEEZSNDATQLASOEWJLVQAYUYHYMAJHJB*

R(2)LS(6) to the TSP gives.

C(8) RP4 KRS(7) SA(SS) RML[GVE] IUT2:2[BF/BG] IUT3 :4[BN/TI][AV/BP] TSP

IMETMMKGYIEXMGUHLVKQHRLLEUMHIGMKJ	MZYNDUSXBNHOCYGLWASXLABKGUQKNVPW
MFNXVQUEXVMGEUMKXHKJMFNVRRXYGVFK*	CRERYORIRPCYPXXEBMPWCRESZACBLZNL*
TGEUHEDEF*VETHGEFEHXXCVKPVGVIXMUF	EQNRWKJM*PNDWYLMUMUSTMESCPQTUAARM
UMCYG?EUMLEUMMHBEUGZHQKME**GGVIXM?E	RZJDNRSLRZZVHQDNOURBXQ**QQTUAA?L
VJNVVMXLOEUMVKNFBBZHEVEFKTHGQHEUMB*	EVZPZSWHKKRXPARMHQLPNBDWYKQMYKH*
MLHKKLEBHKGZVEEMLYKLMBOBYKLEFHKY	RSQSXMJQCYOYMMKWEIWXIRJBEIWLNUBB
KAKFTKXKFJHEVFKOLFMMGXHKQXMCARFTH*	CVSMEASMZPLYNAGWUANTUAYTRXGNSNEV*
IFYEEUVG?EUMCGUFYXLVLEGYIBVMFLFY*	UULLLYPU?LTAINYNNDTKELNVCJYRSMBM*
EUMBMGFZMTUMBMOUFUFAKFTGEUMMOHJEX	MPXLANMSZFRRWXEEYQGCARDNMYKXHPVLO
FJHEVFK?FKXCTTEUVGTHGUVGXHGEZMG*	NXQQEMS?NASIGNLRPQFQVDYNOULBAN*
GHQMOEUVBECMVQUELMQBMMGNVNECGM**	NURKEQDYHMKYPPDLWZRJXRYZYKMLNR**
PMKZVKYEMGGVODFVKENVPMMGJFKLGKF*	MXSSPBBQANNPFMMOILZPDKNROMSXQBM*
BEUGMPMKECGMPMKLMQBMMGMVQUEZVKY*	LKRNZDKSQXNXDZBWRCHXZQKYPDLOPAD*
EMGNFBECECNFYBGMJFKLGTMGEOXHCMBET*	LRYZMIMLZULHNZQWQSKYDXQMGTWXXHMEQ
GXFTXCLMGDHBHEXCGXFTXCEUMBMZHVKG	MKLCHUBFUKXGBUMFJDEMFVIVLVCUGKHVH
FNDHGGHQMMLIBVGEUEHEMKJYZIMBMLE**	JUEKBMCYXLGXGXKXKFUGKFGZBTOZKMM**
UMXFTMBDBHEFNEUMLFFBTHCTHGBMZFP*	EDBKQHEVUKBBDMGXKKMMXIMXDVHJHJM*
MLTVEUEBMMZIXVKQUHKLGVZHLMHVEVK**	TZETKLVEMMKBUJEMKMYLBMEKUHMFMM**
IBMHJUVKEUMYDDMBXMMNEUHKLJFBKMBHK	ENFXGMLEVTMKUHMVFCMCRUMMKJEHCJB
LEUMKTVLMKVKQEUUMUFXXMHXVEEXMVVK**	MVNEIBGDHZZUUMTMKHEHZZHEMQCMMBXI**
GMBEMLEUMJHKLXMHKKLDMMBMLVKEUMUFE	EMMNFFHGNVFEYMEYHMFEBHBMMEKBEV
HVBMGJHDVKQNBFEZEUUMJUHZIMBJHYGML*	IGMLLKVHHUEEZGEMTDVMBFHBQBMVTNMI*
EUMNXHZMEFNXVJAMBIIYEDBGMKEXCLME	HMJELLUVXJMVUGVEMQFMHFGLLBCLXFBV
VXGFNEUMBFFZTVEUVKMMZMBQMLNBFZ***	BAXZXMUHQELUHBBLDKMFUZQNZNMUYUM***
JHRCFYGMHMKCEUVKQR?FIAB*EUMZVGEOR	MVZMUKXGGEEBEPFUT?FIAB*MJLZBVDN
DBKQAGGF*YFOSFQUYXIGFXVNIITNXBPRR	DBKQAGGF*YFQFQUYXIGFXVNIITNXBPRR
ETERGWRGGMASSSTHEWAXYLVHTVKNIKCD*	ETERGWRGGMASSSTHEWAXYLVHTVKNIKCD*
PEEZSNDATQLASOEWJLVQAYUYHYMAJHJB*	PEEZSNDATQLASOEWJLVQAYUYHYMAJHJB*

C(8) KRS(7) SA(SS)	K4 Key	Encryption
IAWI NFBN	PBAV BNTI	MITE ROFB

C(9) RP5 KRS(8) SA(?) RML[EZI] IUT H:V[IL/MN] IUT3 :4[TE/XU] NSP

J	N	F	O	N	N	R	H	P	J	F	Z	N	H	V	I	M	W	R	U	I	R	M	F	V	N	I	J	H	N	R	L	N	K	P	Q	E	V	A	Z	C	Q	I	S	D	P	H	M	X	B	A	Z	M	B	C	R	H	V	U	R	Q	W	T	X		
N	G	Q	Z	W	U	V	F	Z	W	N	H	F	V	N	R	P	I	R	L	N	G	Q	W	Y	Z	P	H	W	G	R	*	D	Y	F	Y	P	S	Y	Z	Y	T	D	P	T	Z	Z	F	C	N	T	X	D	Y	F	K	B	D	C	M	Q	K	W	T	X	
W	F	O	I	H	F	G	F	I	Z	Z	D	W	R	T	W	H	W	J	Z	N	V	G	O	H	F	V	I	F	E	G	*	T	Q	E	X	P	M	N	N	V	A	O	N	F	A	D	T	U	O	V	B	B	Y	N	F	U	Q	Y	X	R	L	N	*		
H	H	W	J	Z	N	?	F	V	N	D	P	H	N	M	F	V	N	N	I	C	F	V	H	K	I	U	R	N	F	**	U	U	O	V	B	B	?	M	Y	K	L	E	Q	Y	A	M	Y	K	K	W	I	U	E	Q	S	V	Y	C	Z	U	*	*			
W	L	Q	W	N	Z	M	S	F	V	N	W	R	Q	G	C	K	I	F	W	G	R	O	I	H	U	I	F	V	N	C	*	F	W	K	T	K	A	X	I	R	Y	Z	T	B	Y	N	M	I	U	M	T	Q	C	E	X	P	R	U	N	P	R	I	*		
N	M	I	R	M	F	C	I	R	H	K	W	F	F	N	M	P	R	M	N	C	U	C	P	P	R	M	F	G	I	R	P	Y	A	U	A	Z	N	L	U	D	P	S	P	N	N	R	X	F	J	X	Z	J	Y	L	C	F	J	X	M	Q	V	C	*		
R	B	R	G	O	R	Z	G	L	I	F	W	G	R	S	M	G	N	H	Z	I	R	U	Z	N	D	B	R	G	O	I	*	D	W	A	N	F	B	A	N	K	T	M	P	Q	B	H	X	V	B	Q	O	V	B	P	O	Y	Z	H	A	Q	F	W	*		
J	G	P	F	F	V	W	H	?	F	V	N	D	H	V	G	P	Z	M	W	F	H	J	P	C	W	N	M	G	P	F	*	V	V	M	M	M	P	T	V	?	M	O	B	J	Q	P	Q	E	O	R	F	M	Q	W	D	L	P	Y	A	N	C	N	*		
F	V	N	C	N	H	G	K	N	O	V	N	C	N	S	O	V	G	B	R	G	O	H	F	V	N	N	S	I	L	F	Z	N	T	Z	M	B	Q	N	A	K	G	Y	X	Z	F	F	P	U	H	D	Y	E	Q	N	P	R	Z	I	T	W	M	S	*		
G	L	I	F	W	G	R	?	G	R	Z	D	O	O	F	V	W	H	O	I	H	V	W	H	Z	I	H	F	K	N	H	*	Q	Z	U	U	F	N	A	?	Q	B	A	J	H	Q	M	Y	T	U	G	U	W	E	P	Q	S	V	M	C	B	Q	*			
H	I	U	N	S	F	V	W	C	F	D	N	W	U	V	F	M	N	U	C	N	N	H	Q	W	Q	F	D	H	N	*	Q	V	Y	R	F	U	E	P	I	N	L	R	P	T	E	M	X	K	Y	L	Z	Y	P	K	P	R	N	M	Q	Y	*	*			
T	N	R	K	W	R	P	F	N	H	H	W	S	E	G	W	R	F	O	W	T	N	H	N	L	G	R	M	H	R	G	*	N	Z	A	A	T	C	C	U	B	O	Q	T	G	N	N	S	J	M	K	T	E	R	O	Y	S	N	A	Z	U	C	N	*		
C	F	V	H	N	T	N	R	F	D	H	N	T	N	R	M	N	U	C	N	N	H	N	W	U	V	F	K	W	R	P	*	M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	*		
F	S	Z	I	D	N	C	F	O	G	F	N	H	Q	G	C	F	D	Q	G	P	C	H	N	L	G	R	M	H	O	N	H	N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M	I	Q	K	X	U	A	R	P	E	Z	U	*	
H	H	Z	G	O	Z	D	M	N	H	E	I	C	I	F	Z	D	H	Z	G	O	Z	D	F	V	N	C	N	K	I	W	R	I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W	*	
G	Q	E	I	H	H	I	U	N	M	N	J	C	W	H	F	V	I	F	N	R	L	P	K	J	N	C	N	M	F	*	L	V	F	R	C	N	D	P	Z	M	H	Z	Z	H	Z	Z	R	G	V	H	R	G	H	K	C	O	S	K	R	N	*	*			
V	N	Z	G	O	N	C	E	I	C	F	G	Q	F	V	N	M	G	G	O	I	D	O	I	H	C	N	K	G	T	*	F	E	C	R	U	I	F	W	V	R	C	I	C	E	N	H	Z	R	R	N	Z	N	J	Z	E	W	I	L	I	N	*	*			
N	M	O	W	F	V	F	C	N	K	J	Z	W	R	U	V	I	R	M	H	W	K	I	M	N	I	F	W	R	D	*	O	K	F	O	R	M	W	F	N	N	R	C	I	V	L	F	N	R	N	P	M	C	N	F	R	V	I	N	G	N	*	*			
J	C	N	I	L	V	W	R	F	V	N	P	E	E	N	C	Z	N	Q	F	V	I	R	M	L	G	C	R	N	C	I	R	F	Q	G	Z	H	N	M	F	W	O	N	R	V	I	N	W	G	D	F	N	D	Y	V	N	N	R	L	F	I	D	L	C	*	
M	F	V	N	R	O	W	M	N	R	W	R	U	F	V	N	V	G	Z	N	I	Z	W	F	F	Z	N	W	W	R	*	N	W	Q	F	J	C	H	E	I	K	V	V	N	O	N	R	I	F	I	K	I	F	N	U	D	N	N	C	Z	J	*	*			
H	N	C	F	N	M	F	V	N	L	I	R	M	Z	N	I	R	M	E	N	N	C	N	M	W	R	F	V	N	V	G	F	F	N	N	Q	G	I	H	Q	W	G	F	P	N	F	P	I	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W	*
I	W	C	N	H	L	I	E	W	R	U	Q	C	G	K	F	V	N	L	V	I	K	J	N	C	L	I	P	H	N	M	*	J	H	N	M	M	R	W	I	I	V	F	F	K	H	F	N	O	E	W	N	C	G	I	U	C	W	N	O	Q	N	J	*	*	
F	V	N	Q	Z	I	K	N	F	G	Q	Z	W	L	B	N	C	J	P	F	E	C	N	H	N	R	F	Z	D	M	N	F	I	N	L	F	M	M	V	W	Z	L	N	W	V	H	W	F	N	U	G	N	I	G	H	M	M	C	D	M	Z	G	C	W	*	
W	Z	H	G	Q	F	V	N	C	G	G	K	O	W	F	V	W	R	N	K	N	C	U	N	M	Q	C	G	K	*	C	B	Z	K	Z	N	V	I	U	F	M	V	I	C	M	E	R	N	G	V	K	U	Q	K	O	N	P	V	N	*	*	*				
F	V	N	K	W	H	F	S	L	I	R	D	G	P	H	N	N	I	R	D	F	V	W	R	U	Y	?	G	J	B	C	*	N	L	M	K	C	W	E	Q	N	W	K	N	V	R	Z	H	H	F	F	C	F	T	M	G	V	O	?	G	J	B	C	*		
P	G	S	G	U	V	P	Z	J	H	G	Z	W	Q	J	J	O	Q	Z	C	T	Y	Y	E	C	R	U	B	H	H	G	*	P	G	S	G	U	V	P	Z	J	H	G	Z	W	Q	J	J	O	Q	Z	C	T	Y	Y	E	C	R	U	B	H	H	G	*		
F	O	F	Y	H	X	Y	H	H	N	B	A	O	I	F	X	B	Z	P	M	W	I	O	W	R	Q	J	R	D	E	*	F	O	F	Y	H	X	Y	H	H	N	B	A	O	I	F	X	B	Z	P	M	W	I	O	W	R	Q	J	R	D	E	*				
T	F	F	K	A	Q	E	B	O	U	M	B	A	S	F	X	L	M	W	U	B	P	V	I	P	N	B	L	I	C	*	T	F	F	K	A	Q	E	B	O	U	M	B	A	S	F	X	L	M	W	U	B	P	V	I	P	N	B	L	I	C	*				

R(2)RS(11) to the TSP gives.

C(9) RP5 KRS(8) SA(?) SML[HNH] IUT2:2[IN/LM] IUT3 :4[UE/CX] TSP

J	N	F	O	N	N	R	H	P	J	F	Z	N	H	V	I	M	W	R	U	I	R	M	F	V	N	I	J	H	N	R	L	N	K	P	Q	E	V	A	Z	C	Q	I	S	D	P	H	M	X	B	A	Z	M	B	C	R	H	V	U	R	Q	W	T	X		
N	G	Q	Z	W	U	V	F	Z	W	N	H	F	V	N	R	P	I	R	L	N	G	Q	W	Y	Z	P	H	W	G	R	L	N	K	P	Q	E	V	A	Z	C	Q	I	S	D	P	H	M	X	B	A	Z	M	B	C	R	H	V	U	R	Q	W	T	X		
W	F	O	I	H	F	G	F	I	Z	Z	D	W	R	T	W	H	W	J	Z	N	V	G	O	H	F	V	I	F	E	G	*	T	Q	E	X	P	M	N	N	V	A	O	N	F	A	D	T	U	O	V	B	B	Y	N	F	U	Q	Y	X	R	L	N	*		
H	H	W	J	Z	N	?	F	V	N	D	P	H	N	M	F	V	N	N	I	C	F	V	H	K	I	U	R	N	F	**	U	U	O	V	B	B	?	M	Y	K	L	E	Q	Y	A	M	Y	K	K	W	I	U	E	Q	S	V	Y	C	Z	U	*	*			
W	L	Q	W	N	Z	M	S	F	V	N	W	R	Q	G	C	K	I	F	W	G	R	O	I	H	U	I	F	V	N	C	*	F	W	K	T	K	A	X	I	R	Y	Z	T	B	Y	N	M	I	U	M	T	Q	C	E	X	P	R	U	N	P	R	I	*		
N	M	I	R	M	F	C	I	R	H	K	W	F	F	N	M	P	R	M	N	C	U	C	P	P	R	M	F	G	I	R	P	Y	A	U	A	Z	N	L	U	D	P	S	P	N	N	R	X	F	J	X	Z	J	Y	L	C	F	J	X	M	Q	V	C	*		
R	B	R	G	O	R	Z	G	L	I	F	W	G	R	S	M	G	N	H	Z	I	R	U	Z	N	D	B	R	G	O	I	*	D	W	A	N	F	B	A	N	K	T	M	P	Q	B	H	X	V	B	Q	O	V	B	P	O	Y	Z	H	A	Q	F	W	*		
J	G	P	F	F	V	W	H	?	F	V	N	D	H	V	G	P	Z	M	W	F	H	J	P	C	W	N	M	G	P	F	*	V	V	M	M	M	P	T	V	?	M	O	B	J	Q	P	Q	E	O	R	F	M	Q	W	D	L	P	Y	A	N	C	N	*		
F	V	N	C	N	H	G	K	N	O	V	N	C	N	S	O	V	G	B	R	G	O	H	F	V	N	N	S	I	L	F	Z	N	T	Z	M	B	Q	N	A	K	G	Y	Y	X	Z	F	F	P	U	H	D	Y	E	Q	N	P	R	Z	I	T	W	M	S	*	
G	L	I	F	W	G	R	?	G	R	Z	D	O	O	F	V	W	H	O	I	H	V	W	H	Z	I	H	F	K	N	H	*	Q	Z	U	U	F	N	A	?	Q	B	A	J	H	Q	M	Y	T	U	G	U	W	E	P	Q	S	V	V	M	C	B	Q	*		
H	I	U	N	S	F	V	E	C	F	D	N	W	U	F	M	N	U	C	N	N	H	Q	W	Q	F	D	H	N	*	Q	V	R	A	F	U	E	P	I	N	L	R	P	T	E	M	X	K	Y	L	Z	Y	P	K	P	R	N	M	Q	Y	*	*				
T	N	R	K	W	R	P	F	N	H	H	W	S	E	G	W	R	F	Q	W	T	N	H	N	L	G	R	M	H	R	G	*	N	Z	A	A	T	C	C	U	B	I	Q	Q	T	G	N	N	S	J	M	K	T	E	R	Q	Y	S	N	A	Z	U	C	N	*	*
C	F	V	H	N	T	N	R	F	D	H	N	T	N	R	M	N	U	C	N	N	H	N	U	V	F	K	W	R	P	*	M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	E	*	*			
C	H	N	L	G	R	M	H	O	N	H	F	S	Z	I	D	N	C	F	O	G	N	H	N	H	Q	G	C	F	D	Q	G	P	I	Q	K	X	U	A	R	P	E	Z	U	N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M	*
H	H	Z	G	O	Z	D	M	N	H	E	I	C	I	F	Z	D	H	Z	G	O	Z	D	F	V	N	C	N	K	I	W	R	I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	G	B	E	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W	*
G	Q	E	I	H	H	I	U	N	M	N	J	C	W	H	F	V	I	F	N	R	L	P	K	J	N	C	N	M	F	*	*	L	V	F	R	C	N	D	P	Z	M	H	Z	Z	H	Z	R	G	V	H	R	G	H	K	C	O	S	K	R	N	*	*			
V	N	Z	G	O	N	C	E	I	C	F	G	Q	F	V	N	M	G	G	C	O	I	D	O	I	H	C	N	K	G	T	*	F	E	C	R	U	I	F	W	V	R	C	I	C	E	N	H	Z	R	R	N	Z	N	J	Z	E	W	I	L	I	N	*			
N	M	O	W	F	V	F	C	N	K	J	Z	W	R	U	V	I	R	M	H	W	K	I	M	N	I	F	W	R	D	*	O	K	F	O	R	M	M	W	F	N	N	R	C	I	V	L	F	N	R	N	P	M	C	N	F	R	V	I	N	G	N	*	*	*	
J	C	N	I	L	W	R	F	V	N	P	E	N	C	Z	N	Q	F	V	I	R	M	L	G	C	R	N	C	I	R	*	F	O	G	Z	H	N	M	F	W	O	N	R	V	I	N	W	G	D	F	N	D	Y	V	N	N	R	L	F	I	D	L	*			
M	F	V	N	R	O	W	M	N	R	U	F	V	N	V	G	Z	N	I	Z	W	F	F	Z	N	W	W	R	*	*	N	W	Q	F	J	C	H	E	I	K	V	V	N	O	N	R	I	F	I	K	I	F	N	U	D	N	N	C	Z	J	*	*				
H	N	C	F	N	M	F	V	N	L	I	R	M	Z	N	I	R	M	E	N	N	C	N	M	W	R	F	V	N	V	G	F	F	N	N	Q	G	I	H	O	W	G	F	P	N	F	I	N	G	F	I	C	K	I	N	N	C	R	C	F	W	*				
I	W	C	N	H	L	I	E	W	R	U	Q	C	G	K	F	V	N	L	V	I	K	J	N	C	L	I	P	H	N	*	J	H	N	M	M	R	W	I	V	F	F	K	H	F	N	O	E	W	N	C	G	I	U	C	W	N	O	Q	N	J	*				
F	V	N	Q	Z	I	K	N	F	G	Q	Z	W	L	B	N	C	J	P	F	E	C	N	H	N	R	F	Z	D	M	N	F	I	N	L	F	M	M	V	W	Z	L	N	W	V	H	W	F	N	U	G	N	I	G	H	M	M	C	D	M	Z	G	C	W	*	
W	Z	H	G	O	F	V	N	C	G	G	K	O	W	F	V	W	R	N	K	N	C	U	N	M	Q	C	G	K	*	*	C	B	Z	K	Z	N	V	I	U	F	M	V	I	C	M	E	R	N	G	V	K	U	Q	K	O	N	P	V	N	*	*	*			
F	V	N	K	W	H	F	S	L	I	R	D	G	P	H	N	N	I	R	D	F	V	W	R	U	Y	?	G	J	B	C	*	N	L	M	K	C	W	E	Q	N	W	K	N	V	R	Z	H	H	F	F	C	F	T	M	E	G	V	O	?	G	J	B	C	*	
P	G	S	G	U	V	P	Z	J	H	G	Z	W	Q	J	J	O	Q	Z	C	T	Y	E	C	R	U	B	H	G	*	P	G	S	G	U	V	P	Z	J	H	G	Z	W	Q	J	J	O	Q	Z	C	T	Y	E	C	R	U	B	H	G	*						
F	O	F	Y	H	X	Y	H	N	B	A	O	I	F	X	B	Z	P	M	W	I	O	W	R	Q	J	R	D	E	*	F	O	F	Y	H	X	Y	H	N	B	A	O	I	F	X	B	Z	P	M	W	I	O	W	R	Q	J	R	D	E	*						
T	F	F	K	A	Q	E	B	O	U	M	B	A	S	F	X	L	M	W	U	B	P	V	P	I	P	N	B	L	I	C	*	T	F	F	K	A	Q	E	B	O	U	M	B	A	S	F	X	L	M	W	U	B	P	V	P	I	P	N	B	L	I	C	*		

C(10) RP5 KRS(9) SA(?) RML[FRQ] IUT H :V[IM/LN] IUT3 :4[UH/SK] NSP

L	Q	G	S	Q	Q	Y	I	T	L	G	K	Q	I	W	J	N	X	Y	V	J	Y	N	G	W	Q	J	L	I	Q	Y	M		
Q	H	U	K	X	V	W	G	K	X	Q	I	G	W	Q	Y	T	J	Y	M	Q	H	U	X	P	K	T	I	X	H	Y	*		
X	G	S	J	I	G	H	G	J	K	K	E	X	Y	O	X	I	X	L	K	Q	W	H	S	I	G	W	J	G	F	H	*		
I	I	X	L	K	Q	?	G	W	Q	E	T	I	Q	N	G	W	Q	Q	J	D	G	W	I	R	J	V	Y	Q	G	*	*		
X	M	U	X	Q	K	N	A	G	W	Q	X	Y	U	H	D	R	J	G	X	H	Y	S	J	I	V	J	G	W	Q	D	*		
Q	N	J	Y	N	G	D	J	Y	I	R	X	G	G	Q	N	T	Y	N	Q	D	V	D	T	T	Y	N	G	H	J	Y	T		
Y	C	Y	H	S	Y	K	H	M	J	G	X	H	Y	A	N	H	Q	I	K	J	Y	V	K	Q	E	C	Y	H	S	J	*		
L	H	T	G	G	W	X	I	?	G	W	Q	E	I	W	H	T	K	N	X	G	I	L	T	D	X	Q	N	H	T	G	*		
G	W	Q	D	Q	I	H	R	Q	S	W	Q	D	Q	A	S	W	H	C	Y	H	S	I	G	W	Q	Q	A	J	M	G	K		
H	M	J	G	X	H	Y	?	H	Y	K	E	S	S	G	W	X	I	S	J	I	W	X	I	K	J	I	G	R	Q	I	*		
I	J	V	Q	A	G	W	X	D	G	E	Q	X	V	W	G	N	Q	V	D	Q	Q	I	U	X	U	G	E	I	Q	*	*		
O	Q	Y	R	X	Y	T	G	O	I	I	X	A	F	H	X	Y	G	U	X	O	Q	I	Q	M	H	Y	N	I	Y	H	*		
D	G	W	I	Q	O	Q	Y	G	E	I	Q	O	Q	Y	N	Q	V	D	Q	Q	I	Q	X	V	W	G	R	X	Y	T	*		
T	D	I	Q	M	H	Y	N	I	S	Q	I	G	A	K	J	E	Q	D	G	S	H	G	Q	I	U	H	D	G	E	U	H		
E	G	W	Q	D	Q	R	J	X	Y	I	I	K	H	S	K	E	N	Q	I	F	J	D	J	G	K	E	I	K	H	S	K		
H	U	F	J	I	I	J	V	Q	N	Q	L	D	X	I	G	W	J	G	Q	Y	M	T	R	L	Q	D	O	N	G	*	*		
W	Q	K	H	S	Q	D	F	J	D	G	H	U	G	W	Q	N	H	H	D	S	J	E	S	J	I	D	Q	R	H	O	*		
Q	N	S	X	G	W	G	D	Q	R	L	K	X	Y	V	W	J	Y	N	I	X	R	J	N	Q	J	G	X	Y	E	*	*		
L	D	Q	J	M	W	X	Y	G	W	Q	T	F	F	Q	D	K	Q	U	G	W	J	Y	N	M	H	D	Y	Q	D	J	Y		
N	G	W	Q	Y	S	X	N	Q	Y	X	Y	V	G	W	Q	W	H	K	Q	J	K	X	G	G	K	Q	X	X	Y	*	*		
I	Q	D	G	Q	N	G	W	Q	M	J	Y	N	K	Q	J	Y	N	F	Q	Q	D	Q	N	X	Y	G	W	Q	H	G			
J	X	D	Q	I	M	J	F	X	Y	V	U	D	H	R	G	W	Q	M	W	J	R	L	Q	D	M	J	T	I	Q	N	*		
G	W	Q	U	K	J	R	Q	G	H	U	K	X	M	C	Q	D	L	T	G	F	D	Q	I	Q	Y	G	K	E	N	Q	G		
X	K	I	H	U	G	W	Q	D	H	H	R	S	X	G	W	X	Y	Q	R	Q	D	V	Q	N	U	D	H	R	*	*	*		
G	W	Q	R	X	I	G	A	M	J	Y	E	H	T	I	Q	J	Y	E	G	W	X	Y	V	P	?	H	L	C	D	*			
T	H	A	H	V	W	T	K	L	I	H	K	X	U	L	S	U	K	D	O	P	P	F	D	Y	V	C	I	I	H	*			
G	S	G	P	I	Z	P	I	I	Q	C	B	B	S	J	G	Z	C	K	T	N	X	J	S	X	Y	U	L	Y	E	F	*		
O	G	G	R	B	U	F	C	S	V	N	C	B	A	G	Z	M	N	X	V	C	T	W	T	J	T	Q	C	M	J	D	*		
Q	R	T	U	F	W	B	K	D	U	J	A	E	T	I	N	Z	C	B	K	N	C	D	Y	I	W	V	Y	U	X	O	Z		
E	P	G	P	T	A	P	L	P	O	E	T	O	K	K	G	D	Q	O	Z	E	P	G	R	C	E	D	N	R	U	N	*		
O	U	F	Z	T	N	Q	Q	W	B	S	Q	G	B	E	O	V	S	W	C	C	P	G	V	U	P	Z	Y	M	Q	*			
V	V	S	W	C	C	?	N	P	R	M	F	U	P	B	N	P	R	R	X	J	V	F	U	A	W	P	D	K	V	*	*		
G	X	R	O	R	B	Z	J	Y	P	K	O	C	P	Q	N	J	V	N	O	U	D	F	Z	T	Y	V	Q	T	Y	J	*		
P	B	V	B	K	Q	M	V	E	T	A	T	Q	Q	Y	Z	G	L	Z	K	L	P	M	D	G	L	Z	N	U	W	D	D		
E	X	B	Q	G	C	B	Q	R	O	N	T	U	C	I	Z	W	C	U	S	W	C	T	S	P	K	I	B	U	G	X	*		
W	W	N	N	N	T	O	W	?	N	S	C	L	U	T	U	F	S	Y	G	N	U	X	E	M	T	P	B	Q	D	Q	*		
Q	O	K	N	C	U	Q	B	R	H	P	P	Z	K	G	T	V	I	E	P	F	U	Q	T	Y	K	J	O	X	N	A			
U	K	V	V	G	Q	B	?	U	C	B	L	I	U	N	P	O	V	H	V	X	F	T	U	A	W	N	D	C	U	*			
U	W	P	Y	G	V	F	T	J	Q	M	Y	T	O	F	N	Z	R	P	M	K	P	T	R	T	Y	Q	N	U	P	*	*		
Q	K	B	B	O	D	D	V	C	U	U	O	H	Q	Q	A	L	N	R	O	F	Y	U	P	A	Q	B	K	V	D	Q	*		
N	Y	P	U	R	F	Y	B	V	K	U	R	F	R	D	Z	P	E	J	K	R	V	Y	T	O	F	N	A	O	C	F	*		
N	J	U	R	Z	V	B	Y	T	F	K	V	Q	I	S	Z	K	K	J	Q	G	V	N	P	T	R	Q	L	Q	N	R	W		
X	L	X	N	E	I	W	Y	J	X	J	Q	Y	N	E	J	W	D	H	W	Y	K	I	D	W	Q	H	M	F	G	Q	H		
M	W	G	Y	D	Q	E	T	K	N	I	K	K	I	K	K	Y	H	W	I	Y	H	I	R	D	S	A	R	Y	Q	*	*		
G	F	D	Y	V	J	G	X	W	Y	D	J	D	F	Q	I	K	Y	Y	Q	K	L	K	F	X	X	J	M	J	Q	*			
S	R	G	S	Y	N	X	G	Q	Q	Y	D	J	W	M	G	Q	Y	Q	T	N	D	Q	G	Y	W	J	Q	H	Q	*	*		
G	U	H	K	I	Q	N	G	X	S	Q	Y	W	J	Q	X	H	E	G	Q	E	P	W	Q	Q	Y	M	G	J	E	M	D		
Q	X	U	G	L	D	I	F	J	R	W	W	Q	S	Q	Y	J	G	J	R	J	G	Q	V	E	Q	D	K	L	*	*			
G	Q	Q	U	H	J	I	U	X	H	G	T	Q	G	T	J	Q	H	G	H	J	D	R	J	Q	D	G	Y	D	G	X			
L	I	Q	N	N	Y	X	J	J	W	G	G	R	I	G	Q	S	F	X	Q	D	H	J	V	D	X	Q	S	U	Q	L	*		
J	Q	M	G	N	N	W	X	K	M	Q	X	W	I	X	G	Q	V	H	Q	J	H	I	N	N	D	E	N	K	H	D	X		
D	C	K	R	K	Q	W	J	V	G	N	W	J	D	N	F	Y	Q	H	W	R	V	U	R	U	Q	T	W	Q	*	*	*		
Q	M	N	R	D	X	F	U	Q	X	R	Q	W	Y	K	I	I	G	D	G	O	N	H	W	S	?	H	L	C	D	*			
T	H	A	H	V	W	T	K	L	I	H	K	X	U	L	S	U	K	D	O	P	P	F	D	Y	V	C	I	I	H	*			
G	S	G	P	I	Z	P	I	I	Q	C	B	B	S	J	G	Z	C	K	T	N	X	J	S	X	Y	U	L	Y	E	F	*		
O	G	G	R	B	U	F	C	S	V	N	C	B	A	G	Z	M	N	X	V	C	T	W	T	J	T	Q	C	M	J	D	*		
Q	R	T	U	F	W	B	K	D	U	J	A	E	T	I	N	Z	C	B	K	N	C	D	Y	I	W	V	Y	U	X	O	Z		
E	P	G	P	T	A	P	L	P	O	E	T	O	K	K	G	D	Q	O	Z	E	P	G	R	C	E	D	N	R	U	N	*		
O	U	F	Z	T	N	Q	Q	W	B	S	Q	G	B	E	O	V	S	W	C	C	P	G	V	U	P	Z	Y	M	Q	*			
V	V	S	W	C	C	?	N	P	R	M	F	U	P	B	N	P	R	R	X	J	V	F	U	A	W	P	D	K	V	*	*		
G	X	R	O	R	B	Z	J	Y	P	K	O	C	P	Q	N	J	V	N	O	U	D	F	Z	T	Y	V	Q	T	Y	J	*		
P	B	V	B	K	Q	M	V	E	T	A	T	Q	Q	Y	Z	G	L	Z	K	L	P	M	D	G	L	Z	N	U	W	D	D		
E	X	B	Q	G	C	B	Q	R	O	N	T	U	C	I	Z	W	C	U	S	W	C	T	S	P	K	I	B	U	G	X	*		
W	W	N	N	N	T	O	W	?	N	S	C	L	U	T	U	F	S	Y	G	N	U	X	E	M	T	P	B	Q	D	Q	*		
Q	O	K	N	C	U	Q	B	R	H	P	P	Z	K	G	T	V	I	E	P	F	U	Q	T	Y	K	J	O	X	N	A			
U	K	V	V	G	Q	B	?	U	C	B	L	I	U	N	P	O	V	H	V	X	F	T	U	A	W	N	D	C	U	*			
U	W	P	Y	G	V	F	T	J	Q	M	Y	T	O	F	N	Z	R	P	M	K	P	T	R	T	Y	Q	N	U	P	*	*		
Q	K	B	B	O	D	D	V	C	U	U	O	H	Q	Q	A	L	N	R	O	F	Y	U	P	A	Q	B	K	V	D	Q	*		
N	Y	P	U	R	F	Y	B	V	K	U	R	F	R	D	Z	P	E	J	K	R	V	Y	T	O	F	N	A	O	C	F	*		
N	J	U	R	Z	V	B	Y	T	F	K	V	Q	I	S	Z	K	K	J	Q	G	V	N	P	T	R	Q	L	Q	N	R	W		
X	L	X	N	E	I	W	Y	J	X	J	Q	Y	N	E	J	W	D	H	W	Y	K	I	D	W	Q	H	M	F	G	Q	H		

R(2)LS(5) to the TSP gives.

C(10) RP5 KRS(9) SA(?) SML[IIQ] IUT2:2[IL/MN] IUT3 :4[KS/CU] TSP

L	Q	G	S	Q	Q	Y	I	T	L	G	K	Q	I	W	J	N	X	Y	V	J	Y	N	G	W	Q	J	L	I	Q	Y	M	
Q	H	U	K	X	V	W	G	K	X	Q	I	G	W	Q	Y	T	J	Y	M	Q	H	U	X	P	K	T	I	X	H	Y	*	
X	G	S	J	I	G	H	G	J	K	K	E	X	Y	O	X	I	X	L	K	Q	W	H	S	I	G	W	J	G	F	H	*	
I	I	X	L	K	Q	?	G	W	Q	E	T	I	Q	N	G	W	Q	Q	J	D	G	W	I	R	J	V	Y	Q	G	*	*	
X	M	U	X	Q	K	N	A	G	W	Q	X	Y	U	H	D	R	J	G	X	H	Y	S	J	I	V	J	G	W	Q	D	*	
Q	N	J	Y	N	G	D	J	Y	I	R	X	G	G	Q	N	T	Y	N	Q	D	V	D	T	T	Y	N	G	H	J	Y	T	
Y	C	Y	H	S	Y	K	H	M	J	G	X	H	Y	A	N	H	Q	I	K	J	Y	V	K	Q	E	C	Y	H	S	J	*	
L	H	T	G	G	W	X	I	?	G	W	Q	E	I	W	H	T	K	N	X	G	I	L	T	D	X	Q	N	H	T	G	*	
G	W	Q	D	Q	I	H	R	Q	S	W	Q	D	Q	A	S	W	H	C	Y	H	S	I	G	W	Q	Q	A	J	M	G	K	
H	M	J	G	X	H	Y	?	H	Y	K	E	S	S	G	W	X	I	S	J	I	W	X	I	K	J	I	G	R	Q	I	*	
I	J	V	Q	A	G	W	X	D	G	E	Q	X	V	W	G	N	Q	V	D	Q	Q	I	U	X	U	G	E	I	Q	*	*	
O	Q	Y	R	X	Y	T	G	O	I	I	X	A	F	H	X	Y	G	U	X	O	Q	I	Q	M	H	Y	N	I	Y	H	*	
D	G	W	I	Q	O	Q	Y	G	E	I	Q	O	Q	Y	N	Q	V	D	Q	Q	I	Q	X	V	W	G	R	X	Y	T	*	
H	Y	N	I	S	Q	I	G	A	K	J	E	Q	D	G	S	H	G	Q	I	U	H	D	G	E	U	H	T	D	I	Q	M	
E	G	W	Q	D	Q	R	J	X	Y	I	I	K	H	S	K	E	N	Q	I	F	J	D	J	G	K	E	I	K	H	S	K	
H	U	F	J	I	I	J	V	Q	N	Q	L	D	X	I	G	W	J	G	Q	Y	M	T	R	L	Q	D	Q	N	G	*	*	
W	O	K	H	S	Q	D	F	J	D	G	H	U	G	W	Q	N	H	H	D	S	J	E	S	J	I	D	Q	R	H	O	*	
Q	N	S	X	G	W	G	D	Q	R	L	K	X	Y	V	W	J	Y	N	I	X	R	J	N	Q	J	G	X	Y	E	*	*	
L	D	Q	J	M	W	X	Y	G	W	Q	T	F	F	D	Q	K	U	G	W	J	Y	N	M	H	D	Y	Q	D	J	*		
N	G	W	Q	Y	S	X	N	Q	Y	X	Y	V	G	W	Q	W	H	K	Q	J	K	X	G	G	K	Q	X	X	Y	*	*	
I	Q	D	G	Q	N	G	W	Q	M	J	Y	N	K	Q	J	Y	N	F	Q	D	Q	N	X	Y	G	W	Q	W	H	G		
J	X	D	Q	I	M	J	F	X	Y	V	U	D	H	R	G	W	Q	M	W	J	R	L	Q	D	M	J	T	I	Q	N	*	
G	W	Q	U	K	J	R	Q	G	H	U	K	X	M	C	Q	D	L	T	G	F	D	Q	I	Q	Y	G	K	E	N	Q	G	
X	K	I	H	U	G	W	Q	D	H	H	R	S	X	G	W	X	Y	Q	R	Q	D	V	Q	N	U	D	H	R	*	*	*	
G	W	Q	R	X	I	G	A	M	J	Y	E	H	T	I	Q	J	Y	E	G	W	X	Y	V	P	?	H	L	C	D	*		
T	H	A	H	V	W	T	K	L	I	H	K	X	U	L	S	U	K	D	O	P	P	F	D	Y	V	C	I	I	H	*		
G	S	G	P	I	Z	P	I	I	Q	C	B	B	S	J	G	Z	C	K	T	N	X	J	S	X	Y	U	L	Y	E	F	*	
O	G	G	R	B	U	F	C	S	V	N	C	B	A	G	Z	M	N	X	V	C	T	W	T	J	T	Q	C	M	J	D	*	
Q	R	T	U	F	W	B	K	D	U	J	A	E	T	I	N	Z	C	B	K	N	C	D	Y	I	W	V	Y	X	O	Z		
E	P	G	P	T	A	P	L	P	O	E	T	O	K	K	G	D	Q	O	Z	E	P	G	R	C	E	D	N	R	U	N	*	
O	U	F	Z	T	N	Q	O	W	B	S	Q	G	B	E	O	V	S	W	C	C	P	Q	G	V	U	P	Z	Y	M	Q	*	
V	V	S	W	C	C	?	N	P	R	M	F	U	P	B	N	P	R	X	J	V	F	U	A	W	P	D	K	V	*	*		
G	X	R	O	R	B	Z	J	Y	P	K	O	C	P	Q	N	J	V	N	O	U	D	F	Z	T	Y	V	Q	N	Y	*		
P	B	V	B	K	O	M	V	E	T	A	T	Q	Q	Y	Z	G	L	Z	K	L	P	M	D	G	L	Z	N	U	W	D		
E	X	B	Q	G	C	B	O	R	O	N	T	Q	C	I	Z	W	C	S	W	C	T	S	P	K	I	B	U	G	X	*		
W	W	N	N	N	T	O	W	?	N	S	C	L	U	T	U	F	S	Y	G	N	U	X	E	M	T	P	B	Q	D	Q	*	
Q	O	K	N	C	U	Q	B	R	H	P	P	Z	K	G	G	T	V	I	E	F	F	U	Q	T	Y	K	J	O	X	N	A	
U	K	V	G	W	Q	B	?	U	C	B	L	I	U	N	P	O	V	H	V	X	F	T	U	A	W	N	D	C	U	*		
U	W	P	Y	G	O	D	T	J	Q	M	Y	T	O	F	N	Z	R	P	M	K	P	T	R	T	Y	Q	N	D	*	*		
Q	K	B	B	O	V	D	V	C	U	O	H	Q	Q	A	L	N	R	O	F	Y	U	P	A	Q	B	K	V	D	Q	*		
N	Y	P	U	R	F	Y	B	V	K	U	K	F	R	D	Z	P	E	J	K	R	V	Y	T	O	F	N	A	O	C	F	*	
V	B	Y	T	F	K	V	Q	I	S	Z	K	K	J	Q	G	V	N	P	T	R	Q	L	Q	N	R	W	N	J	U	R	Z	
X	L	X	N	E	I	W	Y	J	X	J	Q	Y	N	E	J	W	D	H	W	Y	K	I	D	W	Q	H	M	F	G	Q	H	
M	W	G	Y	D	Q	E	T	K	N	I	K	K	I	K	K	Y	H	W	I	Y	H	I	R	D	S	A	R	Y	Q	*	*	
G	F	D	Y	V	J	G	X	W	Y	D	J	D	F	O	I	K	Y	Y	O	K	O	L	K	F	X	X	J	M	J	Q	*	
S	R	G	S	Y	N	X	G	Q	Q	Y	D	J	W	M	G	Q	Y	Q	T	N	D	O	G	Y	W	J	Q	H	Q	*	*	
G	U	H	K	I	Q	N	G	X	S	Q	Y	W	J	Q	X	H	E	G	J	E	Q	E	P	W	Q	Y	M	G	J	E	M	*
Q	X	U	G	L	D	I	F	J	R	W	W	Q	S	O	Y	J	G	R	J	G	Q	V	E	O	Q	D	K	L	*	*		
G	Q	U	H	J	I	U	X	H	G	T	Q	G	T	J	Q	H	G	H	J	D	R	J	Q	D	G	Y	D	G	X			
L	I	Q	N	N	Y	X	J	J	W	G	G	R	I	G	Q	S	F	X	Q	D	H	J	V	D	X	Q	S	U	Q	L	*	
J	Q	M	G	N	N	W	X	K	M	Q	X	W	I	X	G	Q	V	H	Q	J	H	I	N	N	D	E	N	K	H	D	X	
D	C	K	R	K	Q	W	J	V	G	N	W	J	D	N	F	Y	Q	H	W	R	V	U	R	U	Q	T	W	Q	*	*	*	
Q	M	N	R	D	X	F	U	Q	X	R	Q	W	Y	K	I	G	D	G	O	N	H	W	S	?	H	L	C	D	*	*		
T	H	A	H	V	W	T	K	L	I	H	K	X	U	L	S	U	L	S	K	D	O	P	P	F	D	Y	V	C	I	H	*	
G	S	G	P	I	Z	P	I	I	Q	C	B	B	S	J	G	Z	C	K	T	N	X	J	S	X	Y	U	L	Y	E	F	*	
O	G	G	R	B	U	F	C	S	V	N	C	B	A	G	Z	M	N	X	V	C	T	W	T	J	T	Q	C	M	J	D	*	

C(11) RP6 KRS(10) SA(CS) NSP(1) Note. C(11) has 2 NSP positions that show IUT H:V for both the R(1) and R(3) rows that will be discussed in part 4. The 2 NSP positions are shown below for the horizontal case in each instance and finally the vertical instance that is the TSP and resolves both schemes.

M	U	H	A	U	P	J	O	M	H	R	U	J	X	L	Q	Z	P	W	L	P	Q	H	X	U	L	M	J	U	P	N	
U	I	V	R	Z	W	X	H	R	Z	U	J	H	X	U	P	O	L	P	N	U	I	V	Z	T	R	O	J	Z	I	P	*
Z	H	A	L	J	H	I	H	L	R	R	F	Z	P	S	Z	J	Z	M	R	U	X	I	A	J	H	X	L	H	G	I	*
J	J	Z	M	R	U	?	H	X	U	F	O	J	U	Q	H	X	U	U	L	E	H	X	J	Y	L	W	P	U	H	*	*
Y	L	H	Z	I	P	A	L	J	W	L	H	X	U	E	*	Z	N	V	Z	U	R	Q	B	H	X	U	Z	P	V	I	E
W	E	O	O	P	Q	H	I	L	P	O	U	Q	L	P	Q	H	E	L	P	J	Y	Z	H	H	U	Q	O	P	Q	U	E
P	D	P	I	A	P	R	I	N	L	H	Z	I	P	B	Q	I	U	J	R	L	P	W	R	U	F	D	P	I	A	L	*
M	I	O	H	H	X	Z	J	?	H	X	U	F	J	X	I	O	R	Q	Z	H	J	M	O	E	Z	U	Q	I	O	H	*
H	X	U	E	U	J	I	Y	U	A	X	U	E	U	B	A	X	I	D	P	I	A	J	H	X	U	U	B	L	N	H	R
I	N	L	H	Z	I	P	?	I	P	R	F	A	A	H	X	Z	J	A	L	J	X	Z	J	R	L	J	H	Y	U	J	*
J	L	W	U	B	H	X	Z	E	H	F	U	Z	W	X	H	Q	U	W	E	U	J	V	Z	V	H	F	J	U	*	*	
S	U	P	Y	Z	P	O	H	U	J	Z	B	G	I	Z	P	H	V	Z	S	U	J	U	N	I	P	Q	J	P	I	*	
E	H	X	J	U	S	U	P	H	F	J	U	S	U	P	Q	U	W	E	U	U	J	U	Z	W	X	H	Y	Z	P	O	*
H	U	J	V	I	E	H	F	V	I	O	E	J	U	N	I	P	Q	J	A	U	J	H	B	R	L	F	U	E	H	A	I
J	R	I	A	R	F	Q	U	J	G	L	E	L	H	R	F	J	R	I	A	R	F	H	X	U	E	U	Y	L	Z	P	J
I	V	G	L	J	J	L	W	U	Q	U	M	E	Z	J	H	X	L	H	U	P	N	O	Y	M	U	E	U	Q	H	*	*
X	U	R	I	A	U	E	G	L	E	H	I	V	H	X	U	Q	I	I	E	A	L	F	A	L	J	E	U	Y	I	S	*
U	Q	A	Z	H	X	H	E	U	Y	M	R	Z	P	W	X	L	P	Q	J	Z	Y	L	Q	U	L	H	Z	P	F	*	*
M	E	U	L	N	X	Z	P	H	X	U	O	G	G	U	E	R	U	V	H	X	L	P	Q	N	I	E	P	U	E	L	P
Q	H	X	U	P	A	Z	Q	U	P	Z	P	W	H	X	U	X	I	R	U	L	R	Z	H	H	R	U	Z	Z	P	*	*
J	U	E	H	U	Q	H	X	U	N	L	P	Q	R	U	L	P	Q	G	U	U	E	U	Q	Z	P	H	X	U	X	I	H
L	Z	E	U	J	N	L	G	Z	P	W	V	E	I	Y	H	X	U	N	X	L	Y	M	U	E	N	L	O	J	U	Q	*
R	Z	N	D	U	E	M	O	H	G	E	U	J	U	P	H	R	F	Q	U	H	L	H	X	U	V	R	L	Y	U	H	I
Z	P	U	Y	U	E	W	U	Q	V	E	I	*	*	*	Z	R	J	I	V	H	X	U	E	I	I	Y	A	Z	H	X	
H	X	U	Y	Z	J	H	B	N	L	P	F	I	O	J	U	U	L	P	F	H	X	Z	P	W	T	?	I	M	D	E	*
O	I	B	I	W	X	O	R	M	J	I	R	Z	V	M	M	A	V	R	E	S	T	T	G	E	P	W	D	J	J	I	*
H	A	H	T	J	K	T	J	J	U	D	C	C	A	L	H	K	D	R	O	Q	Z	L	A	Z	P	V	M	P	F	G	*
S	H	H	Y	C	V	G	D	A	W	Q	D	C	B	H	K	N	Q	Z	W	D	O	X	O	L	O	U	D	N	L	E	*

U	Y	O	V	G	X	C	R	E	V	L	B	F	O	J	Q	K	D	C	R	Q	D	E	P	J	X	W	P	V	Z	S	K	
F	T	H	T	O	B	T	M	T	S	F	O	S	R	R	H	E	U	S	K	F	T	H	Y	D	F	E	Q	Y	V	Q	*	
S	V	G	K	O	Q	U	U	X	C	A	U	H	C	F	S	W	A	X	D	D	T	U	H	W	V	T	K	P	N	U	*	
W	W	A	X	D	?	Q	T	Y	N	G	V	T	C	Q	T	Y	Y	Z	L	W	G	?	B	X	T	E	R	W	*	*		
L	W	Q	S	V	E	G	K	?	O	P	W	U	O	P	L	*	H	Z	Y	S	Y	C	K	L	P	T	R	S	D	T	U	Q
T	N	E	H	M	K	?	V	X	E	E	T	C	W	C	R	U	N	W	F	?	O	B	O	U	P	K	H	M	K	R	M	
F	Z	C	U	H	D	C	U	Y	S	Q	O	V	D	J	K	X	D	V	A	X	D	O	A	T	R	J	C	V	H	Z	*	
X	X	Q	Q	Q	O	S	X	?	Q	A	D	M	V	O	V	G	A	P	H	Q	V	Z	F	N	O	T	C	U	E	U	*	
U	S	R	Q	D	V	U	C	Y	I	T	T	K	R	H	H	O	W	J	F	T	G	V	U	O	P	R	L	S	Z	Q	B	
V	R	W	H	U	C	?	V	D	C	M	J	V	Q	T	S	W	I	W	Z	G	O	V	B	X	X	Q	E	D	V	*		
V	X	T	P	H	W	G	O	L	U	N	P	O	S	G	Q	K	Y	T	N	R	T	O	Y	O	P	U	Q	V	T	*	*	
U	R	C	C	S	E	E	W	D	V	S	I	U	B	M	Q	Y	S	G	P	V	T	B	U	C	R	W	E	U	*	*		
Q	P	T	V	Y	G	P	C	W	R	V	R	G	Y	E	K	T	F	L	R	Y	W	P	O	S	G	Q	B	S	D	G	*	
Q	T	O	Y	U	M	U	Q	Y	X	Q	L	V	Y	K	W	C	P	O	G	R	W	U	J	A	K	R	R	L	U	H	W	
U	P	Q	F	L	X	E	I	X	P	R	J	E	X	U	I	N	G	H	U	I	Z	M	Z	Q	F	J	X	P	L	Z	L	
N	X	H	P	E	U	F	O	R	Q	J	R	R	J	R	R	P	I	X	J	P	I	J	Y	E	A	B	Y	P	U	*	*	
H	G	E	P	W	L	H	Z	X	P	E	L	E	G	U	J	R	P	P	U	R	M	G	Z	Z	L	N	L	U	*	*		
A	Y	H	A	P	Q	Z	H	U	P	E	L	X	N	H	U	P	O	Q	E	U	H	P	X	L	U	I	U	*	*			
H	V	I	R	J	U	Q	H	Z	A	U	P	X	L	U	Z	I	F	H	U	F	T	X	U	P	N	H	L	F	N	E		
U	Z	V	H	M	E	J	G	L	Y	X	X	U	A	U	P	L	H	L	Y	L	H	U	W	F	U	E	R	M	*	*		
H	U	U	V	I	L	J	V	Z	I	H	O	U	H	O	L	U	I	H	I	L	E	Y	L	U	E	H	P	E	H	Z		
M	J	U	Q	Q	P	Z	L	L	X	H	H	Y	J	H	U	A	G	Z	U	E	I	L	W	E	Z	U	A	V	U	M	*	
Z	X	J	Z	H	U	W	I	U	L	I	J	Q	O	E	F	Q	R	I	E	Z	H	L	U	N	H	Q	X	Z	R	N		
P	U	I	X	Y	W	V	Y	V	U	O	X	U	*	*	*	E	D	R	Y	R	U	X	L	W	H	Q	X	L	E	Q		
U	N	Q	Y	E	Z	G	V	U	Z	Y	U	X	P	R	J	J	H	H	E	H	S	Q	I	X	A	?	I	M	D	E	*	
O	I	B	I	W	X	O	R	M	J	I	R	Z	V	M	M	A	V	R	E	S	T	T	G	E	P	W	D	J	J	I	*	
H	A	H	T	J	K	T	J	J	U	D	C	C	A	L	H	K	D	R	O	Q	Z	L	A	Z	P	V	M	P	F	G	*	
S	H	H	Y	C	V	G	D	A	W	Q	D	C	B	H	K	N	Q	Z	W	D	O	X	O	L	O	U	D	N	L	E	*	

R(2)RS(12) to the NSP(2) gives.

C(11) RP6 KRS(10) SA(CS) NSP(2)

M	U	H	A	U	U	P	J	O	M	H	R	U	J	X	L	Q	Z	P	W	L	P	Q	H	X	U	L	M	J	U	P	N		U	Y	O	V	G	X	C	R	E	V	L	B	F	O	J	Q	K	D	C	R	Q	E	P	J	X	W	P	V	Z	S	K	R		
U	I	V	R	Z	W	X	H	R	Z	U	J	H	X	U	P	O	L	P	N	U	I	V	Z	T	R	O	J	Z	I	P	*		F	T	H	T	O	B	T	M	T	S	F	O	S	R	R	H	E	U	S	K	F	T	H	Y	D	F	E	Q	Y	V	Q	*		
Z	H	A	L	J	H	I	H	L	R	R	F	Z	P	S	Z	J	Z	M	R	U	X	I	A	J	H	X	L	H	G	I	*		S	V	G	K	O	Q	U	U	X	C	A	U	H	C	F	S	W	A	X	D	D	T	U	H	W	V	T	K	P	N	U	*		
J	J	Z	M	R	U	?	H	X	U	F	O	J	U	Q	H	X	U	U	L	E	H	X	J	Y	L	W	P	U	H	*	*		W	X	A	X	D	D	?	Q	T	Y	N	G	V	L	T	C	Q	T	Y	Z	L	W	G	V	B	X	T	E	R	W	*	*		
U	R	Q	B	H	X	U	Z	P	V	I	E	Y	L	H	Z	I	P	A	L	J	W	L	H	X	U	E	*	Z	N	V	Z		Y	C	K	L	P	T	S	R	S	D	T	U	Q	V	L	Q	S	V	G	E	K	O	P	U	O	P	L	H	M	K	R	M		
W	E	O	O	P	Q	H	I	L	P	O	U	Q	L	P	Q	H	E	L	P	J	Y	Z	H	H	U	Q	O	P	Q	U	E		T	N	E	H	M	K	Q	V	X	E	E	T	C	W	C	R	N	W	F	O	B	O	U	O	P	K	H	M	K	R	M			
P	D	P	I	A	P	R	I	N	L	H	Z	I	P	B	Q	I	U	J	R	L	P	W	R	U	F	D	P	I	A	L	*		F	Z	C	U	H	D	C	U	Y	S	Q	O	V	D	J	K	X	D	V	A	X	D	O	A	T	R	J	C	V	H	Z	*		
M	I	O	H	H	X	Z	J	?	H	X	U	F	J	X	I	O	R	Q	Z	H	J	M	O	E	Z	U	Q	I	O	H	*		X	X	Q	Q	Q	O	S	X	?	Q	A	D	M	V	O	V	G	A	P	H	Q	V	Z	F	N	O	T	C	U	E	U	*		
H	X	U	E	U	J	I	Y	U	A	X	U	E	U	B	A	X	I	D	P	I	A	J	H	X	U	U	B	L	N	H	R		U	S	R	Q	D	V	U	C	Y	I	T	T	K	R	H	H	O	W	J	F	T	G	V	U	O	P	R	L	S	Z	Q	B		
I	N	L	H	Z	I	P	?	I	P	R	F	A	A	H	X	Z	J	A	L	J	X	Z	J	R	L	J	H	Y	U	J	*		V	R	W	H	U	C	?	V	D	C	M	J	V	Q	T	S	W	I	W	Z	G	O	V	B	X	X	Q	E	D	V	*			
J	L	W	U	B	H	X	Z	E	H	F	U	Z	W	X	H	Q	U	W	E	U	J	V	Z	V	H	F	J	U	*	*		V	X	T	P	H	W	G	O	L	U	N	P	O	S	G	Q	K	Y	T	N	R	T	O	Y	O	P	U	Q	V	T	*	*			
S	U	P	Y	Z	P	O	H	U	J	Z	B	G	I	Z	P	H	V	Z	S	U	J	U	N	I	P	Q	J	P	I	*		U	R	C	C	S	E	E	W	D	V	S	I	U	B	M	Q	Y	S	G	P	V	T	B	U	C	R	W	E	U	*					
E	H	X	J	U	S	U	P	H	F	J	U	S	U	P	Q	U	W	E	U	U	J	U	Z	W	X	H	Y	Z	P	O	*		Q	T	Y	V	Y	G	P	C	W	R	V	R	G	Y	E	K	T	F	L	Y	G	W	P	O	S	G	O	B	S	D	G	*		
H	U	J	V	I	E	H	F	V	I	O	E	J	U	N	I	P	Q	J	A	U	J	H	B	R	L	F	U	E	H	A	I		Q	T	O	Y	U	M	U	Q	Y	X	Q	L	V	Y	K	W	C	P	O	G	R	W	U	J	A	K	R	R	L	U	H	W		
J	R	I	A	R	F	Q	U	J	G	L	E	L	H	R	F	J	R	I	A	R	F	H	X	U	E	U	Y	L	Z	P	J		U	P	Q	F	L	X	E	I	X	P	R	J	E	X	U	I	N	G	H	U	I	Z	M	Z	Q	F	J	X	P	L	Z	L		
I	V	G	L	J	J	L	W	U	Q	U	M	E	Z	J	H	X	L	H	U	P	N	O	Y	M	U	E	U	Q	H	*	*		N	X	H	P	E	U	F	O	R	Q	J	R	R	J	R	R	P	I	X	J	P	I	J	Y	E	A	B	Y	P	U	*	*		
X	U	R	I	A	U	E	G	L	E	H	I	V	H	X	U	Q	I	I	E	A	L	F	A	L	J	E	U	Y	I	S	*		H	G	E	P	W	L	H	Z	X	P	E	L	E	G	U	J	R	P	P	U	R	U	M	R	G	Z	Z	L	N	L	U	*		
U	Q	A	Z	H	X	H	E	U	Y	M	R	Z	P	W	X	L	P	Q	J	Z	Y	L	Q	U	L	H	Z	P	F	*		A	Y	H	A	P	Q	Z	H	U	U	P	E	L	X	N	H	U	P	O	Q	E	U	H	P	X	L	U	I	U	*	*				
M	E	U	L	N	X	Z	P	H	X	U	O	G	G	U	E	R	U	V	H	L	P	Q	N	I	E	P	U	E	L	*		H	V	I	R	J	U	Q	H	Z	A	U	P	X	L	U	Z	I	F	H	U	F	T	X	U	U	P	N	L	F	N	E				
Q	H	X	U	P	A	Z	Q	U	P	Z	P	W	H	X	U	I	R	U	Z	H	H	R	U	Z	P	*	*	*			U	Z	V	H	M	E	J	G	L	Y	X	X	U	A	U	P	L	H	L	Y	L	H	U	F	U	E	R	M	*	*						
J	U	E	H	U	Q	H	X	U	N	L	P	Q	R	U	L	P	Q	G	U	E	U	R	Z	H	Z	P	H	X	U	X	I	H		H	U	U	V	I	L	J	V	Z	I	H	O	U	H	O	L	U	I	H	I	L	E	Y	L	U	E	H	P	E	H	Z		
L	Z	E	U	J	N	L	G	Z	P	W	V	E	I	Y	H	X	U	N	X	L	Y	M	U	E	N	L	O	J	U	Q	*		M	J	U	Q	P	Z	L	L	X	H	H	Y	J	H	U	A	G	Z	U	E	I	L	W	E	Z	U	A	V	U	M	*			
R	Z	N	D	U	E	M	O	H	G	E	U	J	U	P	H	R	F	Q	U	H	L	H	X	U	V	R	L	Y	U	H	I		Z	X	J	Z	H	U	W	I	U	L	I	J	Q	Q	E	F	Q	R	I	E	Z	H	L	U	N	H	Q	Q	X	Z	R	N		
Y	*	*	*	Z	R	J	I	V	H	X	U	E	I	I	Y	A	Z	H	X	Z	P	U	Y	U	E	W	U	Q	V	E	I		U	*	*	*	E	D	R	Y	R	U	X	L	W	H	Q	X	L	E	Q	G	P	U	I	X	Y	W	V	Y	V	U	O	X		
H	X	U	Y	Z	J	H	B	N	L	P	F	I	O	J	U	U	L	P	F	H	X	Z	P	W	T	?	I	M	D	E	*		U	N	Q	Y	E	Z	G	V	U	Z	Y	U	X	P	R	J	J	H	E	H	S	Q	I	X	A	?	I	M	D	E	*			
O	I	B	I	W	X	O	R	M	J	I	R	Z	V	M	M	A	V	R	E	S	T	G	E	P	W	D	J	J	I	*		O	I	B	I	W	X	O	R	M	J	I	R	Z	V	M	M	A	V	R	E	S	T	G	E	P	W	D	J	J	I	*				
H	A	H	T	J	K	T	J	J	U	D	C	A	L	H	K	D	R	O	Z	Z	L	A	Z	P	V	M	F	F	G	*		H	A	H	T	J	K	T	J	J	U	D	C	A	L	H	K	D	R	O	Z	Z	L	A	Z	P	V	M	F	F	G	*				
S	H	H	Y	C	V	G	D	A	W	Q	D	C	B	H	K	N	Q	Z	W	D	O	X	O	L	O	U	D	N	F	L	E	*		S	H	H	Y	C	V	G	D	A	W	Q	D	C	B	H	K	N	Q	Z	W	D	O	X	O	L	O	U	D	N	F	L	E	*

R(2)RS(15) to the TSP gives.

C(11) RP6 KRS(10) SA(CS) TSP IUT H :V[EL/VD][VO/OS] TSP

M	U	H	A	U	U	P	J	O	M	H	R	U	J	X	L	O	Z	P	W	L	P	O	H	X	U	L	M	J	U	P	N		U	Y	O	V	G	X	C	R	E	V	L	B	F	O	J	O	K	D	C	R	O	D	E	P	J	X	W	P	V	Z	S	K
U	I	V	R	Z	W	X	H	R	Z	U	J	H	X	U	P	O	L	P	N	U	I	V	Z	T	R	O	J	Z	I	P	*		F	T	H	T	O	B	T	M	T	S	F	O	S	R	R	H	E	U	S	K	F	T	H	Y	D	F	E	Q	Y	V	Q	*
Z	H	A	L	J	H	I	H	L	R	R	F	Z	P	S	Z	J	Z	M	R	U	X	I	A	J	H	X	L	H	G	I	*		S	V	G	K	O	Q	U	U	X	C	A	U	H	C	F	S	W	A	X	D	D	T	U	H	W	V	T	K	P	N	U	*
J	J	Z	M	R	U	?	H	X	U	F	O	J	U	Q	H	X	U	U	L	E	H	X	J	Y	L	W	P	U	H	*	*		W	W	A	X	D	D	?	Q	T	Y	N	G	V	T	C	Q	T	Y	Y	Z	L	W	G	V	B	X	T	E	R	W	*	*
P	A	L	J	W	L	H	X	U	E	*	Z	N	V	Z	U	R	Q	B	H	X	U	Z	P	V	I	E	Y	L	H	Z	I		E	G	K	O	P	W	U	O	P	L	*	H	Z	Y	S	Y	C	R	L	P	T	R	S	D	T	U	Q	L	W	Q	S	V
W	E	O	O	P	Q	H	I	L	P	O	U	Q	L	P	Q	H	E	L	P	J	Y	Z	H	H	U	Q	O	P	Q	U	E		T	N	E	H	M	K	Q	V	X	E	E	T	C	W	C	R	U	N	W	F	O	B	U	U	P	K	H	M	K	R	M	
P	D	P	I	A	P	R	I	N	L	H	Z	I	P	B	Q	I	U	J	R	L	P	W	R	U	F	D	P	I	A	L	*		F	Z	C	U	H	D	C	U	Y	S	Q	O	V	D	J	K	X	D	V	A	X	D	O	A	T	R	J	C	V	H	Z	*
M	I	O	H	H	X	Z	J	?	H	X	U	F	J	X	I	O	R	Q	Z	H	J	M	O	E	Z	U	Q	I	O	H	*		X	X	Q	Q	O	S	X	?	Q	A	D	M	V	O	V	G	A	P	H	Q	V	Z	F	N	O	T	C	U	E	U	*	
H	X	U	E	U	J	I	Y	U	A	X	U	E	U	B	A	X	I	D	P	I	A	J	H	X	U	U	B	L	N	H	R		U	S	R	Q	D	V	U	C	Y	I	T	T	K	R	H	H	O	W	J	F	T	G	V	U	O	P	R	L	S	Z	Q	B
I	N	L	H	Z	I	P	?	I	P	R	F	A	A	H	X	Z	J	A	L	J	X	Z	J	R	L	J	H	Y	U	J	*		V	R	W	W	H	U	C	?	V	D	C	M	J	V	Q	T	S	W	I	W	Z	G	O	V	B	X	X	Q	E	D	V	*
J	L	W	U	B	H	X	Z	E	H	F	U	Z	W	X	H	Q	U	W	E	U	U	J	V	Z	V	H	F	J	U	*		V	X	T	P	H	W	G	O	L	U	N	P	O	S	G	Q	K	Y	T	N	R	T	O	Y	O	P	U	Q	V	T	*	*	
S	U	P	Y	Z	P	O	H	U	J	J	Z	B	G	I	Z	P	H	V	Z	S	U	J	U	N	I	P	Q	J	P	I	*		U	R	C	C	S	E	E	W	D	V	S	I	U	U	B	M	Q	Y	S	G	P	V	T	B	U	C	R	W	E	U	*	
E	H	X	J	U	S	U	P	H	F	J	U	S	U	P	Q	U	W	E	U	U	J	U	Z	W	X	H	Y	Z	P	O	*		Q	P	T	V	Y	G	P	C	W	R	V	R	G	Y	E	K	T	F	L	R	Y	W	B	O	S	G	Q	B	S	D	G	*
H	U	J	V	I	E	H	F	V	I	O	E	J	U	N	I	P	Q	J	A	U	J	H	B	R	L	F	U	E	H	A	I		Q	T	O	Y	U	M	U	Q	Y	X	Q	L	V	Y	K	W	C	P	O	G	R	W	U	J	A	K	R	R	L	U	H	W
J	R	I	A	R	F	Q	U	J	G	L	E	L	H	R	F	J	R	I	A	R	F	H	X	U	E	U	Y	L	Z	P	J		U	P	Q	F	L	X	E	I	X	P	R	J	E	X	U	I	N	G	H	U	I	Z	M	Z	Q	F	J	X	P	L	Z	L
I	V	G	L	J	J	L	W	U	Q	U	M	E	Z	J	H	X	L	H	U	P	N	O	Y	M	U	E	U	Q	H	*	*		N	X	H	P	E	U	F	O	R	Q	J	R	R	J	R	R	P	I	X	J	P	I	J	Y	E	A	B	Y	P	U	*	*
X	U	R	I	A	U	E	G	L	E	H	I	V	H	X	U	Q	I	I	E	A	L	F	A	L	J	E	U	Y	I	S	*		H	G	E	P	W	L	H	Z	X	P	E	L	E	G	U	J	R	P	P	U	R	U	M	R	G	Z	Z	L	N	L	U	*
U	Q	A	Z	H	X	H	E	U	Y	M	R	Z	P	W	X	L	P	Q	J	Z	Y	L	Q	U	L	H	Z	P	F	*	*		A	Y	H	A	P	Q	Z	H	U	U	P	E	L	X	N	H	U	P	O	Q	E	U	H	P	X	L	U	I	U	*	*	
M	E	U	L	N	X	Z	P	H	X	U	O	G	G	U	E	R	U	V	H	X	L	P	Q	N	I	E	P	U	E	L	P		H	V	I	R	J	U	Q	H	Z	A	U	P	X	L	U	Z	I	F	H	U	F	T	X	U	P	N	H	L	F	N	E	
Q	H	X	U	P	A	Z	Q	U	P	Z	P	W	H	X	U	X	I	R	U	L	R	Z	H	H	R	U	Z	Z	P	*	*		U	Z	V	H	M	E	J	G	L	Y	X	X	U	A	U	P	L	H	L	Y	L	H	U	W	F	U	E	R	M	*	*	
J	U	E	H	U	Q	H	X	U	N	L	P	Q	R	U	L	P	Q	G	U	E	U	Q	Z	P	H	X	U	X	I	H		H	U	U	V	I	L	J	V	Z	I	H	O	U	H	O	L	U	I	H	I	L	E	Y	L	U	E	H	P	E	H	Z		
L	Z	E	U	J	N	L	G	Z	P	W	V	E	I	Y	H	X	U	N	X	L	Y	M	U	E	N	L	O	J	U	Q	*		M	J	U	Q	Q	P	Z	L	L	X	H	H	Y	J	H	U	A	G	Z	U	E	I	L	W	E	Z	U	A	V	U	M	*
Z	N	D	U	E	M	O	H	G	E	U	J	U	P	H	R	F	Q	U	H	L	H	X	U	V	R	L	Y	U	H	I	V		X	J	Z	H	U	W	I	U	L	I	J	Q	Q	E	F	O	R	I	E	Z	H	L	U	N	H	Q	Q	X	Z	R	N	U
Y	A	Z	H	X	Z	P	U	Y	U	E	W	U	Q	V	E	I	Y	*	*	*	Z	R	J	I	V	H	X	U	E	I	I		X	L	E	Q	G	P	U	I	X	Y	W	Y	V	U	O	X	U	*	*	*	E	D	R	Y	R	U	X	L	W	H	Q	
H	X	U	Y	Z	J	H	B	N	L	P	F	I	O	J	U	U	L	P	F	H	X	Z	P	W	T	?	I	M	D	E	*		U	N	Q	Y	E	Z	G	V	U	Z	Y	U	X	P	R	J	J	H	E	H	S	Q	I	X	A	?	I	M	D	E	*	
O	I	B	I	W	X	O	R	M	J	I	R	Z	V	M	M	A	V	R	E	S	T	T	G	E	P	W	D	J	J	I	*		O	I	B	I	W	X	O	R	M	J	I	R	Z	V	M	M	A	V	R	E	S	T	T	G	E	P	W	D	J	J	I	*
H	A	H	T	J	K	T	J	J	U	D	C	C	A	L	H	K	D	R	O	Q	Z	L	A	Z	P	V	M	P	F	G	*		H	A	H	T	J	K	T	J	J	U	D	C	C	A	L	H	K	D	R	O	Q	Z	L	A	Z	P	V	M	P	F	G	*
S	H	H	Y	C	V	G	D	A	W	Q	D	C	B	H	K	N	Q	Z	W	D	O	X	O	L	O	U	D	N	L	E	*		S	H	H	Y	C	V	G	D	A	W	Q	D	C	B	H	K	N	Q	Z	W	D	O	X	O	L	O	U	D	N	L	E	*

C(11) KRS(10) SA(CS)	K4 Key	Encryption
TJCD IGKU	ELDV VOOS	IANSE MOR

C(12) RP6 KRS(1) SA(SS) IUT H:V [XQ/UT] [UT/QX] IUT3:4[OQ/FI] NSP

N	V	I	B	V	T	L	S	N	I	Y	V	L	Z	M	U	K	T	X	M	T	U	I	Z	V	M	N	L	V	T	Q	
V	J	W	J	K	X	Z	I	Y	K	V	L	I	Z	V	T	S	M	T	Q	V	J	W	K	O	Y	S	L	K	J	T	*
K	I	B	M	L	I	J	I	M	Y	Y	G	K	T	A	K	L	K	N	Y	V	Z	J	B	L	I	Z	M	I	H	J	*
L	L	K	N	Y	V	?	I	Z	V	G	S	L	V	U	I	Z	V	M	F	I	Z	L	P	M	X	T	V	I	*	*	
K	T	W	J	F	P	M	I	K	J	T	B	M	L	X	M	I	Z	V	F	*	K	Q	W	K	V	Y	U	C	I	Z	V
S	V	U	M	T	U	I	F	M	T	L	P	K	I	I	V	U	S	T	U	V	F	X	F	S	S	T	U	I	J	M	T
T	E	T	J	B	T	Y	J	Q	M	I	K	J	T	C	U	J	V	L	Y	M	T	X	Y	V	G	E	T	J	B	M	*
N	J	S	I	I	Z	K	L	?	I	Z	V	G	L	Z	J	S	Y	U	K	I	L	N	S	F	K	V	U	J	S	I	*
I	Z	V	F	V	L	J	P	V	B	Z	V	F	V	C	B	Z	J	E	T	J	B	L	I	Z	V	V	C	M	Q	I	Y
J	Q	M	I	K	J	T	?	J	T	Y	G	B	B	I	Z	K	L	B	M	L	Z	K	L	Y	M	L	I	P	V	L	*
L	M	X	V	C	I	Z	K	F	I	G	V	K	X	Z	I	U	V	X	F	V	L	W	K	W	I	G	L	V	*	*	
A	V	T	P	K	T	S	I	V	L	L	K	C	H	J	K	T	I	W	K	A	V	L	V	Q	J	T	U	L	T	J	*
F	I	Z	L	V	A	V	T	I	G	L	V	A	V	T	U	V	X	F	V	V	L	V	K	X	Z	I	P	K	T	S	*
I	V	L	W	J	F	I	G	W	J	S	F	L	V	Q	J	T	U	L	B	V	L	I	C	Y	M	G	V	F	I	B	J
L	Y	J	B	Y	G	U	V	L	H	M	F	M	I	Y	G	L	Y	J	B	Y	G	I	Z	V	F	V	P	M	K	T	L
J	W	H	M	L	M	X	V	U	V	N	F	K	L	I	Z	M	I	V	T	Q	S	P	N	V	F	V	U	I	*	*	
Z	V	Y	J	B	V	F	H	M	F	I	J	W	I	Z	V	U	J	J	F	B	M	G	B	M	L	F	V	P	J	A	*
V	U	B	K	I	Z	I	F	V	P	N	Y	K	T	X	Z	M	T	U	L	K	P	M	U	V	M	I	K	T	G	*	*
N	F	V	M	Q	Z	K	T	I	Z	V	S	H	H	V	F	Y	V	W	I	Z	M	T	U	Q	J	F	T	V	F	M	T
U	I	Z	V	T	B	K	U	V	T	K	T	X	I	Z	V	Z	J	Y	V	M	Y	K	I	I	Y	V	R	K	T	*	*
L	V	F	I	V	U	I	Z	V	Q	M	T	U	Y	V	M	T	U	H	V	V	F	V	U	K	T	I	Z	V	Z	J	I
M	K	F	V	L	Q	M	H	K	T	X	W	F	J	P	I	Z	V	Q	Z	M	P	N	V	F	Q	M	S	L	V	U	*
M	I	Z	V	W	Y	M	P	V	I	J	W	Y	K	Q	E	V	F	N	S	I	H	F	V	L	V	T	I	Y	G	U	V
P	B	K	I	Z	K	T	V	P	V	F	X	V	U	W	F	J	P	*	*	*	K	Y	L	J	W	I	Z	V	F	J	J
I	Z	V	P	K	L	I	C	Q	M	T	G	J	S	L	V	M	T	G	I	Z	K	T	X	O	?	J	N	E	F	*	
S	J	C	J	X	Z	S	Y	N	L	J	Y	K	W	N	N	B	W	Y	F	A	O	H	F	T	X	E	L	L	J	*	
I	B	I	O	L	R	O	L	L	V	E	D	B	M	I	R	E	Y	S	U	K	M	B	K	T	W	N	T	G	H	*	
A	I	I	P	D	W	H	E	B	X	U	E	D	C	I	R	Q	U	K	X	E	S	Z	S	M	S	V	E	Q	M	F	*
V	P	S	W	H	Z	D	Y	F	W	M	C	G	S	L	U	R	E	D	Y	U	E	F	T	L	Z	X	T	W	K	A	R
G	O	I	O	S	C	O	N	O	A	G	S	A	Y	I	F	V	A	R	G	O	I	E	P	E	G	F	U	P	W	U	*
A	W	H	R	S	U	V	V	Z	D	B	V	I	D	G	A	X	B	Z	E	E	O	V	I	X	W	O	R	T	Q	V	*
X	X	B	Z	E	E	?	U	O	P	Q	H	W	O	D	U	O	P	P	K	M	X	H	W	C	Z	O	F	Y	X	*	*
A	E	O	V	U	M	X	U	A	W	F	H	R	S	T	X	V	S	T	M	*	I	K	P	A	P	D	R	M	T	O	Y
F	O	D	X	D	Y	V	Q	X	G	S	C	S	V	T	R	I	N	R	Y	N	O	Q	F	I	N	R	U	W	Z	F	
G	K	D	V	I	E	D	V	P	A	U	S	W	E	L	R	Z	E	W	B	Z	E	S	B	O	Y	L	D	W	I	R	*
Z	Z	U	U	U	S	A	Z	?	U	B	E	N	W	S	W	H	B	T	I	U	W	K	G	Q	S	O	D	V	F	V	*
V	A	Y	U	E	W	V	D	P	J	O	O	R	Y	I	I	S	X	L	G	O	H	W	V	S	T	Y	M	A	K	U	C
W	Y	X	X	I	V	D	?	W	E	D	N	L	W	U	O	A	X	J	X	K	H	S	W	C	Z	Z	U	F	E	W	*
W	Z	O	T	I	X	H	S	M	V	Q	T	S	A	H	U	R	P	O	Q	Y	O	S	P	S	T	V	U	W	O	*	*
V	Y	D	D	A	F	F	X	E	W	W	A	J	V	V	C	N	U	P	A	H	T	W	O	C	V	D	Y	X	F	V	*
U	T	O	W	P	H	T	D	X	Y	W	H	P	F	R	O	G	M	Y	P	X	T	S	A	H	U	C	A	E	H	*	
U	O	S	P	V	N	V	U	P	Z	U	M	W	P	R	X	D	T	S	H	Y	X	V	L	B	R	Y	Y	M	V	I	X
V	T	U	G	M	Z	F	J	Z	T	Y	L	F	Z	V	J	Q	H	I	V	J	K	N	K	U	G	L	Z	T	M	K	M
Q	Z	I	T	F	V	G	S	Y	U	L	Y	L	Y	T	J	Z	L	T	J	L	P	F	B	C	P	T	V	*	*		
I	H	F	T	X	M	I	K	Z	T	F	M	F	H	V	L	Y	T	T	V	Y	V	N	Y	H	K	K	M	Q	M	V	*
B	P	I	B	T	U	K	I	V	V	T	F	M	Z	Q	I	V	T	V	S	U	F	V	I	T	Z	M	V	J	V	*	*
I	W	J	Y	L	V	U	I	K	B	V	T	Z	M	V	K	J	G	I	V	G	O	Z	V	T	Q	I	M	G	Q	F	
V	K	W	I	N	F	L	H	M	P	Z	Z	V	B	V	T	M	I	M	P	M	I	V	X	G	V	F	Y	N	*	*	
I	V	V	W	J	M	L	W	K	J	I	S	V	I	S	M	V	J	I	M	F	P	M	V	V	F	I	T	F	I	K	
N	L	V	U	U	T	K	M	M	Z	I	I	P	L	I	V	B	H	K	V	F	J	M	X	F	K	V	B	W	V	N	*
I	M	V	Q	I	U	U	Z	K	Y	Q	V	K	Z	L	K	I	V	X	J	V	M	J	L	U	U	F	G	U	Y	J	F
Z	M	F	U	H	T	V	J	Z	P	X	W	P	V	S	Z	V	*	*	*	F	E	Y	P	Y	V	Z	M	X	I	U	
V	Q	U	P	F	K	H	W	V	K	P	V	Z	T	Y	L	L	I	I	F	I	A	U	J	Z	B	?	J	N	E	F	*
S	J	C	J	X	Z	S	Y	N	L	J	Y	K	W	N	N	B	W	Y	F	A	O	H	F	T	X	E	L	L	J	*	
I	B	I	O	L	R	O	L	L	V	E	D	B	M	I	R	E	Y	S	U	K	M	B	K	T	W	N	T	G	H	*	
A	I	I	P	D	W	H	E	B	X	U	E	D	C	I	R	Q	U	K	X	E	S	Z	S	M	S	V	E	Q	M	F	*

R(2)RS(8) to the TSP gives.

C(12) RP6 KRS(11) SA(SS) SML[VVL] RML[IPV] IUT H:V[UT/QX] IUT3 :4[QO/FM] TSP

N	V	I	B	V	T	L	S	N	I	Y	V	L	Z	M	U	K	T	X	M	T	U	I	Z	V	M	N	L	V	T	Q	
V	J	W	J	K	X	Z	I	Y	K	V	L	I	Z	V	T	S	M	T	Q	V	J	W	K	O	Y	S	L	K	J	T	*
K	I	B	M	L	I	J	I	M	Y	Y	G	K	T	A	K	L	K	N	Y	V	Z	J	B	L	I	Z	M	I	H	J	*
L	L	K	N	Y	V	?	I	Z	V	G	S	L	V	U	I	Z	V	M	F	I	Z	L	P	M	X	T	V	I	*	*	
K	V	Y	U	C	I	Z	V	K	T	W	J	F	P	M	I	K	J	T	B	M	L	X	M	I	Z	V	F	*	K	Q	W
S	V	U	M	T	U	I	F	M	T	L	P	K	I	I	V	U	S	T	U	V	F	X	F	S	S	T	U	I	J	M	T
T	E	T	J	B	T	Y	J	Q	M	I	K	J	T	C	U	J	V	L	Y	M	T	X	Y	V	G	E	T	J	B	M	*
N	J	S	I	I	Z	K	L	?	I	Z	V	G	L	Z	J	S	Y	U	K	I	L	N	S	F	K	V	U	J	S	I	*
I	Z	V	F	V	L	J	P	V	B	Z	V	F	V	C	B	Z	J	E	T	J	B	L	I	Z	V	V	C	M	Q	I	Y
J	Q	M	I	K	J	T	?	J	T	Y	G	B	B	I	Z	K	L	B	M	L	Z	K	L	Y	M	L	I	P	V	L	*
L	M	X	V	C	I	Z	K	F	I	G	V	K	X	Z	I	U	V	X	F	V	L	W	K	W	I	G	L	V	*	*	
A	V	T	P	K	T	S	I	V	L	L	K	C	H	J	K	T	I	W	K	A	V	L	V	Q	J	T	U	L	T	J	*
F	I	Z	L	V	A	V	T	I	G	L	V	A	V	T	U	V	X	F	V	L	V	K	X	Z	I	P	K	T	S	*	
I	V	L	W	J	F	I	G	W	J	S	F	L	V	Q	J	T	U	L	B	V	L	I	C	Y	M	G	V	F	I	B	J
L	Y	J	B	Y	G	U	V	L	H	M	F	I	Y	G	L	Y	J	B	Y	G	I	Z	V	F	P	M	K	T	L		
J	W	H	M	L	M	X	V	U	V	N	F	K	L	I	Z	M	I	V	T	Q	S	P	N	V	F	V	U	I	*	*	
Z	V	Y	J	B	V	F	H	M	F	I	J	W	I	Z	V	U	J	J	F	B	M	G	B	M	L	F	V	P	J	A	*
V	U	B	K	I	Z	I	F	V	P	N	Y	K	T	X	Z	M	T	U	L	K	P	M	U	V	M	I	K	T	G	*	
N	F	V	M	Z	K	T	I	Z	V	S	H	F	Y	V	Y	W	I	Z	M	T	U	Q	J	F	T	V	F	M	T		
U	I	Z	V	Q	T	B	K	U	V	T	K	T	X	I	Z	V	Z	J	Y	V	M	T	K	I	I	Y	V	K	T	*	*
L	V	F	I	V	U	I	Z	V	Q	M	T	U	Y	V	Z	M	T	U	H	V	V	F	U	K	T	I	Z	V	Z	J	I
M	K	F	V	L	Q	M	H	K	T	X	W	F	J	P	I	Z	V	Q	Z	M	P	N	V	F	Q	M	S	L	V	U	*
M	I	Z	V	W	Y	M	P	V	I	J	W	Y	K	Q	E	V	F	N	S	I	H	F	V	L	V	T	I	Y	G	U	V
J	W	I	Z	V	F	J	J	P	B	K	I	Z	K	T	V	P	V	F	X	V	U	W	F	J	P	*	*	K	Y	L	
I	Z	V	P	K	L	I	C	Q	M	T	G	J	S	L	V	M	T	G	I	Z	K	T	X	O	?	J	N	E	F	*	
S	J	C	J	O	L	Z	S	N	L	J	Y	K	W	N	B	W	Y	F	A	O	H	F	T	X	E	L	L	J	*		
I	B	I	O	L	R	O	L	L	E	D	B	M	I	R	E	Y	S	U	K	M	B	K	T	W	N	T	G	H	*		
A	I	I	P	D	W	H	E	B	X	U	E	D	C	I	R	Q	U	K	X	E	S	Z	S	M	S	V	E	Q	M	F	*
V	P	S	W	H	Z	D	Y	F	W	M	C	G	S	L	U	R	E	D	Y	U	E	F	T	L	Z	X	T	W	R	A	R
G	O	I	O	S	C	O	N	O	A	G	S	A	Y	I	F	V	A	R	G	O	I	P	E	G	F	U	P	W	U	*	
A	W	H	R	S	U	V	V	Z	D	B	V	I	D	G	A	X	B	Z	E	E	O	V	I	X	W	O	R	T	O	V	*
X	X	B	Z	E	E	?	U	O	P	Q	H	W	O	D	U	O	P	P	K	M	X	H	W	C	Z	O	F	Y	X	*	*
A	P	D	R	M	T	O	Y	A	E	O	V	U	M	X	U	A	W	F	H	R	S	T	X	V	S	T	M	*	I	K	F
F	O	D	X	D	Y	O	X	G	S	C	S	V	T	R	I	N	R	Y	N	O	Q	F	I	N	R	W	Z	F			
G	K	D	V	I	E	D	V	P	A	U	S	W	E	L	R	Z	E	W	B	Z	E	S	B	O	Y	L	D	W	K	*	
Z	Z	U	U	U	S	A	Z	?	U	B	E	N	W	S	W	H	B	T	I	U	W	K	Q	S	O	D	V	F	V	*	
V	A	Y	U	E	W	V	D	P	J	O	O	R	Y	I	I	S	X	L	G	O	H	W	V	S	T	Y	M	A	K	U	
W	Y	X	X	I	V	D	?	W	E	D	N	L	W	U	O	A	X	J	X	K	H	S	W	C	Z	Z	U	F	E	W	*
W	Z	O	T	I	X	H	S	M	V	Q	T	S	A	H	U	R	P	O	Q	Y	O	S	P	S	T	V	U	W	O	*	*
V	Y	D	D	A	F	F	X	E	W	A	J	V	V	C	N	U	P	A	H	T	W	O	C	V	D	Y	X	F	V	*	
U	T	O	W	P	H	T	D	X	Y	W	Y	H	P	F	R	O	G	M	P	X	T	S	A	H	U	C	A	E	H	*	
U	O	S	P	V	N	V	U	P	Z	U	M	W	P	R	X	D	T	S	H	Y	X	V	L	B	R	Y	Y	M	V	I	X
V	T	U	G	M	Z	F	J	Z	T	Y	L	F	Z	V	J	Q	H	I	V	J	K	N	K	U	G	L	Z	T	M	K	M
Q	Z	I	T	F	V	G	S	Y	U	L	Y	L	Y	T	J	Z	L	T	J	L	P	F	B	C	P	T	V	*	*		
I	H	F	T	X	M	I	K	Z	T	F	M	F	H	V	L	Y	T	T	V	Y	V	N	Y	H	K	K	M	Q	M	V	*
B	P	I	B	T	U	K	I	V	T	F	M	Z	Q	I	V	T	S	U	F	V	I	T	Z	M	V	J	V	*	*		
I	W	J	Y	L	V	I	K	B	V	T	Z	M	V	K	J	G	I	V	S	O	Z	V	T	Q	I	M	G	Q	F		
V	K	W	I	N	F	L	H	M	P	Z	Z	V	B	M	T	M	I	M	P	M	I	V	X	G	V	F	Y	N	*	*	
I	V	V	W	J	M	L	W	K	P	J	S	V	I	S	M	V	J	I	J	M	F	P	M	V	V	F	I	T	F	I	K
N	L	V	U	U	T	K	M	M	Z	I	I	P	L	I	V	B	H	K	V	F	J	M	X	F	K	V	B	W	N	*	
I	M	V	Q	I	U	U	Z	K	Y	Q	V	K	Z	L	K	I	V	X	J	V	M	J	L	U	U	F	G	U	Y	J	F
P	Y	V	Z	M	X	I	U	Z	M	F	U	H	T	V	J	Z	P	X	W	P	W	S	Z	V	*	*	F	E	Y		
V	Q	U	P	F	K	H	W	V	R	P	V	Z	T	Y	L	I	I	F	I	A	U	J	Z	B	?	J	N	E	F	*	
S	J	C	J	X	Z	S	Y	N	L	J	Y	K	W	N	B	W	Y	F	A	O	H	F	T	X	E	L	L	J	*		
I	B	I	O	L	R	O	L	L	E	D	B	M	I	R	E	Y	S	U	K	M	B	K	T	W	N	T	G	H	*		
A	I	I	P	D	W	H	E	B	X	U	E	D	C	I	R	Q	U	K	X	E	S	Z	S	M	S	V	E	Q	M	F	*

Part 3

Method Derivation

Initial Break

The method for decrypting K4 is given by two crib examples that make use of the plain text in order to walk the solver (albeit cryptically) through the steps necessary. Subsequent use of these steps against rail positions 2-6 show the consistent use of two algorithms and three IUT schemes that can be traced back directly to these initial set of cryptic instructions. The method by which these two cribs are discovered will unfold gradually, but we first need to establish the initial break and define the first cryptogram from which everything else will follow.

Cryptogram 1

As already mentioned in the section on rail positions in part 1, the 4 letter marker RE,V,F appears in Rail Position 1 when a left shift of 13 is applied to row 1 as shown below.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

Examining the distribution of the Y's in the second row of the cipher text we see that there is a consecutive repeat of the YY pattern. This arises as the distance between the Y's are 9,9,4 and 6 respectively as shown below. The U and N have been highlighted in the cylinder rendition below as transformation of row 2 by a left shift of 10 will leave the same letter relationship *invariant under transformation*.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

If we now apply a left shift of 10 to row 2 then we are left with the following arrangement.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		Y	Q	T	Q	U	X	Q	B	Q	V
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

As has already been pointed out earlier, the **encryption** of UNYY CKB with OBKRUOX will give KRYPTOS when using the vigenere scheme, so the blocks above actually represent the entry point into the overall solve and give the first cryptogram, which has a full 8 letters of UNYYCKBF giving KRYPTOS L (when encrypted with OBKRUOXO). The order in which the letters are read off is, UNYY for the first four and CKBF for the final four. We will refine this result later (*as the intended result is UNYY CKBD*), but now that we have our target row alignment and letters we can examine the plain text and reciprocal rows to extract the additional information required in order to construct the first crib.

Extraction of the first crib

If we widen the scope of the previous row transformations to include a view of these same operations on the plain text cylinder and additionally include counter-symmetric operations on the reciprocal rows, then the presence of new structural information can be shown that will provide the basis for the first crib.

Examining the plain text cylinder after the initial left shift of 13 characters for Rail Position 1 immediately causes the following relationship of note to appear, which is the word HOT running in both directions in the vertical plane.

This may at first glance appear to be a trivial coincidence, but it will gradually become apparent that this an essential part of the first crib,

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N	
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	

Another aspect of this arrangement worthy of note is the presence of the anagram HOLE in row 3.

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N	
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	

If we now apply the left shift of 10 to row 2 to match the operations we made against the cipher text cylinder we can see that the relationship that defined HOT is destroyed, but the anagram of the word HOLE reappears, albeit in a different spatial relationship as two LBA's that are read vertically rather than horizontally.

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		E	O	F	L	I	G	H	T	L	I
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	

It will be seen that these two observations for the occurrence of HOT and the persistence of HOLE will actually come to define the basis for the entire K4 solve. We have already used a very specific term to describe the spatial relationship between the letters UN and YY in the cipher text, which was that they were *'invariant under transformation'*.

The general principle of invariance under transformation is well known in puzzle solving circles as it's the same geometric principle that Ernő Rubik was trying to explain to his architectural students before stumbling across the concept of the cube that now bears his name. In cryptographic terms it is also the reason why transposition ciphers can easily be broken as their underlying statistical properties are also invariant under transformation.

As the reciprocal rows for Rail Position 1 are all contained within the full 3 rows of K4 the last 3 rows of both the cipher and plain text cylinders will always be identical, which means for the purposes of Rail Position 1 we only need to consider 3 views simultaneously. We already have the transformed rows for the first 3 rows of the cipher and plain text cylinders, so we now only need to examine the reciprocal rows (26, 27 and 28) when transformed counter-symmetrically. This operation gives the following result.

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
A	W	I	N	F	B	N	Y	P		T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I
G	K	U	H	U	A	U	E	K	C	A	R		V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

The most striking feature of this operation is the presence of the aligned double digraph DI at the end of rows 27 and 28.

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
A	W	I	N	F	B	N	Y	P		T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I
G	K	U	H	U	A	U	E	K	C	A	R		V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

Another less obvious feature is also apparent, which is the mirroring of the letters ALN on the plain text cylinder in rows 1-3. Discarding rows 4 – 25 and displaying these two rows in closer proximity better displays this relationship.

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		E	O	F	L	I	G	H	T	L	I
I	T	W	A	S	T	O	T	A	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O		

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
A	W	I	N	F	B	N	Y	P		T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I
G	K	U	H	U	A	U	E	K	C	A	R		V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

This symmetrical relationship is important as it has a wider implications when viewed across the entirety of the plain text cylinder. This relationship actually defines an X on the plain text cylinder when additional diagonal letters are projected as an extension to these first 3 as shown below.

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E	
E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N	*	E	O	F	L	I	G	H	T	L	I	
I	T	W	A	S	T	O	T	A	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	*		
S	S	I	B	L	E	?	T	H	E	Y	U	S	E	D	T	H	E	E	A	R	T	H	S	M	A	G	N	E	T	*		
I	C	F	I	E	L	D	X	T	H	E	I	N	F	O	R	M	A	T	I	O	N	W	A	S	G	A	T	H	E	R	*	
E	D	A	N	D	T	R	A	N	S	M	I	T	T	E	D	U	N	D	E	R	G	R	U	N	D	T	O	A	N	U		
N	K	N	O	W	N	L	O	C	A	T	I	O	N	X	D	O	E	S	L	A	N	G	L	E	Y	K	N	O	W	A	*	
B	O	U	T	H	I	S	?	T	H	E	Y	S	H	O	U	L	D	I	T	S	B	U	R	I	E	D	O	U	T	*		
T	H	E	R	E	S	O	M	E	W	H	E	R	E	X	W	H	O	K	N	O	W	S	T	H	E	E	X	A	C	T	I	
O	C	A	T	I	O	N	?	O	N	L	Y	W	W	T	H	I	S	W	A	S	H	I	S	L	A	S	T	M	E	S	*	
S	A	G	E	X	T	H	I	R	T	Y	E	I	G	H	T	D	E	G	R	E	E	S	F	I	F	T	Y	S	E	*	*	
V	E	N	M	I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	*	
R	T	H	S	E	V	E	N	T	Y	S	E	V	E	N	D	E	G	R	E	E	S	E	I	G	H	T	M	I	N	U	*	
T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	W	E	S	T	X	L	A	Y	E	R	T	W	O	
S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S	
O	F	P	A	S	S	A	G	E	D	E	B	R	I	S	T	H	A	T	E	N	C	U	M	B	E	R	E	D	T	*	*	
H	E	L	O	W	E	R	P	A	R	T	O	F	T	H	E	D	O	O	R	W	A	Y	W	A	S	R	E	M	O	V	*	
E	D	W	I	T	H	T	R	E	M	B	L	I	N	G	H	A	N	D	S	I	M	A	D	E	A	T	I	N	Y	*	*	
B	R	E	A	C	H	I	N	T	H	E	U	P	P	E	R	L	E	F	T	H	A	N	D	C	O	R	N	E	R	A	N	
D	T	H	E	N	W	I	D	E	N	I	N	G	T	H	E	H	O	L	E	A	L	I	T	T	L	E	I	I	N	*	*	
S	E	R	T	E	D	T	H	E	C	A	N	D	L	E	A	N	D	P	E	E	R	E	D	I	N	T	H	E	H	O	T	
A	I	R	E	S	C	A	P	I	N	G	F	R	O	M	T	H	E	C	H	A	M	B	E	R	C	A	U	S	E	D	*	
T	H	E	F	L	A	M	E	T	O	F	L	I	C	K	E	R	B	U	T	P	R	E	S	E	N	T	L	Y	D	E	T	
I	L	S	O	F	T	H	E	R	O	O	M	W	I	T	H	I	N	E	M	E	R	G	E	D	F	R	O	M	*	*	*	
T	H	E	M	I	S	T	X	C	A	N	Y	O	U	S	E	E	A	N	Y	T	H	I	N	G	Q	?	O	B	K	R	*	
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	*	
A	W	I	N	F	B	N	Y	P	*	T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	
G	K	U	H	U	A	U	E	K	C	A	R	*	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	

The remarkable outcome of note in this new relationship is not the X itself, but the implications that arise from the positioning of the letters WL that sit at the intersection of the arms in row 15. When deriving the first cryptogram it was shown that the distribution of the Y's in row 2 allowed for a transformation that retained an invariant spatial relationship. It will be seen from the above that a left shift of 16 on row 15 will result in a swap of the letters WL on the right of row 15 with the letters WL on the left. By virtue of the cylinders 32 letter width this transformation can repeat indefinitely.

If we explore this relationship further and sketch an identical X projection about the left hand instance of the letters WL then we see that this relationship has distinct markers that deem its existence non-coincidental.

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N	*	E	O	F	L	I	G	H	T	L	I
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	*
S	S	I	B	L	E	?	T	H	E	Y	U	S	E	D	T	H	E	E	A	R	T	H	S	M	A	G	N	E	T	*	*
I	C	F	I	E	L	D	X	T	H	E	I	N	F	O	R	M	A	T	I	O	N	W	A	S	G	A	T	H	E	R	*
E	D	A	N	D	T	R	A	N	S	M	I	T	T	E	D	U	N	D	E	R	G	R	U	U	N	D	T	O	A	N	U
N	K	N	O	W	N	L	O	C	A	T	I	O	N	X	D	O	E	S	L	A	N	G	L	E	Y	K	N	O	W	A	*
B	O	U	T	H	I	S	?	T	H	E	Y	S	H	O	U	L	D	I	T	S	B	U	R	I	E	D	O	U	T	*	
T	H	E	R	E	S	O	M	E	W	H	E	R	E	X	W	H	O	K	N	O	W	S	T	H	E	E	X	A	C	T	L
O	C	A	T	I	O	N	?	O	N	L	Y	W	W	T	H	I	S	W	A	S	H	I	S	L	A	S	T	M	E	S	*
S	A	G	E	X	T	H	I	R	T	Y	E	I	G	H	T	D	E	G	R	E	E	S	F	I	F	T	Y	S	E	*	*
V	E	N	M	I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	*
R	T	H	S	E	V	E	N	T	Y	S	E	V	E	N	D	E	G	R	E	E	S	E	I	G	H	T	M	I	N	U	*
T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	W	E	S	T	X	L	A	Y	E	R	T	W	O
S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S
O	F	P	A	S	S	A	G	E	D	E	B	R	I	S	T	H	A	T	E	N	C	U	M	B	E	R	E	D	T	*	*
H	E	L	O	W	E	R	P	A	R	T	O	F	T	H	E	D	O	O	R	W	A	Y	W	A	S	R	E	M	O	V	*
E	D	W	I	T	H	T	R	E	M	B	L	I	N	G	H	A	N	D	S	I	M	A	D	E	A	T	I	N	Y	*	*
B	R	E	A	C	H	I	N	T	H	E	U	P	P	E	R	L	E	F	T	H	A	N	D	C	O	R	N	E	R	A	N
D	T	H	E	N	W	I	D	E	N	I	N	G	T	H	E	H	O	L	E	A	L	I	T	T	L	E	I	I	N	*	*
S	E	R	T	E	D	T	H	E	C	A	N	D	L	E	A	N	D	P	E	E	R	E	D	I	N	T	H	E	H	O	T
A	I	R	E	S	C	A	P	I	N	G	F	R	O	M	T	H	E	C	H	A	M	B	E	R	C	A	U	S	E	D	*
T	H	E	F	L	A	M	E	T	O	F	L	I	C	K	E	R	B	U	T	P	R	E	S	E	N	T	L	Y	D	E	T
I	L	S	O	F	T	H	E	R	O	O	M	W	I	T	H	I	N	E	M	E	R	G	E	D	F	R	O	M	*	*	*
T	H	E	M	I	S	T	X	C	A	N	Y	O	U	S	E	E	A	N	Y	T	H	I	N	G	Q	?	O	B	K	R	*
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	*
A	W	I	N	F	B	N	Y	P	*	T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I
G	K	U	H	U	A	U	E	K	C	A	R	*	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

If it is not already apparent, the letters WL and D that straddle the intersection of this second X correspond to the slightly raised letters YA and R on the original sculpture. This may well be an artistic intervention to imply that a RAY is passing through this position between the raised letters.



Also of interest are the two blocks of four letters directly above and below the WL position, which are SE/FO and AS/OW as shown below.

O	C	A	T	I	O	N	?	O	N	L	Y	W	W	T	H	I	S	W	A	S	H	I	S	L	A	S	T	M	E	S	*
S	A	G	E	X	T	H	I	R	T	Y	E	I	G	H	T	D	E	G	R	E	E	S	F	I	F	T	Y	S	E	*	*
V	E	N	M	I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	*
R	T	H	S	E	V	E	N	T	Y	S	E	V	E	N	D	E	G	R	E	E	S	E	I	G	H	T	M	I	N	U	*
T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	W	E	S	T	X	L	A	Y	E	R	T	W	O
S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S
O	F	P	A	S	S	A	G	E	D	E	B	R	I	S	T	H	A	T	E	N	C	U	M	B	E	R	E	D	T	*	*
H	E	L	O	W	E	R	P	A	R	T	O	F	T	H	E	D	O	O	R	W	A	Y	W	A	S	R	E	M	O	V	*
E	D	W	I	T	H	T	R	E	M	B	L	I	N	G	H	A	N	D	S	I	M	A	D	E	A	T	I	N	Y	*	*
B	R	E	A	C	H	I	N	T	H	E	U	P	P	E	R	L	E	F	T	H	A	N	D	C	O	R	N	E	R	A	N
D	T	H	E	N	W	I	D	E	N	I	N	G	T	H	E	H	O	L	E	A	L	I	T	T	L	E	I	I	N	*	*

These two blocks are anagrams of the words FORCES and SHADOW, each deficient in the RC and HD digraphs respectively. Another feature of note is SOS written diagonally across rows 13 to 15.

The use of the raised letters R, A, and Y in the corresponding plain text also suggests we may be dealing with a phenomena associated with light. If the arms of the X are symbolic of light paths focusing then WL may well be a cryptic nod to the concept of wavelength. The interchangeability of the positions of WL may also be intended to express (cryptically) the notion of a polarisation filter. As will be seen subsequently, this is not a guess and is borne out by the operation required in order to discover the next cryptogram and its associated crib. The idea that rotating row 15 by 16 spaces left or right will lead to Invariance under transformation for the 4 letter blocks FO/WL and WE/WL is also a factor that adds additional weight of the emerging role of IUT in the cipher scheme.

We saw previously that there is nothing in the original derivation of cryptogram C(1) that would lead us to believe the intended result is anything other than UNYY CKBF, which would give a C(1) result of KRYPTOS L.

Contemplation of this 'L' after a number of other avenues had been exhausted prompted the question of whether this lone 'L' was connected with the extraneous L on the tableau panels. After some thought it appeared that it might suggest that the tableau or something else should shift one letter to the right. The application of a global right shift through the Kryptos alphabet of KRS(n) has been discussed in part 1, so we shall now apply KRS(1) to all the cipher and plain text simultaneously to replicate how this assertion was originally tested.

Note that the row alignment positions for Rail Position 1 have been reset back to their original settings (along with the reciprocal rows).

C	F	O	X	F	F	Q	A	V	C	O	M	F	A	I	B	E	J	Q	H	B	Q	E	O	I	F	B	C	A	F	Q	D
F	S	G	M	J	H	I	O	M	J	F	A	O	I	F	Q	V	B	Q	D	F	S	G	J	U	M	V	A	J	S	Q	*
J	O	X	B	A	O	S	O	B	M	M	P	J	Q	W	J	A	J	C	M	F	I	S	X	A	O	I	B	O	T	S	*
A	A	J	C	M	F	?	O	I	F	P	V	A	F	E	O	I	F	F	B	Y	O	I	A	N	B	H	Q	F	O	*	*
J	D	G	J	F	M	E	Z	O	I	F	J	Q	G	S	Y	N	B	O	J	S	Q	X	B	A	H	B	O	I	F	Y	*
F	E	B	Q	E	O	Y	B	Q	A	N	J	O	O	F	E	V	Q	E	F	Y	H	Y	V	V	Q	E	O	S	B	Q	V
Q	R	Q	S	X	Q	M	S	D	B	O	J	S	Q	Z	E	S	F	A	M	B	Q	H	M	F	P	R	Q	S	X	B	*
C	S	V	O	O	I	J	A	?	O	I	F	P	A	I	S	V	M	E	J	O	A	C	V	Y	J	F	E	S	V	O	*
O	I	F	Y	F	A	S	N	F	X	I	F	Y	F	Z	X	I	S	R	Q	S	X	A	O	I	F	F	Z	B	D	O	M
S	D	B	O	J	S	Q	?	S	Q	M	P	X	X	O	I	J	A	X	B	A	I	J	A	M	B	A	O	N	F	A	*
A	B	H	F	Z	O	I	J	Y	O	P	F	J	H	I	O	E	F	H	Y	F	F	A	G	J	G	O	P	A	F	*	*
W	F	Q	N	J	Q	V	O	F	A	A	J	Z	T	S	J	O	G	J	W	F	A	F	D	S	Q	E	A	Q	S	*	*
Y	O	I	A	F	W	F	Q	O	P	A	F	W	F	Q	E	F	H	Y	F	F	A	F	J	H	I	O	N	J	Q	V	*
O	F	A	G	S	Y	O	P	G	S	V	Y	A	F	D	S	Q	E	A	X	F	A	O	Z	M	B	P	F	Y	O	X	S
A	M	S	X	M	P	E	F	A	T	B	Y	B	O	M	P	A	M	S	X	M	P	O	I	F	Y	F	N	B	J	Q	A
S	G	T	B	A	A	B	H	F	E	F	C	Y	J	A	O	I	B	O	F	Q	D	V	N	C	F	Y	F	E	O	*	*
I	F	M	S	X	F	Y	T	B	Y	O	S	G	O	I	F	E	S	S	Y	X	B	P	X	B	A	Y	F	N	S	W	*
F	E	X	J	O	I	O	Y	F	N	C	M	J	Q	H	I	B	O	E	A	J	N	B	E	F	B	O	J	O	P	*	*
C	Y	F	B	D	I	J	Q	O	I	F	V	T	T	F	Y	M	F	G	O	I	B	Q	E	D	S	Y	Q	F	Y	B	Q
E	O	I	F	Q	X	J	E	F	Q	J	Q	H	O	I	F	I	S	M	F	B	M	J	O	O	M	F	J	J	Q	*	*
A	F	Y	O	F	E	O	I	F	D	B	Q	E	M	F	B	Q	E	T	F	F	Y	F	E	J	Q	O	I	F	I	S	O
B	J	Y	F	A	D	B	T	J	Q	H	G	Y	S	N	O	I	F	D	I	B	N	C	F	Y	D	B	V	A	F	E	*
O	I	F	G	M	B	N	F	O	S	G	M	J	D	R	F	Y	C	V	O	T	Y	F	A	F	Q	O	M	P	E	F	O
J	M	A	S	G	O	I	F	Y	S	S	N	X	J	O	I	J	Q	F	N	F	Y	H	F	E	G	Y	S	N	*	*	*
O	I	F	N	J	A	O	Z	D	B	Q	P	S	V	A	F	F	B	Q	P	O	I	J	Q	H	U	?	S	C	R	Y	*
V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W	U	U	T	Y	Q	H	R	A	A	S	*
O	X	O	U	A	L	U	A	A	F	R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T	*
W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B	Y	*
F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J	W	L
P	U	O	U	V	Z	U	C	U	W	P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G	E	*
W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R	U	F	O	H	G	U	L	Q	D	F	*
H	H	X	I	R	R	?	E	U	N	D	T	G	U	K	E	U	N	N	J	B	H	T	G	Z	I	U	Y	M	H	*	*
O	J	N	W	N	K	L	B	Q	U	M	W	R	U	F	E	B	H	E	W	G	Y	T	L	V	Q	H	F	V	Q	B	*
U	K	H	K	M	F	D	H	P	V	Z	V	F	F	Q	L	O	C	L	M	C	U	D	Y	O	C	L	E	G	I	Y	Y
P	J	K	F	O	R	K	F	N	W	E	V	G	R	A	L	I	R	G	X	I	R	V	X	U	M	A	K	G	O	J	*
I	I	E	E	E	V	W	I	?	E	X	R	C	G	V	G	T	X	Q	O	E	G	J	P	D	V	U	K	F	Y	F	*
F	W	M	E	R	G	F	K	N	S	U	U	L	M	O	O	V	H	A	P	U	T	G	F	V	Q	M	B	W	J	E	Z
G	M	H	H	O	F	K	?	G	R	K	C	A	G	E	U	W	H	S	H	J	T	V	G	Z	I	I	E	Y	R	G	*
G	I	U	Q	O	H	T	V	B	F	D	Q	V	W	T	E	L	N	U	D	M	U	V	N	V	Q	F	E	G	U	*	*
F	M	K	K	W	Y	Y	H	R	G	G	W	S	F	F	Z	C	E	N	W	T	Q	G	U	Z	F	K	M	H	Y	F	*
E	Q	U	G	N	T	Q	K	H	M	G	M	T	N	Y	L	U	P	B	M	N	H	Q	V	W	T	E	Z	W	R	T	*
E	U	V	N	F	C	F	E	N	I	E	B	G	N	L	H	K	Q	V	T	M	H	F	A	X	L	M	M	B	F	O	H
F	Q	E	P	B	I	Y	S	I	Q	M	A	Y	I	F	S	D	T	O	F	S	J	C	J	E	P	A	I	O	B	J	B
D	I	O	Q	Y	F	P	V	M	E	A	M	M	A	M	M	Q	S	I	A	Q	S	A	N	Y	X	Z	N	Q	F	*	*
O	T	Y	Q	H	B	O	J	I	Q	Y	B	Y	T	F	A	M	Q	Q	F	M	F	C	M	T	J	J	B	D	B	F	*
X	N	O	X	Q	E	J	O	F	F	O	Y	B	I	D	O	F	O	F	V	E	Y	F	O	Q	I	B	F	S	F	*	*
O	G	S	M	A	F	E	O	J	X	F	Q	I	B	F	J	S	P	O	F	P	U	I	F	F	Q	D	O	B	P	D	Y
F	J	G	O	C	Y	A	T	B	N	I	I	F	X	F	Q	B	O	B	N	B	O	F	H	P	F	F	Y	M	C	*	*
O	F	F	G	S	B	A	G	J	S	O	V	F	O	V	B	F	S	O	S	B	Y	N	B	F	F	Y	O	Q	Y	O	J
C	A	F	E	E	Q	J	B	B	I	O	O	N	A	O	F	X	T	J	F	Y	S	B	H	Y	J	F	X	G	F	C	*
B	F	D	O	E	E	I	J	M	D	F	J	I	A	J	O	F	H	S	F	B	S	A	E	E	Y	P	E	M	S	Y	J
Y	R	M	N	M	F	I	B	H	O	E	I	B	Y	E	T	Q	F	S	I	N	H	G	N	G	F	V	I	F	*	*	*
F	D	E	N	Y	J	T	G	F	J	N	F	I	Q	M	A	A	O	O	Y	O	W	E	S	I	X	?	S	C	R	Y	*
V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W	U	U	T	Y	Q	H	R	A	A	S	*
O	X	O	U	A	L	U	A	A	F	R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T	*
W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B	Y	*

Cryptogram 2

We are initially guided in our discovery of cryptogram 2 by the use of the double X configuration that was previously deduced from the steps taken in the construction of the first cryptic crib.

If we apply the same double X overlay on the transformed text above then it becomes apparent (after some significant trial and error) that the discovery of the second cryptogram has to be done via manipulation of the reciprocal rows for Rail Position 1. To clarify this point, it means that the crib we have been given exists in rows 26,27 and 28, but that the cryptogram that we are expected to retrieve C(2) is actually still in the original Rail Position. It is also found during this process that we need to apply a simple symmetric transform rather than the previous counter-symmetric operation to the reciprocal rows. With those considerations in mind we now apply a right shift of 2 to row 28, a left shift of 10 to row 27 and finally a right shift of 11 to row 26.

The results of these operations on the cipher text cylinder are shown relative to the double X pattern outlined previously.

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G	E	*	P	U	O	U	V	Z	U	C	U	W
U	F	O	H	G	U	L	Q	D	F	*	W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R
H	H	X	I	R	R	?	E	U	N	D	T	G	U	K	E	U	N	N	J	B	H	T	G	Z	I	U	Y	M	H	*	*
O	J	N	W	N	K	L	B	Q	U	M	W	R	U	F	E	B	H	E	W	G	Y	T	L	V	Q	H	F	V	Q	B	*
U	K	H	K	M	F	D	H	P	V	Z	V	F	F	Q	L	O	C	L	M	C	U	D	Y	O	C	L	E	G	I	Y	Y
P	J	K	F	O	R	K	F	N	W	E	V	G	R	A	L	I	R	G	X	I	R	V	X	U	M	A	K	G	O	J	*
I	I	E	E	E	V	W	I	?	E	X	R	C	G	V	G	T	X	Q	O	E	G	J	P	D	V	U	K	F	Y	F	*
F	W	M	E	R	G	F	K	N	S	U	U	L	M	O	O	V	H	A	P	U	T	G	F	V	Q	M	B	W	J	E	Z
G	M	H	H	O	F	K	?	G	R	K	C	A	G	E	U	W	H	S	H	J	T	V	G	Z	I	I	E	Y	R	G	*
G	I	U	Q	O	H	T	V	B	F	D	Q	V	W	T	E	L	N	U	D	M	U	V	N	V	Q	F	E	G	U	*	*
F	M	K	K	W	Y	Y	H	R	G	G	W	S	F	F	Z	C	E	N	W	T	Q	G	U	Z	F	K	M	H	Y	F	*
E	Q	U	G	N	T	Q	K	H	M	G	M	T	N	Y	L	U	P	B	M	N	H	Q	V	W	T	E	Z	W	R	T	*
E	U	V	N	F	C	F	E	N	I	E	B	G	N	L	H	K	Q	V	T	M	H	F	A	X	L	M	M	B	F	O	H
F	Q	E	P	B	I	Y	S	I	Q	M	A	Y	I	F	S	D	T	O	F	S	J	C	J	E	P	A	I	Q	B	J	B
D	I	O	Q	Y	F	P	V	M	E	A	M	M	A	M	M	Q	S	I	A	Q	S	A	N	Y	X	Z	N	Q	F	*	*
O	T	Y	Q	H	B	O	J	I	Q	Y	B	Y	T	F	A	M	Q	Q	F	M	F	C	M	T	J	J	B	D	B	F	*
X	N	O	X	Q	E	J	O	F	F	Q	Y	B	I	D	O	F	Q	F	V	E	Y	F	O	Q	I	B	F	S	F	*	*
O	G	S	M	A	F	E	O	J	X	F	Q	I	B	F	J	S	P	O	F	P	U	I	F	F	Q	D	O	B	P	D	Y
F	J	G	O	C	Y	A	T	B	N	I	I	F	X	F	Q	B	O	B	N	B	O	F	H	P	F	F	Y	M	C	*	*
O	F	F	G	S	B	A	G	J	S	O	V	F	O	V	B	F	S	O	S	B	Y	N	B	F	F	Y	O	Q	Y	O	J
C	A	F	E	E	Q	J	B	B	I	O	N	A	O	F	X	T	J	F	Y	S	B	H	Y	J	F	X	G	F	C	*	*
B	F	D	O	E	E	I	J	M	D	F	J	I	A	J	O	F	H	S	F	B	S	A	E	E	Y	P	E	M	S	Y	J
Y	R	M	N	M	F	I	B	H	O	E	I	B	Y	E	T	Q	F	S	I	N	H	G	N	G	F	V	I	F	*	*	*
F	D	E	N	Y	J	T	G	F	J	N	F	I	Q	M	A	A	O	O	Y	O	W	E	S	I	X	?	S	C	R	Y	*
U	U	T	Y	Q	H	R	A	A	S	*	V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T	*	O	X	O	U	A	L	U	A	A	F
Y	*	W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

It will be seen immediately that the letters ALN have been retained in the same filter position and have now been joined by the letters DCI in the second filter position, which is believed to be representative of CANDIL light passing through the two (hypothesised) polarising filters. The second cryptogram is presented without a full introduction at this point, but the order in which the letters are read is LWFR PR(R)G. The R in the second grouping of 4 is bracketed as it represents the position of

the spelling mistake in K1. Encrypting LWFR PRRG against the next 8 letters of K4, which are GHUL BSOL gives rise to the result TEAM EAST for cryptogram 2.

One further observation of note before looking more deeply into the first two cribs is that 7 of the 8 letters comprising cryptogram 2 appear to have been lifted directly from K4 itself starting at position 21, with the single spelling mistake also being echoed in the K1 plain text for that same position.

E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R

All of the phenomena described up to this point have been engineered in order to cement the significance of these two end results for cryptograms 1 and 2 by adding tangible methods of corroboration. These initial two cryptograms and their associated cribs are all we are given in the way of help in order to complete the remainder of the solves for Rail Positions 2 to 6. We obviously have the plain text clues for EAST NORTH EAST and BERLIN CLOCK in their relative positions, but these can only be used for verification purposes and do not in themselves give anything away about the underlying method. It is necessary therefore to re-examine everything we have already been given in these two cribs to see if it's possible to extract any more information about the underlying method.

In depth re-examination of the first two cryptograms and their associated cribs

The significance of some of the initial structures discovered in the plain text cylinder under the transforms for C(1) were passed over quite quickly in the previous treatment, so we should now address the question of letter selection and whether the crib examples can give us any further pointers towards how we can identify the intended letters without recourse to support from these large scale visual structures.

Two distinct schemes per rail position

The first and most obvious conclusion to draw first is that there are two distinct alignments for each rail position. Irrespective of the required KRS(1) operation after the first block of 8, C(2) is telling us that the second alignment scheme needs to be solved via the rails 'reciprocal rows' rather than in the primary rail itself. Given that we have already established the presence of 6 rails positions this would imply that we are capable of generating at least 96 letters of plain text via this scheme.

Invariance under transformation

If we step back to the initial left shift of 13 applied to row 1 in the plain text cylinder then we will remember the vertical striping of the letters HOT running in both directions.

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N	
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	

It was this initial configuration that allowed us to spot the persistence of the UN YY grouping in the cipher text as shown again here.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

This appears to be an important observation as the characters that remain constant have become the first 4 of the intended cryptogram.

We were initially led to the solve of cryptogram 2 by the visual guidance given by the double X crib, but looking in greater detail at the cipher cylinder for Rail Position 1 and the reciprocal rows after the first global right shift we find a very similar mechanism to that just described for C(1). The H and N characters share the same relationship as H and T in cryptogram 1.

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G	E		P	U	O	U	V	Z	U	C	U	W
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T		O	X	O	U	A	L	U	A	A	F
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

On closer inspection we find there is only one symmetrical shift pattern of row 2 that provides a complete mirroring between the rail position and its reciprocal rows, which is found to be for the letters HUN as shown here.

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
E		P	U	O	U	V	Z	U	C	U	W	P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
T		O	X	O	U	A	L	U	A	A	F	R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

The next observation under this row alignment is critical as it shows that the use of the double X as a visual crib was purely a guide pointing us at this configuration. If we look at the relationship between the digraphs AS and MV then we can see that there is an equivalent persistence of a 4 letter block identical to that of the UNYY configuration in C(1).

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
E		P	U	O	U	V	Z	U	C	U	W	P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
T		O	X	O	U	A	L	U	A	A	F	R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

A left shift of 12 in row 27 with a symmetrical change in the reciprocal row 2 brings us back to the relationship first discovered via the crib.

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G	E		P	U	O	U	V	Z	U	C	U	W
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T		O	X	O	U	A	L	U	A	A	F
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

There is another and much more subtle aspect of this transformation that is harder to spot, but will be seen to be relevant in later solves. If we examine the plain and cipher text cylinders simultaneously for this same transformation in Rail Position 1 then it can be seen that there is a similar relationship that occurs in the top half of the cylinder under the same transformation. All three views will be shown simultaneously to show what is happening. These initial positions that form our starting points for a given rail alignment shall be referred to here on in as the ‘normalised symmetry position’ or NSP. The * character will designate the reciprocal rows for any given position. KRS(n) will be used to designate how much the text has been right shifted through the Kryptos alphabet. The ‘transformed symmetry position’ will be referred to as the TSP.

NSP for Rail Position 1 KRS(1).

PT

Q	D	C	F	O	X	F	F	Q	A	V	C	O	M	F	A	I	B	E	J	Q	H	B	Q	E	O	I	F	B	C	A	F
Q		F	S	G	M	J	H	I	O	M	J	F	A	O	I	F	Q	V	B	Q	D	F	S	G	J	U	M	V	A	J	S
I	S	X	A	O	I	B	O	T	S		J	O	X	B	A	O	S	O	B	M	M	P	J	Q	W	J	A	J	C	M	F

CT

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
E		P	U	O	U	V	Z	U	C	U	W	P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

* CT/PT

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
T		O	X	O	U	A	L	U	A	A	F	R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

TSP for Rail Position 1 KRS(1).

PT

Q	D	C	F	O	X	F	F	Q	A	V	C	O	M	F	A	I	B	E	J	Q	H	B	Q	E	O	I	F	B	C	A	F
F	A	O	I	F	Q	V	B	Q	D	F	S	G	J	U	M	V	A	J	S	Q		F	S	G	M	J	H	I	O	M	J
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

CT

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G	E		P	U	O	U	V	Z	U	C	U	W
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

* CT/PT

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T		O	X	O	U	A	L	U	A	A	F
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

Mirroring and row striping

There does not appear to be any symmetry under RS1 for Rail Position 1 like the clean ALN relationship that we saw under KRS(0). What we do see however is a feature under KRS(0) that appears to be echoed to some degree under RS1 for the cipher text cylinder. This feature is the presence of the vertical row stripe ABC. We first see it here in Rail Position 1 under KRS(0).

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		Y	Q	T	Q	U	X	Q	B	Q	V
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

We next see this feature in *Rail Position 1 under KRS(1) as shown here.

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T		O	X	O	U	A	L	U	A	A	F
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

The most interesting aspect of this feature is that if we read the rows as 2, 3 and then finally 1 the relationship reverts back to ALN and ABC.

There also appears to be signature in the reciprocal rows of the plain text cylinder that is noteworthy in the TSP.

* CT/*PT

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T		O	X	O	U	A	L	U	A	A	F
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

The role of digraphs

This next piece of analysis was not discussed in part 2 so as not to add confusion, but is left in as part of this more detailed analysis as there may well be some additional spatial information imparted through the use of digraphs.

If we go back to the results for Cryptogram 1 as displayed on the plain text cylinder then it can be seen that there does appear to be some significant positional information conveyed through the use of digraphs. Looking in more detail at the most obvious digraphs we can note the following relationships.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		Y	Q	T	Q	U	X	Q	B	Q	V
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		E	O	F	L	I	G	H	T	L	I
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
A	W	I	N	F	B	N	Y	P		T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I
G	K	U	H	U	A	U	E	K	C	A	R		V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

The spatial relationships here are extremely interesting as there is a separation on the cylinder between the diagonal TL and AN of 9 letters, which is the same as the separation of the diagonal HE letters that cross the location of our first 4 letter block.

In analysing the role played by digraphs another aspect of the IUT scheme became apparent at this stage, which is the role played by the IUT 3:4. In C(2) we can see that the cipher text for the NSP position has the following digraph relationships prior to transformation.

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
E		P	U	O	U	V	Z	U	C	U	W	P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

We know that we see 3 of these letters again as part of the 4 that make our PR/RG LBA's. Whilst this was noted, it wasn't really cemented as a meaningful part of the scheme until the implications of C(9)'s start and end positions were fully appraised.

In terms of the 3rd type of IUT transformation we see in the cryptograms from C(7) onwards, which we have described as IUT H:V then the best example is given by the previous scenario in the plain text of C(1) where the word HOLE could be read horizontally in the NSP and then as 2 LBA's in the vertical plane from the corresponding TSP position.

Temporal IUT

There may be some evidence for a form of temporal IUT as seen in this example from C(2), where a relationship exists between the PT and CT when viewed as a superposition of the NSP and TSP.

PT NSP

Q	D	C	F	O	X	F	F	Q	A	V	C	O	M	F	A	I	B	E	J	Q	H	B	Q	E	O	I	F	B	C	A	F
Q		F	S	G	M	J	H	I	O	M	J	F	A	O	I	F	Q	V	B	Q	D	F	S	G	J	U	M	V	A	J	S
I	S	X	A	O	I	B	O	T	S		J	O	X	B	A	O	S	O	B	M	M	P	J	Q	W	J	A	J	C	M	F

CT NSP

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
E		P	U	O	U	V	Z	U	C	U	W	P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

PT TSP

Q	D	C	F	O	X	F	F	Q	A	V	C	O	M	F	A	I	B	E	J	Q	H	B	Q	E	O	I	F	B	C	A	F
F	A	O	I	F	Q	V	B	Q	D	F	S	G	J	U	M	V	A	J	S	Q		F	S	G	M	J	H	I	O	M	J
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

CT TSP

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G	E		P	U	O	U	V	Z	U	C	U	W
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

The start position for the MV block and its locking counterpart in the plaintext appear to be relevant positional markers in this instance, but evidence for a similar relationship has only been identified in one additional cryptogram so far.

Now that we have established how the algorithms and IUT scheme were initially found with the help of the visual cribs, we shall revisit each of the cryptograms in order to see if any additional information can be extracted.

For the purposes of making results easy to replicate a transformed version of the text for the rail position and reciprocal rows are given for both the cipher and plain text cylinders with the correct amount of KRS(n) applied.

Rows are additionally shown in their original alignments as the 'initial row position' or IRP.

Part 4

Cryptogram Solves

C(1 – 6)

Detailed Analysis

Cryptogram 1

C(1) RP1 IRP KRS(0)

PT

B	E	T	W	E	E	N	S	U	B	T	L	E	S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N	
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	

*PT

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	

CT

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

*CT

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	

C(1) RP1 KRS(0) SA(CS) RML[HOT] IUT2 :2[UN/YY] IUT3 :4[DK/B] NSP

PT

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N	
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	

*PT

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	
G	K	U	H	U	A	U	E	K	C	A	R		V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

CT

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

*CT

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	
G	K	U	H	U	A	U	E	K	C	A	R		V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

R(2)LS(10) to the TSP gives.

C(1) RP1 KRS(0) SA(CS) IUT[UN/YY] SML[ANL] TSP

PT

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		E	O	F	L	I	G	H	T	L	I
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	

*PT

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
A	W	I	N	F	B	N	Y	P		T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I
G	K	U	H	U	A	U	E	K	C	A	R		V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

CT

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		Y	Q	T	Q	U	X	Q	B	Q	V
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

*CT

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
A	W	I	N	F	B	N	Y	P		T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I
G	K	U	H	U	A	U	E	K	C	A	R		V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

C(1) RP1 KRS(0) SA(CS) IUT2 :2[UN/YY] IUT3 :4[DK/B] TSP Digraph Analysis

PT

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E
E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		E	O	F	L	I	G	H	T	L	I
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	

PT

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
A	W	I	N	F	B	N	Y	P		T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I
G	K	U	H	U	A	U	E	K	C	A	R		V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

CT

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		Y	Q	T	Q	U	X	Q	B	Q	V
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

*CT

U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	
A	W	I	N	F	B	N	Y	P		T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I
G	K	U	H	U	A	U	E	K	C	A	R		V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I

Notes. The first 4 letters are given to us by the IUT 2:2 configuration where UN/YY is invariant. At this stage we see that IUT3:4 is actually DK/B, which means the intended 2 LBA's must be CK/BD.

Whilst we have already extracted our required 8 letters, the spatial relationships given by the digraph analysis do remain intriguing.

Letter Selection

RP1 CT

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y
Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		Y	Q	T	Q	U	X	Q	B	Q	V
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	

Results

C(1) KRS(0) SA(CS)	K4 Key	Encryption
UNYY CKBD	OBKR UOXO	KRYP TOSI

Cryptogram 2

C(2) RP1 IRP KRS(1)

PT

C	F	O	X	F	F	Q	A	V	C	O	M	F	A	I	B	E	J	Q	H	B	Q	E	O	I	F	B	C	A	F	Q	D
F	S	G	M	J	H	I	O	M	J	F	A	O	I	F	Q	V	B	Q	D	F	S	G	J	U	M	V	A	J	S	Q	
J	O	X	B	A	O	S	O	B	M	M	P	J	Q	W	J	A	J	C	M	F	I	S	X	A	O	I	B	O	T	S	

CT

F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J	W	L
P	U	O	U	V	Z	U	C	U	W	P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G	E	
W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R	U	F	O	H	G	U	L	Q	D	F	

*CT / PT

V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W	U	U	T	Y	Q	H	R	A	A	S	
O	X	O	U	A	L	U	A	A	F	R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T	
W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B	Y	

C(2) RP1 KRS(1) SA(SS) SML[HUN] IUT2:2[AS/MV] IUT3:4[G/RR][WLF] NSP

PT

Q	D	C	F	O	X	F	F	Q	A	V	C	O	M	F	A	I	B	E	J	Q	H	B	Q	E	O	I	F	B	C	A	F
Q		F	S	G	M	J	H	I	O	M	J	F	A	O	I	F	Q	V	B	Q	D	F	S	G	J	U	M	V	A	J	S
I	S	X	A	O	I	B	O	T	S		J	O	X	B	A	O	S	O	B	M	M	P	J	Q	W	J	A	J	C	M	F

CT

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
E		P	U	O	U	V	Z	U	C	U	W	P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

*CT/PT

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
T		O	X	O	U	A	L	U	A	A	F	R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

R(2)LS(12) to the TSP gives.

C(2) RP1 KRS(1) SA(SS) RML[HLG] IUT2 :2[AS/MV] IUT4 :4[PR/RG][LW/FR] TSP

PT

Q	D	C	F	O	X	F	F	Q	A	V	C	O	M	F	A	I	B	E	J	Q	H	B	Q	E	O	I	F	B	C	A	F
F	A	O	I	F	Q	V	B	Q	D	F	S	G	J	U	M	V	A	J	S	Q		F	S	G	M	J	H	I	O	M	J
I	S	X	A	O	I	B	O	T	S		J	O	X	B	A	O	S	O	B	M	M	P	J	Q	W	J	A	J	C	M	F

CT

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G	E		P	U	O	U	V	Z	U	C	U	W
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

*CT/PT

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T		O	X	O	U	A	L	U	A	A	F
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

RP1 TSP KRS(1) SA(SS) IUT[AS/MV] Digraph Analysis

PT

Q	D	C	F	O	X	F	F	Q	A	V	C	O	M	F	A	I	B	E	J	Q	H	B	Q	E	O	I	F	B	C	A	F
F	A	O	I	F	Q	V	B	Q	D	F	S	G	J	U	M	V	A	J	S	Q		F	S	G	M	J	H	I	O	M	J
I	S	X	A	O	I	B	O	T	S		J	O	X	B	A	O	S	O	B	M	M	P	J	Q	W	J	A	J	C	M	F

*PT

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T		O	X	O	U	A	L	U	A	A	F
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

CT

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G	E		P	U	O	U	V	Z	U	C	U	W
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

*CT

U	U	T	Y	Q	H	R	A	A	S		V	S	Z	S	H	I	V	M	C	A	S	M	J	G	C	C	X	G	M	Y	W
R	K	K	X	B	O	L	R	M	V	E	J	B	X	J	Q	G	C	Q	P	T		O	X	O	U	A	L	U	A	A	F
Y		W	O	O	N	K	G	T	R	X	H	E	R	K	Z	O	L	D	E	J	H	R	V	I	V	B	V	F	R	D	B

Notes. Whilst there is no interesting additional data to be gleaned from digraphs in this instance we do have the interesting result with regards the matching up with the K4 cipher text. The presence of the single mistake being in the same location as the misspelt Q in K1 is deemed non-coincidental.

Letter Selection

CT

W	L	F	N	V	G	T	I	K	M	Y	G	B	Z	P	V	A	E	L	R	K	M	E	R	Y	Q	A	I	H	Q	G	J
P	V	W	M	M	O	Y	F	W	L	P	U	O	N	R	P	Y	E	N	G	E		P	U	O	U	V	Z	U	C	U	W
U	F	O	H	G	U	L	Q	D	F		W	G	T	L	V	E	F	F	I	K	X	F	O	K	P	W	H	X	I	R	R

Results

C(2) KRS(1) SA(SS)	K4 Key	Encryption
LWFR PRRG	GHUL BSOL	TEAM EAST

Cryptogram 3

C(3) RP2 IRP KRS(2)

PT

U	Y	U	A	Z	U	N	A	E	C	S	L	A	U	K	F	A	G	B	N	C	U	I	N	G	T	Y	U	A	Z	C	
D	A	W	S	S	J	L	B	?	S	J	G	T	B	J	A	W	N	F	L	S	B	D	W	P	L	G	F	A	W	S	
S	J	G	P	G	B	A	Q	G	Z	J	G	P	G	K	Z	J	A	Y	U	A	Z	B	S	J	G	G	K	C	E	S	N

*PT

F	S	J	G	U	Z	L	F	G	U	L	U	I	S	J	G	J	A	N	G	C	N	L	S	S	N	G	L	L	U		
B	G	P	S	G	F	S	J	G	E	C	U	F	N	G	C	U	F	O	G	G	P	G	F	L	U	S	J	G	J	A	S
C	L	P	G	B	E	C	O	L	U	I	H	P	A	Q	S	J	G	E	J	C	Q	D	G	P	E	C	W	B	G	F	

CT

T	L	R	G	S	Y	R	G	Q	X	F	W	H	Y	B	M	J	Y	H	Z	J	Y	W	Z	V	N	B	R	H	S	L	
J	J	F	F	F	W	X	J	?	F	Z	Y	D	H	W	H	O	Z	U	S	F	H	L	T	E	W	V	R	G	P	G	
G	X	N	F	Y	H	G	R	Q	A	V	V	M	N	S	S	W	I	B	T	V	O	H	G	W	U	N	C	X	L	F	K

*CT

G	L	H	S	D	P	B	O	C	Q	J	J	G	Z	G	U	C	S	C	Q	C	S	G	I	T	G	G	P	N	D		
S	G	G	H	A	C	B	H	L	A	S	W	G	S	W	C	G	A	S	A	C	P	Q	C	G	G	P	S	U	P	S	L
D	B	G	F	F	U	L	C	C	J	S	S	Q	B	S	G	Z	O	L	G	P	A	C	I	P	L	G	Z	H	G	D	

C(3) RP2 KRS(2) SA(CS) IUT2:2[GZ/OJ] IUT3:4[WFN] RML[NFW] NSP

PT

U	Y	U	A	Z	U	N	A	E	C	S	L	A	U	K	F	A	G	B	N	C	U	I	N	G	T	Y	U	A	Z	C	
S	J	G	T	B	J	A	W	N	F	L	S	B	D	W	P	L	G	F	A	W	S		D	A	W	S	S	J	L	B	?
S	J	G	G	K	C	E	S	N	S	J	G	P	G	B	A	Q	G	Z	J	G	P	G	K	Z	J	A	Y	U	A	Z	B

*Rail Position 2 PT

U	L	U	I	S	J	G	J	A	N	G	C	N	L	S	S	N	G	L	L	U			F	S	J	G	U	Z	L	F	G
F	L	U	S	J	G	J	A	S	B	G	P	S	G	F	S	J	G	E	C	U	F	N	G	C	U	F	O	G	G	P	G
C	L	P	G	B	E	C	O	L	U	I	H	P	A	Q	S	J	G	E	J	C	Q	D	G	P	E	C	W	B	G	F	

Rail Position 2 CT

T	L	R	G	S	Y	R	G	Q	X	F	W	H	Y	B	M	J	Y	H	Z	J	Y	W	Z	V	N	B	R	H	S	L	
F	Z	Y	D	H	W	H	O	Z	U	S	F	H	L	T	E	W	V	R	G	P	G		J	J	F	F	F	W	X	J	?
G	W	U	N	C	X	L	F	K	G	X	N	F	Y	H	G	R	Q	A	V	V	M	N	S	S	W	I	B	T	V	O	H

*Rail Position 2 CT

Q	J	J	G	Z	G	U	C	S	C	Q	C	S	G	I	T	G	G	P	N	D			G	L	H	S	D	P	B	O	C
C	G	G	P	S	U	P	S	L	S	G	G	H	A	C	B	H	L	A	S	W	G	S	W	C	G	A	S	A	C	P	Q
D	B	G	F	F	U	L	C	C	J	S	S	Q	B	S	G	Z	O	L	G	P	A	C	I	P	L	G	Z	H	G	D	

R(2)LS(4) to the TSP gives.

C(3) RP2 KRS(2) SA(CS) IUT[GZ/OJ] IUT3:4[FN/WE] SML[GHG] TSP

PT

U	Y	U	A	Z	U	N	A	E	C	S	L	A	U	K	F	A	G	B	N	C	U	I	N	G	T	Y	U	A	Z	C	
B	J	A	W	N	F	L	S	B	D	W	P	L	G	F	A	W	S		D	A	W	S	S	J	L	B	?	S	J	G	T
S	J	G	G	K	C	E	S	N	S	J	G	P	G	B	A	Q	G	Z	J	G	P	G	K	Z	J	A	Y	U	A	Z	B

*PT

U	L	U	I	S	J	G	J	A	N	G	C	N	L	S	S	N	G	L	L	U			F	S	J	G	U	Z	L	F	G
G	G	P	G	F	L	U	S	J	G	J	A	S	B	G	P	S	G	F	S	J	G	E	C	U	F	N	G	C	U	F	O
C	L	P	G	B	E	C	O	L	U	I	H	P	A	Q	S	J	G	E	J	C	Q	D	G	P	E	C	W	B	G	F	

CT

T	L	R	G	S	Y	R	G	Q	X	F	W	H	Y	B	M	J	Y	H	Z	J	Y	W	Z	V	N	B	R	H	S	L	
H	W	H	O	Z	U	S	F	H	L	T	E	W	V	R	G	P	G		J	J	F	F	F	W	X	J	?	F	Z	Y	D
G	W	U	N	C	X	L	F	K	G	X	N	F	Y	H	G	R	Q	A	V	V	M	N	S	S	W	I	B	T	V	O	H

*CT

Q	J	J	G	Z	G	U	C	S	C	Q	C	S	G	I	T	G	G	P	N	D			G	L	H	S	D	P	B	O	C
A	C	P	Q	C	G	G	P	S	U	P	S	L	S	G	G	H	A	C	B	H	L	A	S	W	G	S	W	C	G	A	S
D	B	G	F	F	U	L	C	C	J	S	S	Q	B	S	G	Z	O	L	G	P	A	C	I	P	L	G	Z	H	G	D	

C(3) RP2 TSP KRS(2) SA(CS) IUT[GZ/OJ] Digraph Analysis

PT

U	Y	U	A	Z	U	N	A	E	C	S	L	A	U	K	F	A	G	B	N	C	U	I	N	G	T	Y	U	A	Z	C	
B	J	A	W	N	F	L	S	B	D	W	P	L	G	F	A	W	S		D	A	W	S	S	J	L	B	?	S	J	G	T
S	J	G	G	K	C	E	S	N	S	J	G	P	G	B	A	Q	G	Z	J	G	P	G	K	Z	J	A	Y	U	A	Z	B

*PT

U	L	U	I	S	J	G	J	A	N	G	C	N	L	S	S	N	G	L	L	U			F	S	J	G	U	Z	L	F	G
G	G	P	G	F	L	U	S	J	G	J	A	S	B	G	P	S	G	F	S	J	G	E	C	U	F	N	G	C	U	F	O
C	L	P	G	B	E	C	O	L	U	I	H	P	A	Q	S	J	G	E	J	C	Q	D	G	P	E	C	W	B	G	F	

CT

T	L	R	G	S	Y	R	G	Q	X	F	W	H	Y	B	M	J	Y	H	Z	J	Y	W	Z	V	N	B	R	H	S	L	
H	W	H	O	Z	U	S	F	H	L	T	E	W	V	R	G	P	G		J	J	F	F	F	W	X	J	?	F	Z	Y	D
G	W	U	N	C	X	L	F	K	G	X	N	F	Y	H	G	R	Q	A	V	V	M	N	S	S	W	I	B	T	V	O	H

*CT

Q	J	J	G	Z	G	U	C	S	C	Q	C	S	G	I	T	G	G	P	N	D			G	L	H	S	D	P	B	O	C
A	C	P	Q	C	G	G	P	S	U	P	S	L	S	G	G	H	A	C	B	H	L	A	S	W	G	S	W	C	G	A	S
D	B	G	F	F	U	L	C	C	J	S	S	Q	B	S	G	Z	O	L	G	P	A	C	I	P	L	G	Z	H	G	D	

Notes. The IUT[3:4] doubles up with the RML in this instance. The position of the PL/GP and GF/GB blocks are highlighted here as the placement is of interest.

Letter Selection

CT

T	L	R	G	S	Y	R	G	Q	X	F	W	H	Y	B	M	J	Y	H	Z	J	Y	W	Z	V	N	B	R	H	S	L	
H	W	H	O	Z	U	S	F	H	L	T	E	W	V	R	G	P	G		J	J	F	F	F	W	X	J	?	F	Z	Y	D
G	W	U	N	C	X	L	F	K	G	X	N	F	Y	H	G	R	Q	A	V	V	M	N	S	S	W	I	B	T	V	O	H

Results

C(3) KRS(2) SA(CS)	K4 Key	Encryption
IFBB WFLR	FNWE GZJO	ROON DEAS

Cryptogram 4

C(4) RP2 IRP KRS(3)

PT

V	P	V	B	K	V	Q	B	F	D	A	M	B	V	R	G	B	H	C	Q	D	V	J	Q	H	O	P	V	B	K	D	
E	B	X	A	A	L	M	C	?	A	L	H	O	C	L	B	X	Q	G	M	A	C	E	X	T	M	H	G	B	X	A	
A	L	H	T	H	C	B	U	H	K	L	H	T	H	R	K	L	B	P	V	B	K	C	A	L	H	H	R	D	F	A	Q

*PT

G	A	L	H	V	K	M	G	H	V	M	V	J	A	L	H	L	B	Q	H	D	Q	M	A	A	Q	H	M	M	V		
C	H	T	A	H	G	A	L	H	F	D	V	G	Q	H	D	V	G	S	H	H	T	H	G	M	V	A	L	H	L	B	A
D	M	T	H	C	F	D	S	M	V	J	I	T	B	U	A	L	H	F	L	D	U	E	H	T	F	D	X	C	H	G	

CT

O	M	Y	H	A	P	Y	H	U	Z	G	X	I	P	C	N	L	P	I	K	L	P	X	K	W	Q	C	Y	I	A	M	
L	L	G	G	G	X	Z	L	?	G	K	P	E	I	X	I	S	K	V	A	G	I	M	O	F	X	W	Y	H	T	H	
H	Z	Q	G	P	I	H	Y	U	B	W	W	N	Q	A	A	X	J	C	O	W	S	I	H	X	V	Q	D	Z	M	G	R

*CT

H	M	I	A	E	T	C	S	D	U	L	L	H	K	H	V	D	A	D	U	D	A	H	J	O	H	H	T	Q	E		
A	H	H	I	B	D	C	I	M	B	A	X	H	A	X	D	H	B	A	B	D	T	U	D	H	H	T	A	V	T	A	M
E	C	H	G	G	V	M	D	D	L	A	A	U	C	A	H	K	S	M	H	T	B	D	J	T	M	H	K	I	H	E	

C(4) RP2 KRS(3) SA(SS) IUT2:2[DH/BA] IUT3:4[BEX][XAP] RML[HHT] NSP

PT

A	M	B	V	R	G	B	H	C	Q	D	V	J	Q	H	O	P	V	B	K	D		V	P	V	B	K	V	Q	B	F	D
A	C	E	X	T	M	H	G	B	X	A		E	B	X	A	A	L	M	C	?	A	L	H	O	C	L	B	X	Q	G	M
A	L	H	T	H	C	B	U	H	K	L	H	T	H	R	K	L	B	P	V	B	K	C	A	L	H	H	R	D	F	A	Q

*PT

G	A	L	H	V	K	M	G	H	V	M	V	J	A	L	H	L	B	Q	H	D	Q	M	A	A	Q	H	M	M	V		
H	T	H	G	M	V	A	L	H	L	B	A	C	H	T	A	H	G	A	L	H	F	D	V	G	Q	H	D	V	G	S	H
J	I	T	B	U	A	L	H	F	L	D	U	E	H	T	F	D	X	C	H	G		D	M	T	H	C	F	D	S	M	V

CT

G	X	I	P	C	N	L	P	I	K	L	P	X	K	W	Q	C	Y	I	A	M		O	M	Y	H	A	P	Y	H	U	Z
G	I	M	O	F	X	W	Y	H	T	H		L	L	G	G	G	X	Z	L	?	G	K	P	E	I	X	I	S	K	V	A
H	Z	Q	G	P	I	H	Y	U	B	W	W	N	Q	A	A	X	J	C	O	W	S	I	H	X	V	Q	D	Z	M	G	R

* CT

H	M	I	A	E	T	C	S	D	U	L	L	H	K	H	V	D	A	D	U	D	A	H	J	O	H	H	T	Q	E		
D	T	U	D	H	H	T	A	V	T	A	M	A	H	H	I	B	D	C	I	M	B	A	X	H	A	X	D	H	B	A	B
A	A	U	C	A	H	K	S	M	H	T	B	D	J	T	M	H	K	I	H	E		E	C	H	G	G	V	M	D	D	L

R(2)RS(4) to the TSP gives.

C(4) RP2 KRS(3) SA(SS) IUT2:2[DH/BA] IUT3:4[BZ/EX][PU/XA] RML[HHA][HVD] TSP

PT

A	M	B	V	R	G	B	H	C	Q	D	V	J	Q	H	O	P	V	B	K	D		V	P	V	B	K	V	Q	B	F	D
X	Q	G	M	A	C	E	X	T	M	H	G	B	X	A		E	B	X	A	A	L	M	C	?	A	L	H	O	C	L	B
A	L	H	T	H	C	B	U	H	K	L	H	T	H	R	K	L	B	P	V	B	K	C	A	L	H	H	R	D	F	A	Q

*PT

G	A	L	H	V	K	M	G	H	V	M	V	J	A	L	H	L	B	Q	H	D	Q	M	A	A	Q	H	M	M	V		
V	G	S	H	H	T	H	G	M	V	A	L	H	L	B	A	C	H	T	A	H	G	A	L	H	F	D	V	G	Q	H	D
J	I	T	B	U	A	L	H	F	L	D	U	E	H	T	F	D	X	C	H	G		D	M	T	H	C	F	D	S	M	V

CT

G	X	I	P	C	N	L	P	I	K	L	P	X	K	W	Q	C	Y	I	A	M		O	M	Y	H	A	P	Y	H	U	Z
S	K	V	A	G	I	M	O	F	X	W	Y	H	T	H		L	L	G	G	G	X	Z	L	?	G	K	P	E	I	X	I
H	Z	Q	G	P	I	H	Y	U	B	W	W	N	Q	A	A	X	J	C	O	W	S	I	H	X	V	Q	D	Z	M	G	R

*CT

H	M	I	A	E	T	C	S	D	U	L	L	H	K	H	V	D	A	D	U	D	A	H	J	O	H	H	T	Q	E		
H	B	A	B	D	T	U	D	H	H	T	A	V	T	A	M	A	H	H	I	B	D	C	I	M	B	A	X	H	A	X	D
A	A	U	C	A	H	K	S	M	H	T	B	D	J	T	M	H	K	I	H	E		E	C	H	G	G	V	M	D	D	L

C(4) RP2 KRS(3) SA(SS) IUT[DH/BA] TSP Digraph Analysis

PT

A	M	B	V	R	G	B	H	C	Q	D	V	J	Q	H	O	P	V	B	K	D		V	P	V	B	K	V	Q	B	F	D
X	Q	G	M	A	C	E	X	T	M	H	G	B	X	A		E	B	X	A	A	L	M	C	?	A	L	H	O	C	L	B
A	L	H	T	H	C	B	U	H	K	L	H	T	H	R	K	L	B	P	V	B	K	C	A	L	H	H	R	D	F	A	Q

* PT

G	A	L	H	V	K	M	G	H	V	M	V	J	A	L	H	L	B	Q	H	D	Q	M	A	A	Q	H	M	M	V		
V	G	S	H	H	T	H	G	M	V	A	L	H	L	B	A	C	H	T	A	H	G	A	L	H	F	D	V	G	Q	H	D
J	I	T	B	U	A	L	H	F	L	D	U	E	H	T	F	D	X	C	H	G		D	M	T	H	C	F	D	S	M	V

CT

G	X	I	P	C	N	L	P	I	K	L	P	X	K	W	Q	C	Y	I	A	M		O	M	Y	H	A	P	Y	H	U	Z
S	K	V	A	G	I	M	O	F	X	W	Y	H	T	H		L	L	G	G	G	X	Z	L	?	G	K	P	E	I	X	I
H	Z	Q	G	P	I	H	Y	U	B	W	W	N	Q	A	A	X	J	C	O	W	S	I	H	X	V	Q	D	Z	M	G	R

* CT

H	M	I	A	E	T	C	S	D	U	L	L	H	K	H	V	D	A	D	U	D	A	H	J	O	H	H	T	Q	E		
H	B	A	B	D	T	U	D	H	H	T	A	V	T	A	M	A	H	H	I	B	D	C	I	M	B	A	X	H	A	X	D
A	A	U	C	A	H	K	S	M	H	T	B	D	J	T	M	H	K	I	H	E		E	C	H	G	G	V	M	D	D	L

Notes. The HM/HH and MVFL digraph blocks appear to be in play here, but again we are simply committing the observation to the record.

Letter Selection

CT

G	X	I	P	C	N	L	P	I	K	L	P	X	K	W	Q	C	Y	I	A	M		O	M	Y	H	A	P	Y	H	U	Z
S	K	V	A	G	I	M	O	F	X	W	Y	H	T	H		L	L	G	G	G	X	Z	L	?	G	K	P	E	I	X	I
H	Z	Q	G	P	I	H	Y	U	B	W	W	N	Q	A	A	X	J	C	O	W	S	I	H	X	V	Q	D	Z	M	G	R

Results

C(3) KRS(2) SA(SS)	K4 Key	Encryption
VQQP RENGK	BZEX PUXA	TNOR THEA

Cryptogram 5

C(5) RP3 IRP KRS(4)

PT

D	E	L	I	Y	B	M	N	O	B	S	I	N	L	M	B	H	I	L	O	I	I	D	J	N	J	B	S	D	I		
K	I	W	V	N	W	Z	B	I	D	D	N	Y	A	C	N	W	B	J	N	K	I	D	I	G	C	W	H	D	W	C	
O	B	M	D	I	K	I	W	B	S	D	I	K	I	W	H	I	L	O	I	I	D	I	N	L	M	B	V	N	W	Z	

*PT

C	J	A	E	D	D	E	L	I	H	I	F	O	N	D	B	M	E	B	I	W	G	Z	V	F	I	O	I	H	B		
M	I	U	C	R	I	O	A	E	O	B	C	J	B	M	I	H	C	C	O	R	E	S	R	E	D	O	I	V	C	K	
I	H	R	N	B	M	B	O	I	V	F	U	N	W	L	M	E	W	H	D	N	V	E	H	I	E	B	N	W	S		

CT

J	M	X	W	B	L	A	Z	E	I	G	W	Z	K	A	H	Q	V	X	G	U	X	Z	V	Z	W	I	H	J	X		
I	U	P	P	K	O	O	L	T	J	J	K	C	I	I	Y	F	H	V	K	A	W	J	X	Y	I	P	U	L	O	I	
H	W	X	J	V	A	W	P	L	U	J	U	A	V	O	Q	X	S	E	U	V	L	W	Z	K	A	H	Y	K	T	A	

*CT

G	M	B	W	O	I	S	Z	U	H	D	U	U	D	U	U	W	C	M	D	W	C	D	V	O	R	Y	V	W	I		
B	A	O	W	L	E	B	N	M	W	O	E	O	A	I	D	U	W	W	I	U	I	F	U	A	N	N	E	G	E	I	
R	V	B	R	W	H	N	B	I	I	W	O	E	M	G	B	I	W	I	Z	H	O	I	B	W	M	E	I	C	I		

C(5) RP3 KRS(4) SA(CS) IUT2:2[XK/YP] IUT3:4[AWOL] RML[QOW] NSP

PT

D	E	L	I	Y	B	M	N	O	B	S	I	N	L	M	B	H	I	L	O	I	I	D	J	N	J	B	S	D	I		
A	C	N	W	B	J	N	K	I	D	I	G	C	W	H	D	W	C		K	I	W	V	N	W	Z	B	I	D	D	N	Y
I	N	L	M	B	V	N	W	Z		O	B	M	D	I	K	I	W	B	S	D	I	K	I	W	H	I	L	O	I	I	D

*PT

I	F	O	N	D	B	M	E	B	I	W	G	Z	V	F	I	O	I	H	B			C	J	A	E	D	D	E	L	I	H
O	R	E	S	R	E	D	O	I	V	C	K		M	I	U	C	R	I	O	A	E	O	B	C	J	B	M	I	H	C	C
I	H	R	N	B	M	B	O	I	V	F	U	N	W	L	M	E	W	H	D	N	V	E	H	I	E	B	N	W	S		

CT

J	M	X	W	B	L	A	Z	E	I	G	W	Z	K	A	H	Q	V	X	G	U	X	Z	V	Z	W	I	H	J	X		
I	I	Y	F	H	V	K	A	W	J	X	Y	I	P	U	L	O	I		I	U	P	P	K	O	O	L	T	J	J	K	C
W	Z	K	A	H	Y	K	T	A		H	W	X	J	V	A	W	P	L	U	J	U	A	V	O	Q	X	S	E	U	V	L

*CT

D	U	U	D	U	U	W	C	M	D	W	C	D	V	O	R	Y	V	W	I			G	M	B	W	O	I	S	Z	U	H
I	U	I	F	U	A	N	N	E	G	E	I		B	A	O	W	L	E	B	N	M	W	O	E	O	A	I	D	U	W	W
R	V	B	R	W	H	N	B	I	I	W	O	E	M	G	B	I	W	I	Z	H	O	I	B	W	M	E	I	C	I		

R(2)LS(9) to the TSP gives.

C(5) RP3 KRS(4) SA(CS) IUT2:2[XK/YP] SML[IHW] IUT3:4[OW/AL] TSP

PT

D	E	L	I	Y	B	M	N	O	B	S	I	N	L	M	B	H	I	L	O	I	I	D	J	N	J	B	S	D	I		
A	C	N	W	B	J	N	K	I	D	I	G	C	W	H	D	W	C		K	I	W	V	N	W	Z	B	I	D	D	N	Y
I	N	L	M	B	V	N	W	Z		O	B	M	D	I	K	I	W	B	S	D	I	K	I	W	H	I	L	O	I	I	D

*PT

I	F	O	N	D	B	M	E	B	I	W	G	Z	V	F	I	O	I	H	B			C	J	A	E	D	D	E	L	I	H
O	R	E	S	R	E	D	O	I	V	C	K		M	I	U	C	R	I	O	A	E	O	B	C	J	B	M	I	H	C	C
I	H	R	N	B	M	B	O	I	V	F	U	N	W	L	M	E	W	H	D	N	V	E	H	I	E	B	N	W	S		

CT

J	M	X	W	B	L	A	Z	E	I	G	W	Z	K	A	H	Q	V	X	G	U	X	Z	V	Z	W	I	H	J	X		
J	X	Y	I	P	U	L	O	I		I	U	P	P	K	O	O	L	T	J	J	K	C	I	I	Y	F	H	V	K	A	W
W	Z	K	A	H	Y	K	T	A		H	W	X	J	V	A	W	P	L	U	J	U	A	V	O	Q	X	S	E	U	V	L

*CT

D	U	U	D	U	U	W	C	M	D	W	C	D	V	O	R	Y	V	W	I			G	M	B	W	O	I	S	Z	U	H
O	E	O	A	I	D	U	W	W	I	U	I	F	U	A	N	N	E	G	E	I		B	A	O	W	L	E	B	N	M	W
R	V	B	R	W	H	N	B	I	I	W	O	E	M	G	B	I	W	I	Z	H	O	I	B	W	M	E	I	C	I		

C(5) RP3 KRS(4) SA(CS) IUT2 :2[DH/BA] TSP Digraph Analysis

PT

D	E	L	I	Y	B	M	N	O	B	S	I	N	L	M	B	H	I	L	O	I	I	D	J	N	J	B	S	D	I		
A	C	N	W	B	J	N	K	I	D	I	G	C	W	H	D	W	C		K	I	W	V	N	W	Z	B	I	D	D	N	Y
I	N	L	M	B	V	N	W	Z		O	B	M	D	I	K	I	W	B	S	D	I	K	I	W	H	I	L	O	I	I	D

*PT

I	F	O	N	D	B	M	E	B	I	W	G	Z	V	F	I	O	I	H	B			C	J	A	E	D	D	E	L	I	H
O	R	E	S	R	E	D	O	I	V	C	K		M	I	U	C	R	I	O	A	E	O	B	C	J	B	M	I	H	C	C
I	H	R	N	B	M	B	O	I	V	F	U	N	W	L	M	E	W	H	D	N	V	E	H	I	E	B	N	W	S		

CT

J	M	X	W	B	L	A	Z	E	I	G	W	Z	K	A	H	Q	V	X	G	U	X	Z	V	Z	W	I	H	J	X		
J	X	Y	I	P	U	L	O	I		I	U	P	P	K	O	O	L	T	J	J	K	C	I	I	Y	F	H	V	K	A	W
W	Z	K	A	H	Y	K	T	A		H	W	X	J	V	A	W	P	L	U	J	U	A	V	O	Q	X	S	E	U	V	L

*CT

D	U	U	D	U	U	W	C	M	D	W	C	D	V	O	R	Y	V	W	I			G	M	B	W	O	I	S	Z	U	H
O	E	O	A	I	D	U	W	W	I	U	I	F	U	A	N	N	E	G	E	I		B	A	O	W	L	E	B	N	M	W
R	V	B	R	W	H	N	B	I	I	W	O	E	M	G	B	I	W	I	Z	H	O	I	B	W	M	E	I	C	I		

Notes. This is interesting as the AWLO LBA's are read in a diagonal manner for the first time rather than horizontally. It is believed that this is the start of the transition to the vertical reads for all LBA's post C(6).

Letter Selection

CT

J	M	X	W	B	L	A	Z	E	I	G	W	Z	K	A	H	Q	V	X	G	U	X	Z	V	Z	W	I	H	J	X		
J	X	Y	I	P	U	L	O	I		I	U	P	P	K	O	O	L	T	J	J	K	C	I	I	Y	F	H	V	K	A	W
W	Z	K	A	H	Y	K	T	A		H	W	X	J	V	A	W	P	L	U	J	U	A	V	O	Q	X	S	E	U	V	L

Results

C(5) KRS(4) SA(CS)	K4 Key	Encryption
SSOT WTQS	XKPY AWLO	STBS TREE

Cryptogram 6

C(6) RP3 IRP KRS(5)

PT

E	F	M	J	P	C	N	Q	S	C	A	J	Q	M	N	C	I	J	M	S	J	J	E	L	Q	L	C	A	E	J		
R	J	X	W	Q	X	K	C	J	E	E	Q	P	B	D	Q	X	C	L	Q	R	J	E	J	H	D	X	I	E	X	D	
S	C	N	E	J	R	J	X	C	A	E	J	R	J	X	I	J	M	S	J	J	E	J	Q	M	N	C	W	Q	X	K	

*PT

D	L	B	F	E	E	F	M	J	I	J	G	S	Q	E	C	N	F	C	J	X	H	K	W	G	J	S	J	I	C		
N	J	V	D	Y	J	S	B	F	S	C	D	L	C	N	J	I	D	D	S	Y	F	A	Y	F	E	S	J	W	D	R	
J	I	Y	Q	C	N	C	S	J	W	G	V	Q	X	M	N	F	X	I	E	Q	W	F	I	J	F	C	Q	X	A		

CT

L	N	Z	X	C	M	B	K	F	J	H	X	K	R	B	I	U	W	Z	H	V	Z	K	W	K	X	J	I	L	Z		
J	V	T	T	R	S	S	M	O	L	L	R	D	J	J	P	G	I	W	R	B	X	L	Z	P	J	T	V	M	S	J	
I	X	Z	L	W	B	X	T	M	V	L	V	B	W	S	U	Z	A	F	V	W	M	X	K	R	B	I	P	R	O	B	

*CT

H	N	C	X	S	J	A	K	V	I	E	V	V	E	V	V	X	D	N	E	X	D	E	W	S	Y	P	W	X	J		
C	B	S	X	M	F	C	Q	N	X	S	F	S	B	J	E	V	X	X	J	V	J	G	V	B	Q	Q	F	H	F	J	
Y	W	C	Y	X	I	Q	C	J	J	X	S	F	N	H	C	J	X	J	K	I	S	J	C	X	N	F	J	D	J		

C(6) RP3 KRS(5) SA(SS) IUT2:2[VV/VB] RML[VVY] IUT3:4[WSZ][H(W)O] NSP

PT

Q	L	C	A	E	J			E	F	M	J	P	C	N	Q	S	C	A	J	Q	M	N	C	I	J	M	S	J	J	E	L
P	B	D	Q	X	C	L	Q	R	J	E	J	H	D	X	I	E	X	D		R	J	X	W	Q	X	K	C	J	E	E	Q
S	C	N	E	J	R	J	X	C	A	E	J	R	J	X	I	J	M	S	J	J	E	J	Q	M	N	C	W	Q	X	K	

*PT

D	L	B	F	E	E	F	M	J	I	J	G	S	Q	E	C	N	F	C	J	X	H	K	W	G	J	S	J	I	C		
L	C	N	J	I	D	D	S	Y	F	A	Y	F	E	S	J	W	D	R		N	J	V	D	Y	J	S	B	F	S	C	D
J	F	C	Q	X	A			J	I	Y	Q	C	N	C	S	J	W	G	V	Q	X	M	N	F	X	I	E	Q	W	F	I

CT

K	X	J	I	L	Z			L	N	Z	X	C	M	B	K	F	J	H	X	K	R	B	I	U	W	Z	H	V	Z	K	W
D	J	J	P	G	I	W	R	B	X	L	Z	P	J	T	V	M	S	J		J	V	T	T	R	S	S	M	O	L	L	R
I	X	Z	L	W	B	X	T	M	V	L	V	B	W	S	U	Z	A	F	V	W	M	X	K	R	B	I	P	R	O	B	

*CT

H	N	C	X	S	J	A	K	V	I	E	V	V	E	V	V	X	D	N	E	X	D	E	W	S	Y	P	W	X	J		
S	B	J	E	V	X	X	J	V	J	G	V	B	Q	Q	F	H	F	J		C	B	S	X	M	F	C	Q	N	X	S	F
X	N	F	J	D	J			Y	W	C	Y	X	I	Q	C	J	J	X	S	F	N	H	C	J	X	J	K	I	S	J	C

R(2)RS(3) to the TSP gives.

C(6) RP3 KRS(5) SA(SS) IUT2 :2[VV/VB] IUT3:4[HW/OR][ZS/OS] RML[WJJ] TSP

PT

Q	L	C	A	E	J			E	F	M	J	P	C	N	Q	S	C	A	J	Q	M	N	C	I	J	M	S	J	J	E	L
E	E	Q	P	B	D	Q	X	C	L	Q	R	J	E	J	H	D	X	I	E	X	D		R	J	X	W	Q	X	K	C	J
S	C	N	E	J	R	J	X	C	A	E	J	R	J	X	I	J	M	S	J	J	E	J	Q	M	N	C	W	Q	X	K	

*PT

D	L	B	F	E	E	F	M	J	I	J	G	S	Q	E	C	N	F	C	J	X	H	K	W	G	J	S	J	I	C		
S	C	D	L	C	N	J	I	D	D	S	Y	F	A	Y	F	E	S	J	W	D	R		N	J	V	D	Y	J	S	B	F
J	F	C	Q	X	A			J	I	Y	Q	C	N	C	S	J	W	G	V	Q	X	M	N	F	X	I	E	Q	W	F	I

CT

K	X	J	I	L	Z			L	N	Z	X	C	M	B	K	F	J	H	X	K	R	B	I	U	W	Z	H	V	Z	K	W
L	L	R	D	J	J	P	G	I	W	R	B	X	L	Z	P	J	T	V	M	S	J		J	V	T	T	R	S	S	M	O
I	X	Z	L	W	B	X	T	M	V	L	V	B	W	S	U	Z	A	F	V	W	M	X	K	R	B	I	P	R	O	B	

*CT

H	N	C	X	S	J	A	K	V	I	E	V	V	E	V	V	X	D	N	E	X	D	E	W	S	Y	P	W	X	J		
X	S	F	S	B	J	E	V	X	X	J	V	J	G	V	B	Q	Q	F	H	F	J		C	B	S	X	M	F	C	Q	N
X	N	F	J	D	J			Y	W	C	Y	X	I	Q	C	J	J	X	S	F	N	H	C	J	X	J	K	I	S	J	C

C(6) RP3 TSP KRS(5) SA(SS) IUT2:2[VV/VB] Digraph Analysis

PT

Q	L	C	A	E	J			E	F	M	J	P	C	N	Q	S	C	A	J	Q	M	N	C	I	J	M	S	J	J	E	L
E	E	Q	P	B	D	Q	X	C	L	Q	R	J	E	J	H	D	X	I	E	X	D		R	J	X	W	Q	X	K	C	J
S	C	N	E	J	R	J	X	C	A	E	J	R	J	X	I	J	M	S	J	J	E	J	Q	M	N	C	W	Q	X	K	

*PT

D	L	B	F	E	E	F	M	J	I	J	G	S	Q	E	C	N	F	C	J	X	H	K	W	G	J	S	J	I	C		
S	C	D	L	C	N	J	I	D	D	S	Y	F	A	Y	F	E	S	J	W	D	R		N	J	V	D	Y	J	S	B	F
J	F	C	Q	X	A			J	I	Y	Q	C	N	C	S	J	W	G	V	Q	X	M	N	F	X	I	E	Q	W	F	I

CT

K	X	J	I	L	Z			L	N	Z	X	C	M	B	K	F	J	H	X	K	R	B	I	U	W	Z	H	V	Z	K	W
L	L	R	D	J	J	P	G	I	W	R	B	X	L	Z	P	J	T	V	M	S	J		J	V	T	T	R	S	S	M	O
I	X	Z	L	W	B	X	T	M	V	L	V	B	W	S	U	Z	A	F	V	W	M	X	K	R	B	I	P	R	O	B	

*CT

H	N	C	X	S	J	A	K	V	I	E	V	V	E	V	V	X	D	N	E	X	D	E	W	S	Y	P	W	X	J		
X	S	F	S	B	J	E	V	X	X	J	V	J	G	V	B	Q	Q	F	H	F	J		C	B	S	X	M	F	C	Q	N
X	N	F	J	D	J			Y	W	C	Y	X	I	Q	C	J	J	X	S	F	N	H	C	J	X	J	K	I	S	J	C

Notes. This is the last cryptogram before we flip to reading all LBA's in the vertical plane. IT is not known at this stage whether this cryptogram is simply wrong or whether the puzzle maker has flipped the W in the IUT3:4 scheme to an M to imply a reversal of sorts. The final 4 letters for the ZS and OS LBA's are now read in the fully vertical configuration.

Letter Selection

CT

K	X	J	I	L	Z			L	N	Z	X	C	M	B	K	F	J	H	X	K	R	B	I	U	W	Z	H	V	Z	K	W
L	L	R	D	J	J	P	G	I	W	R	B	X	L	Z	P	J	T	V	M	S	J		J	V	T	T	R	S	S	M	O
I	X	Z	L	W	B	X	T	M	V	L	V	B	W	S	U	Z	A	F	V	W	M	X	K	R	B	I	P	R	O	B	

Results

C(6) KRS(5) SA(SS)	K4 Key	Encryption
JQSS EKZZ	HWOR ZSOS	TLEA DSTO

Reflections on BERLIN

If we briefly skip ahead and examine the relationship between the IUT scheme as we understand it so far and the plain text for BERLIN CLOCK then we can see that we are in slightly different territory. We know that the rail markers have changed from a horizontal configuration into a vertical one and we know from the previous 2 cryptograms that our read order appears to be transitioning to the vertical plane. Whether or not this has made a difference to the general scheme or not is yet to be seen, but as we have the plain text for positions 64 through to 75 then we are in a position to check our reasoning. Interestingly a coincidence exists by which Rail 5 was actually in the same position as Rail 5 in the previous papers scheme and rather amazingly this allowed for a resolution of the string ERLIN CLO in the correct position. A re-examination of the evidence now that we have the specific rail markers and the fact that they are vertical markers rather than horizontal leads to the following observation for the exact same rows under the same degree of right shifting as the original paper. The only thing that has been altered is the row alignment and read direction, which is now in the vertical plane.

C(9) RP5 KRS(8) CT with relevant row shifts applied to achieve TSP.

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
I	Q	K	X	U	A	R	P	E	Z	U	N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M
I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W

The K4 string for Cryptogram 9 is YPVT TMZF, which encrypted with CXUE ILMN gives ERLI NCLO. This would imply that the change of markers to the vertical direction have also indicated a change of direction in which we should be reading off the letter blocks. This was perhaps hinted at in Cryptogram 5 with the read order of the letters AWLO.

It was mentioned previously that Rail 5 has no reciprocal symmetry position, but there are some intriguing properties of this row alignment that show how the principle of ‘invariance under transformation’ has been utilised for this case. For this first case (i.e. what would have been the CS case in the symmetry scheme) we can see that the letter relationship IL MN has two positions in which it is invariant in the horizontal plane as shown here.

IUT[NM/IL] position 1

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
I	Q	K	X	U	A	R	P	E	Z	U	N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M
G	W	J	W	M	D	H	V	R	I	W	I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N

IUT[NM/IL] position 2

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
I	Q	K	X	U	A	R	P	E	Z	U	N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M
W	I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I

There is also an interesting glimpse of how the complexity of the puzzle may be progressing as we move into these final rails. We have already discussed the invariance under transformation of the ILMN letters, but now when we look at what may be happening in terms of our expected row symmetry we find some unexpected differences. Taking the proposed TSP for cryptogram 9 we can identify the following top and bottom row letters for the rail that are of interest, which are:

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
I	Q	K	X	U	A	R	P	E	Z	U	N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M
I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W

Further analysis shows that there are only two possible settings for row symmetry in this position, which are for ENI and EZI as shown here.

ENI

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
U	M	Y	P	K	N	J	N	M	K	V	M	I	Q	K	X	U	A	R	P	E	Z	U	N	H	O	X	Z	Z	I	N	F
I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W

EZI

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M	I	Q	K	X	U	A	R	P	E	Z	U
I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W

We would appear to be dealing with the case for EZI in this instance as there is yet a third configuration of ILNM that reveals itself here in the bottom left hand corner as shown here.

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M	I	Q	K	X	U	A	R	P	E	Z	U
I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W

This invariance type whereby letters that are read in the horizontal plane for the NSP and then in the vertical plane for the TSP are classified as the IUT H:V type.

Degrees of freedom

The scientific method would typically urge caution at this point as we should always be vigilant for false positives given the number of numerical coincidences that abound in this universe. The question we should always ask is, 'no matter how good these results are, what are the statistical chances of them being coincidental'. An answer to this question was sought through programmatic means and the answer was much higher than might be expected for comfort. The occurrence of symmetrical letter occurrences in matching groups of three within each rail position are actually much higher than intuition would suggest.

The puzzle maker would appear to have reduced at least some of the available degrees of freedom by the selection of configurations that give rise to consistent phenomena in the NSP and TSP positions, but there is still a sense that the smoking gun is still at large in terms of being sure that each NSP and TSP position is unique. What is needed is a programmatic way to select these alignments in such a way where we can be certain that no other possibilities exist. It may well be that we need to make an additional observation to close that gap or conduct a re-appraisal of the data already presented. Irrespective of these insecurities about the uniqueness of the NSP and TSP positions, we shall now continue to C(7) and beyond.

Part 5

Cryptogram Solves C(7 – 12) Detailed Analysis

Cryptogram 7

C(7) RP4 IRP KRS(6)

PT

U	W	F	E	M	D	Q	L	A	E	E	X	P	U	D	Q	U	Z	L	X	L	A	N	L	J	M	A	E	X			
D	Q	L	X	U	F	D	T	I	G	Z	B	E	R	F	L	L	G	Z	B	D	Q	U	Z	N	K	?	E	H	S	A	
R	E	T	E	N	Q	R	W	H	F	E	W	U	M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E	

*PT

U	D	P	G	F	D	E	D	G	W	W	B	U	Z	Y	U	F	U	H	W	L	Q	E	P	F	D	Q	G	D	C	E	
F	F	U	H	W	L	?	D	Q	L	B	R	F	L	J	D	Q	L	L	G	A	D	Q	F	X	G	N	Z	L	D		
U	I	M	U	L	W	J	T	D	Q	L	U	Z	M	E	A	X	G	D	U	E	Z	P	G	F	N	G	D	Q	L	A	

CT

A	S	W	X	W	L	Q	G	N	D	J	Q	G	A	J	C	Z	L	E	Q	X	N	M	X	M	L	R	Q	L			
L	I	J	X	A	U	C	M	L	U	X	L	Q	Z	W	F	F	D	D	A	D	Y	J	E	Q	P	?	E	H	S	A	
R	E	T	E	N	Q	R	W	H	F	E	W	U	M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E	

*CT

Y	M	C	V	R	J	L	L	Q	O	P	L	D	O	B	Y	N	P	Q	S	S	K	L	D	N	M	K	V	Z	I	L	
N	N	P	Q	S	S	?	J	K	X	I	C	M	K	O	J	K	X	X	U	G	N	C	M	T	Q	K	A	W	N		
D	U	X	Y	X	O	V	G	Z	K	W	Y	S	K	L	J	G	N	J	Y	M	A	C	V	R	Z	N	L	R	Z	G	

C(7) RP4 KRS(6) SA(CS) IUT H:V [RH/XU] IUT3:4[PMW] SML[NJL] NSP

PT

U	W	F	E	M	D	Q	L	A	E	E	X	P	U	D	Q	U	Z	L	X	L	A	N	L	J	M	A	E	X			
Z	B	E	R	F	L	L	G	Z	B	D	Q	U	Z	N	K	?	E	H	S	A		D	Q	L	X	U	F	D	T	I	G
M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E		R	E	T	E	N	Q	R	W	H	F	E	W	U

*PT

W	L	Q	E	P	F	D	Q	G	D	C	E		U	D	P	G	F	D	E	D	G	W	W	B	U	Z	Y	U	F	U	H
Q	F	X	G	N	Z	L	D			F	F	U	H	W	L	?	D	Q	L	B	R	F	L	J	D	Q	L	L	G	A	D
U	I	M	U	L	W	J	T	D	Q	L	U	Z	M	E	A	X	G	D	U	E	Z	P	G	F	N	G	D	Q	L	A	

CT

A	S	W	X	W	L	Q	G	N	D	J	Q	G	A	J	C	Z	L	E	Q	X	N	M	X	M	L	R	Q	L			
X	L	Q	Z	W	F	F	D	D	A	D	Y	J	E	Q	P	?	E	H	S	A		L	I	J	X	A	U	C	M	L	U
M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E		R	E	T	E	N	Q	R	W	H	F	E	W	U

*CT

S	S	K	L	D	N	M	K	V	Z	I	L		Y	M	C	V	R	J	L	L	Q	O	P	L	D	O	B	Y	N	P	Q
C	M	T	Q	K	A	W	N			N	N	P	Q	S	S	?	J	K	X	I	C	M	K	O	J	K	X	X	U	G	N
D	U	X	Y	X	O	V	G	Z	K	W	Y	S	K	L	J	G	N	J	Y	M	A	C	V	R	Z	N	L	R	Z	G	

R(2)RS(8) to the TSP gives.

C(7) RP4 KRS(6) SA(CS) IUT H :V[RH/XU] IUT3 :4[MW/PN] RML[LXY] TSP

PT

U	W	F	E	M	D	Q	L	A	E	E	X	P	U	D	Q	U	Z	L	X	L	A	N	L	J	M	A	E	X			
L	X	U	F	D	T	I	G	Z	B	E	R	F	L	L	G	Z	B	D	Q	U	Z	N	K	?	E	H	S	A		D	Q
M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E		R	E	T	E	N	Q	R	W	H	F	E	W	U

*PT

W	L	Q	E	P	F	D	Q	G	D	C	E		U	D	P	G	F	D	E	D	G	W	W	B	U	Z	Y	U	F	U	H
		F	F	U	H	W	L	?	D	Q	L	B	R	F	L	J	D	Q	L	L	G	A	D	Q	F	X	G	N	Z	L	D
U	I	M	U	L	W	J	T	D	Q	L	U	Z	M	E	A	X	G	D	U	E	Z	P	G	F	N	G	D	Q	L	A	

CT

A	S	W	X	W	L	Q	G	N	D	J	Q	G	A	J	C	Z	L	E	Q	X	N	M	X	M	L	R	Q	L			
J	X	A	U	C	M	L	U	X	L	Q	Z	W	F	F	D	D	A	D	Y	J	E	Q	P	?	E	H	S	A		L	I
M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E		R	E	T	E	N	Q	R	W	H	F	E	W	U

*CT

S	S	K	L	D	N	M	K	V	Z	I	L		Y	M	C	V	R	J	L	L	Q	O	P	L	D	O	B	Y	N	P	Q
		N	N	P	Q	S	S	?	J	K	X	I	C	M	K	O	J	K	X	X	U	G	N	C	M	T	Q	K	A	W	N
D	U	X	Y	X	O	V	G	Z	K	W	Y	S	K	L	J	G	N	J	Y	M	A	C	V	R	Z	N	L	R	Z	G	

C(7) RP4 KRS(6) SA(CS) IUT H :V[RH/XU] IUT3 :4[MW/PN] RML[LXY] TSP

Digraph Analysis

PT

U	W	F	E	M	D	Q	L	A	E	E	X	P	U	D	Q	U	Z	L	X	L	A	N	L	J	M	A	E	X			
L	X	U	F	D	T	I	G	Z	B	E	R	F	L	L	G	Z	B	D	Q	U	Z	N	K	?	E	H	S	A		D	Q
M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E		R	E	T	E	N	Q	R	W	H	F	E	W	U

*PT

W	L	Q	E	P	F	D	Q	G	D	C	E		U	D	P	G	F	D	E	D	G	W	W	B	U	Z	Y	U	F	U	H
		F	F	U	H	W	L	?	D	Q	L	B	R	F	L	J	D	Q	L	L	G	A	D	Q	F	X	G	N	Z	L	D
U	I	M	U	L	W	J	T	D	Q	L	U	Z	M	E	A	X	G	D	U	E	Z	P	G	F	N	G	D	Q	L	A	

CT

A	S	W	X	W	L	Q	G	N	D	J	Q	G	A	J	C	Z	L	E	Q	X	N	M	X	M	L	R	Q	L			
J	X	A	U	C	M	L	U	X	L	Q	Z	W	F	F	D	D	A	D	Y	J	E	Q	P	?	E	H	S	A		L	I
M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E		R	E	T	E	N	Q	R	W	H	F	E	W	U

*CT

S	S	K	L	D	N	M	K	V	Z	I	L		Y	M	C	V	R	J	L	L	Q	O	P	L	D	O	B	Y	N	P	Q
		N	N	P	Q	S	S	?	J	K	X	I	C	M	K	O	J	K	X	X	U	G	N	C	M	T	Q	K	A	W	N
D	U	X	Y	X	O	V	G	Z	K	W	Y	S	K	L	J	G	N	J	Y	M	A	C	V	R	Z	N	L	R	Z	G	

Notes. We can see that there has been an immediate jump in the complexity of the IUT scheme. Here the horizontal relationship XU/RH transforms into XU/RH in the vertical plane from rail 3 to rail 1. It will be noticed that the RML is now present in the TSP rather than the NSP.

Letter Selection

CT

A	S	W	X	W	L	Q	G	N	D	J	Q	G	A	J	C	Z	L	E	Q	X	N	M	X	M	L	R	Q	L			
J	X	A	U	C	M	L	U	X	L	Q	Z	W	F	F	D	D	A	D	Y	J	E	Q	P	?	E	H	S	A		L	I
M	H	H	P	M	W	A	Y	K	K	C	A	Z	N	S	F	F	E		R	E	T	E	N	Q	R	W	H	F	E	W	U

Results

C(7) KRS(6) SA(CS)	K4 Key	Encryption
WATJ KLUD	MWPN RHXU	ITAC RONO

Cryptogram 8

C(8) RP4 IRP KRS(7)

PT

V	X	G	F	N	E	U	M	B	F	F	Z	T	V	E	U	V	K	M	Z	M	B	Q	M	L	N	B	F	Z			
E	U	M	Z	V	G	E	O	J	H	K	C	F	Y	G	M	M	H	K	C	E	U	V	K	Q	R	?	F	I	A	B	
Y	F	O	F	Q	U	Y	X	I	G	F	X	V	N	I	I	T	N	X	B	P	R	R	D	B	K	Q	A	G	G	F	

*PT

V	E	T	H	G	E	F	E	H	X	X	C	V	K	P	V	G	V	I	X	M	U	F	T	G	E	U	H	E	D	F	
G	G	V	I	X	M	?	E	U	M	C	Y	G	M	L	E	U	M	M	H	B	E	U	G	Z	H	Q	K	M	E		
V	J	N	V	M	X	L	O	E	U	M	V	K	N	F	B	Z	H	E	V	F	K	T	H	G	Q	H	E	U	M	B	

CT

B	A	X	Z	X	M	U	H	Q	E	L	U	H	B	L	D	K	M	F	U	Z	Q	N	Z	N	M	Y	U	M			
M	J	L	Z	B	V	D	N	M	V	Z	M	U	K	X	G	G	E	E	B	E	P	L	F	U	T	?	F	I	A	B	
Y	F	O	F	Q	U	Y	X	I	G	F	X	V	N	I	I	T	N	X	B	P	R	R	D	B	K	Q	A	G	G	F	

*CT

P	N	D	W	Y	L	M	M	U	S	T	M	E	S	C	P	Q	T	U	A	A	R	M	E	Q	N	R	W	K	J	M	
Q	Q	T	U	A	A	?	L	R	Z	J	D	N	R	S	L	R	Z	Z	V	H	Q	D	N	O	U	R	B	X	Q		
E	V	Z	P	Z	S	W	H	K	R	X	P	A	R	M	L	H	Q	L	P	N	B	D	W	Y	K	Q	M	Y	K	H	

C(8) RP4 KRS(7) SA(SS) IUT2:2[BF/BG] RML[LZM] IUT3:4[TNI][ABV] NSP

PT

V	X	G	F	N	E	U	M	B	F	F	Z	T	V	E	U	V	K	M	Z	M	B	Q	M	L	N	B	F	Z			
Z	V	G	E	O	J	H	K	C	F	Y	G	M	M	H	K	C	E	U	V	K	Q	R	?	F	I	A	B		E	U	M
D	B	K	Q	A	G	G	F		Y	F	O	F	Q	U	Y	X	I	G	F	X	V	N	I	I	T	N	X	B	P	R	R

*PT

T	G	E	U	H	E	D	F		V	E	T	H	G	E	F	E	H	X	X	C	V	K	P	V	G	V	I	X	M	U	F
I	X	M	?	E	U	M	C	Y	G	M	L	E	U	M	M	H	B	E	U	G	Z	H	Q	K	M	E			G	G	V
V	J	N	V	M	X	L	O	E	U	M	V	K	N	F	B	Z	H	E	V	F	K	T	H	G	Q	H	E	U	M	B	

CT

B	A	X	Z	X	M	U	H	Q	E	L	U	H	B	L	D	K	M	F	U	Z	Q	N	Z	N	M	Y	U	M			
Z	B	V	D	N	M	V	Z	M	U	K	X	G	G	E	E	B	E	P	L	F	U	T	?	F	I	A	B		M	J	L
D	B	K	Q	A	G	G	F		Y	F	O	F	Q	U	Y	X	I	G	F	X	V	N	I	I	T	N	X	B	P	R	R

*CT

E	Q	N	R	W	K	J	M		P	N	D	W	Y	L	M	M	U	S	T	M	E	S	C	P	Q	T	U	A	A	R	M
U	A	A	?	L	R	Z	J	D	N	R	S	L	R	Z	Z	V	H	Q	D	N	O	U	R	B	X	Q			Q	Q	T
E	V	Z	P	Z	S	W	H	K	R	X	P	A	R	M	L	H	Q	L	P	N	B	D	W	Y	K	Q	M	Y	K	H	

R(2)LS(5) to the TSP gives.

C(8) RP4 KRS(7) SA(SS) IUT2:2[BF/BG] RML[EGV] IUT3:4[BN/TI][AV/BP] TSP

PT

V	X	G	F	N	E	U	M	B	F	F	Z	T	V	E	U	V	K	M	Z	M	B	Q	M	L	N	B	F	Z			
J	H	K	C	F	Y	G	M	M	H	K	C	E	U	V	K	Q	R	?	F	I	A	B		E	U	M	Z	V	G	E	O
D	B	K	Q	A	G	G	F		Y	F	O	F	Q	U	Y	X	I	G	F	X	V	N	I	I	T	N	X	B	P	R	R

*PT

T	G	E	U	H	E	D	F		V	E	T	H	G	E	F	E	H	X	X	C	V	K	P	V	G	V	I	X	M	U	F
U	M	C	Y	G	M	L	E	U	M	M	H	B	E	U	G	Z	H	Q	K	M	E			G	G	V	I	X	M	?	E
V	J	N	V	M	X	L	O	E	U	M	V	K	N	F	B	Z	H	E	V	F	K	T	H	G	Q	H	E	U	M	B	

CT

B	A	X	Z	X	M	U	H	Q	E	L	U	H	B	L	D	K	M	F	U	Z	Q	N	Z	N	M	Y	U	M			
M	V	Z	M	U	K	X	G	G	E	E	B	E	P	L	F	U	T	?	F	I	A	B		M	J	L	Z	B	V	D	N
D	B	K	Q	A	G	G	F		Y	F	O	F	Q	U	Y	X	I	G	F	X	V	N	I	I	T	N	X	B	P	R	R

*CT

E	Q	N	R	W	K	J	M		P	N	D	W	Y	L	M	M	U	S	T	M	E	S	C	P	Q	T	U	A	A	R	M
R	Z	J	D	N	R	S	L	R	Z	Z	V	H	Q	D	N	O	U	R	B	X	Q			Q	Q	T	U	A	A	?	L
E	V	Z	P	Z	S	W	H	K	R	X	P	A	R	M	L	H	Q	L	P	N	B	D	W	Y	K	Q	M	Y	K	H	

C(8) RP4 KRS(7) SA(SS) IUT2:2[BF/BG] RML[EGV] IUT3:4[BN/TI][AV/BP] TSP

Digraph Analysis

PT

V	X	G	F	N	E	U	M	B	F	F	Z	T	V	E	U	V	K	M	Z	M	B	Q	M	L	N	B	F	Z			
J	H	K	C	F	Y	G	M	M	H	K	C	E	U	V	K	Q	R	?	F	I	A	B		E	U	M	Z	V	G	E	O
D	B	K	Q	A	G	G	F		Y	F	O	F	Q	U	Y	X	I	G	F	X	V	N	I	I	T	N	X	B	P	R	R

*PT

T	G	E	U	H	E	D	F		V	E	T	H	G	E	F	E	H	X	X	C	V	K	P	V	G	V	I	X	M	U	F
U	M	C	Y	G	M	L	E	U	M	M	H	B	E	U	G	Z	H	Q	K	M	E			G	G	V	I	X	M	?	E
V	J	N	V	M	X	L	O	E	U	M	V	K	N	F	B	Z	H	E	V	F	K	T	H	G	Q	H	E	U	M	B	

CT

B	A	X	Z	X	M	U	H	Q	E	L	U	H	B	L	D	K	M	F	U	Z	Q	N	Z	N	M	Y	U	M			
M	V	Z	M	U	K	X	G	G	E	E	B	E	P	L	F	U	T	?	F	I	A	B		M	J	L	Z	B	V	D	N
D	B	K	Q	A	G	G	F		Y	F	O	F	Q	U	Y	X	I	G	F	X	V	N	I	I	T	N	X	B	P	R	R

*CT

E	Q	N	R	W	K	J	M		P	N	D	W	Y	L	M	M	U	S	T	M	E	S	C	P	Q	T	U	A	A	R	M
R	Z	J	D	N	R	S	L	R	Z	Z	V	H	Q	D	N	O	U	R	B	X	Q			Q	Q	T	U	A	A	?	L
E	V	Z	P	Z	S	W	H	K	R	X	P	A	R	M	L	H	Q	L	P	N	B	D	W	Y	K	Q	M	Y	K	H	

Notes. This cryptogram is interesting as the puzzle maker has made sure the TSP makes full use of the 8 available digraphs for this alignment. The PN digraphs were highlighted in the NSP as they hint at an interesting alternative method of derivation for this alignment.

Letter Selection

CT

B	A	X	Z	X	M	U	H	Q	E	L	U	H	B	L	D	K	M	F	U	Z	Q	N	Z	N	M	Y	U	M			
M	V	Z	M	U	K	X	G	G	E	E	B	E	P	L	F	U	T	?	F	I	A	B		M	J	L	Z	B	V	D	N
D	B	K	Q	A	G	G	F		Y	F	O	F	Q	U	Y	X	I	G	F	X	V	N	I	I	T	N	X	B	P	R	R

Results

C(8) KRS(7) SA(SS)	K4 Key	Encryption
IAWI NFBN	PBAV BNTI	MITE ROFB

Cryptogram 9

As mentioned previously the rows do not sit cleanly either side of the axis of symmetry for rail position 5 so the reciprocal rows are omitted for Cryptograms 9 and 10.

C(9) RP5 IRP KRS(8)

PT

C	F	V	H	N	T	N	R	F	D	H	N	T	N	R	M	N	U	C	N	N	H	N	W	U	V	F	K	W	R	P	
F	N	H	Q	G	C	F	D	Q	G	P	C	H	N	L	G	R	M	H	O	N	H	F	S	Z	I	D	N	C	F	O	G
H	Z	G	O	Z	D	M	N	H	E	I	C	I	F	Z	D	H	Z	G	O	Z	D	F	V	N	C	N	K	I	W	R	H

CT

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
M	Y	P	K	N	J	N	M	K	V	M	I	Q	K	X	U	A	R	P	E	Z	U	N	H	O	X	Z	Z	I	N	F	U
N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W	I

C(9) RP5 KRS(8) RML[EZI] IUT H:V[IL/MN] IUT3:4[XUE] NSP

PT

C	F	V	H	N	T	N	R	F	D	H	N	T	N	R	M	N	U	C	N	N	H	N	W	U	V	F	K	W	R	P	
F	S	Z	I	D	N	C	F	O	G	F	N	H	Q	G	C	F	D	Q	G	P	C	H	N	L	G	R	M	H	O	N	H
H	H	Z	G	O	Z	D	M	N	H	E	I	C	I	F	Z	D	H	Z	G	O	Z	D	F	V	N	C	N	K	I	W	R

CT

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M	I	Q	K	X	U	A	R	P	E	Z	U
I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W

R(2)RS(11) to the TSP gives.

C(9) RP5 KRS(8) RML[HNH][NND] IUT H:V[IL/MN] IUT3:4[CX/UE] TSP

PT

C	F	V	H	N	T	N	R	F	D	H	N	T	N	R	M	N	U	C	N	N	H	N	W	U	V	F	K	W	R	P	
C	H	N	L	G	R	M	H	O	N	H	F	S	Z	I	D	N	C	F	O	G	F	N	H	Q	G	C	F	D	Q	G	P
H	H	Z	G	O	Z	D	M	N	H	E	I	C	I	F	Z	D	H	Z	G	O	Z	D	F	V	N	C	N	K	I	W	R

CT

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
I	Q	K	X	U	A	R	P	E	Z	U	N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M
I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W

Letter Selection

CT

M	R	Y	Q	K	E	R	A	U	Z	Q	Z	E	K	C	X	Y	D	I	Z	K	U	R	P	T	E	M	S	T	B	E	
I	Q	K	X	U	A	R	P	E	Z	U	N	H	O	X	Z	Z	I	N	F	U	M	Y	P	K	N	J	N	M	K	V	M
I	N	R	M	D	I	V	C	G	V	R	Z	H	C	V	N	G	L	E	F	N	G	W	J	W	M	D	H	V	R	I	W

RP2 TSP KRS(3) SA(SS) IUT[DH/BA] Results

C(9) KRS(8) SA(?)	K4 Key	Encryption
YPVT TMZF	CXUE ILMN	ERLI NCLO

Cryptogram 10

As mentioned previously the rows do not sit cleanly either side of the axis of symmetry for rail position 5 so the reciprocal rows are omitted for Cryptograms 9 and 10.

C(10) RP5 IRP KRS(9)

PT

D	G	W	I	Q	O	Q	Y	G	E	I	Q	O	Q	Y	N	Q	V	D	Q	Q	I	Q	X	V	W	G	R	X	Y	T	
G	Q	I	U	H	D	G	E	U	H	T	D	I	Q	M	H	Y	N	I	S	Q	I	G	A	K	J	E	Q	D	G	S	H
I	K	H	S	K	E	N	Q	I	F	J	D	J	G	K	E	I	K	H	S	K	E	G	W	Q	D	Q	R	J	X	Y	I

CT

N	Y	P	U	R	F	Y	B	V	K	U	K	F	R	D	Z	P	E	J	K	R	V	Y	T	O	F	N	A	O	C	F	
N	P	T	R	Q	L	Q	N	R	W	N	J	U	R	Z	V	B	Y	T	F	K	V	Q	I	S	Z	K	K	J	Q	G	V
Q	Y	N	E	J	W	D	H	W	Y	K	I	D	W	Q	H	M	F	G	Q	H	X	L	X	N	E	I	W	Y	J	X	J

C(10) RP5 KRS(9) RML[FRQ] IUT H:V[IL/MN] IUT3:4[SUK] NSP

PT

D	G	W	I	Q	O	Q	Y	G	E	I	Q	O	Q	Y	N	Q	V	D	Q	Q	I	Q	X	V	W	G	R	X	Y	T	
T	D	I	Q	M	H	Y	N	I	S	Q	I	G	A	K	J	E	Q	D	G	S	H	G	Q	I	U	H	D	G	E	U	H
E	G	W	Q	D	Q	R	J	X	Y	I	I	K	H	S	K	E	N	Q	I	F	J	D	J	G	K	E	I	K	H	S	K

CT

N	Y	P	U	R	F	Y	B	V	K	U	K	F	R	D	Z	P	E	J	K	R	V	Y	T	O	F	N	A	O	C	F	
N	J	U	R	Z	V	B	Y	T	F	K	V	Q	I	S	Z	K	K	J	Q	G	V	N	P	T	R	Q	L	Q	N	R	W
X	L	X	N	E	I	W	Y	J	X	J	Q	Y	N	E	J	W	D	H	W	Y	K	I	D	W	Q	H	M	F	G	Q	H

R(2)LS(5) to the TSP gives.

C(10) RP5 KRS(9) RML[HNH] IUT H:V[IL/MN] IUT3:4[SK/CU] TSP

PT

D	G	W	I	Q	O	Q	Y	G	E	I	Q	O	Q	Y	N	Q	V	D	Q	Q	I	Q	X	V	W	G	R	X	Y	T	
H	Y	N	I	S	Q	I	G	A	K	J	E	Q	D	G	S	H	G	Q	I	U	H	D	G	E	U	H	T	D	I	Q	M
E	G	W	Q	D	Q	R	J	X	Y	I	I	K	H	S	K	E	N	Q	I	F	J	D	J	G	K	E	I	K	H	S	K

CT

N	Y	P	U	R	F	Y	B	V	K	U	K	F	R	D	Z	P	E	J	K	R	V	Y	T	O	F	N	A	O	C	F	
V	B	Y	T	F	K	V	Q	I	S	Z	K	K	J	Q	G	V	N	P	T	R	Q	L	Q	N	R	W	N	J	U	R	Z
X	L	X	N	E	I	W	Y	J	X	J	Q	Y	N	E	J	W	D	H	W	Y	K	I	D	W	Q	H	M	F	G	Q	H

Letter Selection

CT

N	Y	P	U	R	F	Y	B	V	K	U	K	F	R	D	Z	P	E	J	K	R	V	Y	T	O	F	N	A	O	C	F	
V	B	Y	T	F	K	V	Q	I	S	Z	K	K	J	Q	G	V	N	P	T	R	Q	L	Q	N	R	W	N	J	U	R	Z
X	L	X	N	E	I	W	Y	J	X	J	Q	Y	N	E	J	W	D	H	W	Y	K	I	D	W	Q	H	M	F	G	Q	H

RP2 TSP KRS(3) SA(SS) IUT[DH/BA] Results

C(10) KRS(9) SA(?)	K4 Key	Encryption
PKWG DKZX	SKCU LIMN	CKSB RILL

Cryptogram 11

As mentioned at the head of this document. Rail position 6 is taken as being 1 row down from the position that its marker would imply in order to break the symmetry relationship with Rail Position 4.

C(11) RP6 IRP KRS(10)

PT

J	J	Z	M	R	U	?	H	X	U	F	O	J	U	Q	H	X	U	U	L	E	H	X	J	Y	L	W	P	U	H		
Z	N	V	Z	U	R	Q	B	H	X	U	Z	P	V	I	E	Y	L	H	Z	I	P	A	L	J	W	L	H	X	U	E	
U	Q	L	P	Q	H	E	L	P	J	Y	Z	H	H	U	Q	O	P	Q	U	E	W	E	O	O	P	Q	H	I	L	P	O

*PT

H	X	U	V	R	L	Y	U	H	I	V	R	Z	N	D	U	E	M	O	H	G	E	U	J	U	P	H	R	F	Q	U	H
Z	R	J	I	V	H	X	U	E	I	I	Y	A	Z	H	X	Z	P	U	Y	U	E	W	U	Q	V	E	I	Y			
H	X	U	Y	Z	J	H	B	N	L	P	F	I	O	J	U	U	L	P	F	H	X	Z	P	W	T	?	I	M	D	E	

CT

W	W	A	X	D	D	?	Q	T	Y	N	G	V	T	C	Q	T	Y	Y	Z	L	W	G	V	B	X	T	E	R	W		
H	Z	Y	S	Y	C	K	L	P	T	R	S	D	T	U	Q	L	W	Q	S	V	E	G	K	O	P	W	U	O	P	L	
T	C	W	C	R	U	N	W	F	O	B	O	U	U	P	K	H	M	K	R	M	T	N	E	H	M	K	Q	V	X	E	E

*CT

L	U	N	H	Q	Q	X	Z	R	N	U	Z	X	J	Z	H	U	W	I	U	L	I	J	Q	Q	E	F	Q	R	I	E	Z
E	D	R	Y	R	U	X	L	W	H	Q	X	L	E	Q	G	P	U	I	X	Y	W	V	Y	V	U	O	X	U			
U	N	Q	Y	E	Z	G	V	U	Z	Y	U	X	P	R	J	J	H	H	E	H	S	Q	I	X	A	?	I	M	D	E	

Cryptogram 11 presents a very interesting challenge. As can be seen from the plain text, it is believed that we are aiming for the word BRILLIANCE, which is achievable from this rail position under the current KRS(10), but that would require reading the LBA's in the horizontal plane. We find instead that we are given a hybrid IUT H:V type transformation to resolve, whereby there are 2 NSP positions that resolve in the horizontal plane and one TSP that resolves both schemes. This only occurs in the cipher text, so we shall temporarily dispense with the other views until evaluating the TSP itself.

C(11) RP6 KRS(10) IUT2:2[[DV/EL] NSP 1

CT

W	W	A	X	D	D	?	Q	T	Y	N	G	V	T	C	Q	T	Y	Y	Z	L	W	G	V	B	X	T	E	R	W		
L	W	Q	S	V	E	G	K	O	P	W	U	O	P	L		H	Z	Y	S	Y	C	K	L	P	T	R	S	D	T	U	Q
T	N	E	H	M	K	Q	V	X	E	E	T	C	W	C	R	U	N	W	F	O	B	O	U	U	P	K	H	M	K	R	M

R(2)RS(12) to the NSP 2 gives.

C(11) RP6 KRS(10) IUT2:2[[SO/VO] NSP 2

CT

W	W	A	X	D	D	?	Q	T	Y	N	G	V	T	C	Q	T	Y	Y	Z	L	W	G	V	B	X	T	E	R	W		
Y	C	K	L	P	T	R	S	D	T	U	Q	L	W	Q	S	V	E	G	K	O	P	W	U	O	P	L		H	Z	Y	S
T	N	E	H	M	K	Q	V	X	E	E	T	C	W	C	R	U	N	W	F	O	B	O	U	U	P	K	H	M	K	R	M

R(2)RS(15) to the **TSP** gives.

RP6 KRS(10) IUT H:V[EL/DV][VO/OS]

PT

J	J	Z	M	R	U	?	H	X	U	F	O	J	U	Q	H	X	U	U	L	E	H	X	J	Y	L	W	P	U	H		
P	A	L	J	W	L	H	X	U	E		Z	N	V	Z	U	R	Q	B	H	X	U	Z	P	V	I	E	Y	L	H	Z	I
W	E	O	O	P	Q	H	I	L	P	O	U	Q	L	P	Q	H	E	L	P	J	Y	Z	H	H	U	Q	O	P	Q	U	E

*PT

R	Z	N	D	U	E	M	O	H	G	E	U	J	U	P	H	R	F	Q	U	H	L	H	X	U	V	R	L	Y	U	H	I
Y	A	Z	H	X	Z	P	U	Y	U	E	W	U	Q	V	E	I	Y				Z	R	J	I	V	H	X	U	E	I	I
H	X	U	Y	Z	J	H	B	N	L	P	F	I	O	J	U	U	L	P	F	H	X	Z	P	W	T	?	I	M	D	E	

CT

W	W	A	X	D	D	?	Q	T	Y	N	G	V	T	C	Q	T	Y	Y	Z	L	W	G	V	B	X	T	E	R	W		
E	G	K	O	P	W	U	O	P	L		H	Z	Y	S	Y	C	K	L	P	T	R	S	D	T	U	Q	L	W	Q	S	V
T	N	E	H	M	K	Q	V	X	E	E	T	C	W	C	R	U	N	W	F	O	B	O	U	U	P	K	H	M	K	R	M

*CT

Z	X	J	Z	H	U	W	I	U	L	I	J	Q	Q	E	F	Q	R	I	E	Z	H	L	U	N	H	Q	Q	X	Z	R	N
X	L	E	Q	G	P	U	I	X	Y	W	V	Y	V	U	O	X	U				E	D	R	Y	R	U	X	L	W	H	Q
U	N	Q	Y	E	Z	G	V	U	Z	Y	U	X	P	R	J	J	H	H	E	H	S	Q	I	X	A	?	I	M	D	E	

Notes. This is perhaps the most challenging cryptogram to solve as we are initially wrong footed by the alternate spelling and then finally by the lack of our usual RML or SML support to imply the alignment may be correct.

Letter Selection

CT

W	W	A	X	D	D	?	Q	T	Y	N	G	V	T	C	Q	T	Y	Y	Z	L	W	G	V	B	X	T	E	R	W		
E	G	K	O	P	W	U	O	P	L		H	Z	Y	S	Y	C	K	L	P	T	R	S	D	T	U	Q	L	W	Q	S	V
T	N	E	H	M	K	Q	V	X	E	E	T	C	W	C	R	U	N	W	F	O	B	O	U	U	P	K	H	M	K	R	M

Results

C(11) KRS(10) SA(CS)	K4 Key	Encryption
TJCD IGKU	ELDV VOOS	IANS EMOR

Cryptogram 12

C(12) RP6 IRP KRS(11)

PT

L	L	K	N	Y	V	?	I	Z	V	G	S	L	V	U	I	Z	V	V	M	F	I	Z	L	P	M	X	T	V	I		
K	Q	W	K	V	Y	U	C	I	Z	V	K	T	W	J	F	P	M	I	K	J	T	B	M	L	X	M	I	Z	V	F	
V	U	M	T	U	I	F	M	T	L	P	K	I	I	V	U	S	T	U	V	F	X	F	S	S	T	U	I	J	M	T	S

*PT

I	Z	V	W	Y	M	P	V	I	J	W	Y	K	Q	E	V	F	N	S	I	H	F	V	L	V	T	I	Y	G	U	V	M
K	Y	L	J	W	I	Z	V	F	J	J	P	B	K	I	Z	K	T	V	P	V	F	X	V	U	W	F	J	P			
I	Z	V	P	K	L	I	C	Q	M	T	G	J	S	L	V	V	M	T	G	I	Z	K	T	X	O	?	J	N	E	F	

CT

X	X	B	Z	E	E	?	U	O	P	Q	H	W	O	D	U	O	P	P	K	M	X	H	W	C	Z	O	F	Y	X		
I	K	P	A	P	D	R	M	T	O	Y	A	E	O	V	U	M	X	U	A	W	F	H	R	S	T	X	V	S	T	M	
O	D	X	D	Y	V	Q	X	G	S	C	S	V	V	T	R	I	N	R	Y	N	O	Q	F	I	N	R	U	W	Z	F	F

*CT

M	V	Q	I	U	U	Z	K	Y	Q	V	K	Z	L	K	I	V	X	J	V	M	J	L	U	U	F	G	U	Y	J	F	I
F	E	Y	P	Y	V	Z	M	X	I	U	Z	M	F	U	H	T	V	J	Z	P	X	W	P	W	V	S	Z	V			
V	Q	U	P	F	K	H	W	V	K	P	V	Z	T	Y	L	L	I	I	F	I	A	U	J	Z	B	?	J	N	E	F	

C(12) RP6 KRS(11) SA(SS) IUT H:V[UX/QT][UQ/TX] IUT3:4[OQF] NSP

PT

L	L	K	N	Y	V	?	I	Z	V	G	S	L	V	U	I	Z	V	V	M	F	I	Z	L	P	M	X	T	V	I		
K	T	W	J	F	P	M	I	K	J	T	B	M	L	X	M	I	Z	V	F		K	Q	W	K	V	Y	U	C	I	Z	V
S	V	U	M	T	U	I	F	M	T	L	P	K	I	I	V	U	S	T	U	V	F	X	F	S	S	T	U	I	J	M	T

*PT

M	I	Z	V	W	Y	M	P	V	I	J	W	Y	K	Q	E	V	F	N	S	I	H	F	V	L	V	T	I	Y	G	U	V
P	B	K	I	Z	K	T	V	P	V	F	X	V	U	W	F	J	P				K	Y	L	J	W	I	Z	V	F	J	J
I	Z	V	P	K	L	I	C	Q	M	T	G	J	S	L	V	V	M	T	G	I	Z	K	T	X	O	?	J	N	E	F	

CT

X	X	B	Z	E	E	?	U	O	P	Q	H	W	O	D	U	O	P	P	K	M	X	H	W	C	Z	O	F	Y	X		
A	E	O	V	U	M	X	U	A	W	F	H	R	S	T	X	V	S	T	M		I	K	P	A	P	D	R	M	T	O	Y
F	O	D	X	D	Y	V	Q	X	G	S	C	S	V	V	T	R	I	N	R	Y	N	O	Q	F	I	N	R	U	W	Z	F

*CT

I	M	V	Q	I	U	U	Z	K	Y	Q	V	K	Z	L	K	I	V	X	J	V	M	J	L	U	U	F	G	U	Y	J	F
Z	M	F	U	H	T	V	J	Z	P	X	W	P	W	V	S	Z	V				F	E	Y	P	Y	V	Z	M	X	I	U
V	Q	U	P	F	K	H	W	V	K	P	V	Z	T	Y	L	L	I	I	F	I	A	U	J	Z	B	?	J	N	E	F	

R(2)RS(8) to the TSP gives.

C(12) RP6 KRS(11) SA(SS) IUT H:V[UX/QT][UQ/TX] IUT3:4[OQ/FM] RML[VPI][VVL]TSP

PT

L	L	K	N	Y	V	?	I	Z	V	G	S	L	V	U	I	Z	V	V	M	F	I	Z	L	P	M	X	T	V	I		
K	V	Y	U	C	I	Z	V	K	T	W	J	F	P	M	I	K	J	T	B	M	L	X	M	I	Z	V	F		K	Q	W
S	V	U	M	T	U	I	F	M	T	L	P	K	I	I	V	U	S	T	U	V	F	X	F	S	S	T	U	I	J	M	T

*PT

M	I	Z	V	W	Y	M	P	V	I	J	W	Y	K	Q	E	V	F	N	S	I	H	F	V	L	V	T	I	Y	G	U	V
J	W	I	Z	V	F	J	J	P	B	K	I	Z	K	T	V	P	V	F	X	V	U	W	F	J	P				K	Y	L
I	Z	V	P	K	L	I	C	Q	M	T	G	J	S	L	V	V	M	T	G	I	Z	K	T	X	O	?	J	N	E	F	

CT

X	X	B	Z	E	E	?	U	O	P	Q	H	W	O	D	U	O	P	P	K	M	X	H	W	C	Z	O	F	Y	X		
A	P	D	R	M	T	O	Y	A	E	O	V	U	M	X	U	A	W	F	H	R	S	T	X	V	S	T	M		I	K	P
F	O	D	X	D	Y	V	Q	X	G	S	C	S	V	V	T	R	I	N	R	Y	N	O	Q	F	I	N	R	U	W	Z	F

*CT

I	M	V	Q	I	U	U	Z	K	Y	Q	V	K	Z	L	K	I	V	X	J	V	M	J	L	U	U	F	G	U	Y	J	F
P	Y	V	Z	M	X	I	U	Z	M	F	U	H	T	V	J	Z	P	X	W	P	W	V	S	Z	V				F	E	Y
V	Q	U	P	F	K	H	W	V	K	P	V	Z	T	Y	L	L	I	I	F	I	A	U	J	Z	B	?	J	N	E	F	

**C(12) RP6 KRS(11) SA(SS) IUT H:V[UX/QT][UQ/TX] IUT3:4[OQ/FM] RML[VPI][VVL] TSP
Digraph Analysis**

PT

L	L	K	N	Y	V	?	I	Z	V	G	S	L	V	U	I	Z	V	V	M	F	I	Z	L	P	M	X	T	V	I		
K	V	Y	U	C	I	Z	V	K	T	W	J	F	P	M	I	K	J	T	B	M	L	X	M	I	Z	V	F		K	Q	W
S	V	U	M	T	U	I	F	M	T	L	P	K	I	I	V	U	S	T	U	V	F	X	F	S	S	T	U	I	J	M	T

*PT

M	I	Z	V	W	Y	M	P	V	I	J	W	Y	K	Q	E	V	F	N	S	I	H	F	V	L	V	T	I	Y	G	U	V
J	W	I	Z	V	F	J	J	P	B	K	I	Z	K	T	V	P	V	F	X	V	U	W	F	J	P				K	Y	L
I	Z	V	P	K	L	I	C	Q	M	T	G	J	S	L	V	V	M	T	G	I	Z	K	T	X	O	?	J	N	E	F	

CT

X	X	B	Z	E	E	?	U	O	P	Q	H	W	O	D	U	O	P	P	K	M	X	H	W	C	Z	O	F	Y	X		
A	P	D	R	M	T	O	Y	A	E	O	V	U	M	X	U	A	W	F	H	R	S	T	X	V	S	T	M		I	K	P
F	O	D	X	D	Y	V	Q	X	G	S	C	S	V	V	T	R	I	N	R	Y	N	O	Q	F	I	N	R	U	W	Z	F

*CT

I	M	V	Q	I	U	U	Z	K	Y	Q	V	K	Z	L	K	I	V	X	J	V	M	J	L	U	U	F	G	U	Y	J	F
P	Y	V	Z	M	X	I	U	Z	M	F	U	H	T	V	J	Z	P	X	W	P	W	V	S	Z	V				F	E	Y
V	Q	U	P	F	K	H	W	V	K	P	V	Z	T	Y	L	L	I	I	F	I	A	U	J	Z	B	?	J	N	E	F	

Notes. This is the final cryptogram and perhaps the most satisfying to complete as it embodies all previous elements of the IUT scheme.

Letter Selection

CT

X	X	B	Z	E	E	?	U	O	P	Q	H	W	O	D	U	O	P	P	K	M	X	H	W	C	Z	O	F	Y	X		
A	P	D	R	M	T	O	Y	A	E	O	V	U	M	X	U	A	W	F	H	R	S	T	X	V	S	T	M		I	K	P
F	O	D	X	D	Y	V	Q	X	G	S	C	S	V	V	T	R	I	N	R	Y	N	O	Q	F	I	N	R	U	W	Z	F

C(12) RP6 KRS(3) SA(SS) IUT[DH/BA] TSP

Results

C(12) KRS(11) SA(SS)	K4 Key	Encryption
HUAU EKCA	OQFM UTXQ	NING STAR

Part 6

Conclusions

Conclusions

Putting all 12 individual cryptograms together we can see that we are left with the following composite message. As can be seen there are multiple spelling mistakes and the author takes on board that some solves may have been completed incorrectly despite matching the requirements for the proposed scheme, but high confidence is placed in a number of the solves beyond those for the known plain text.

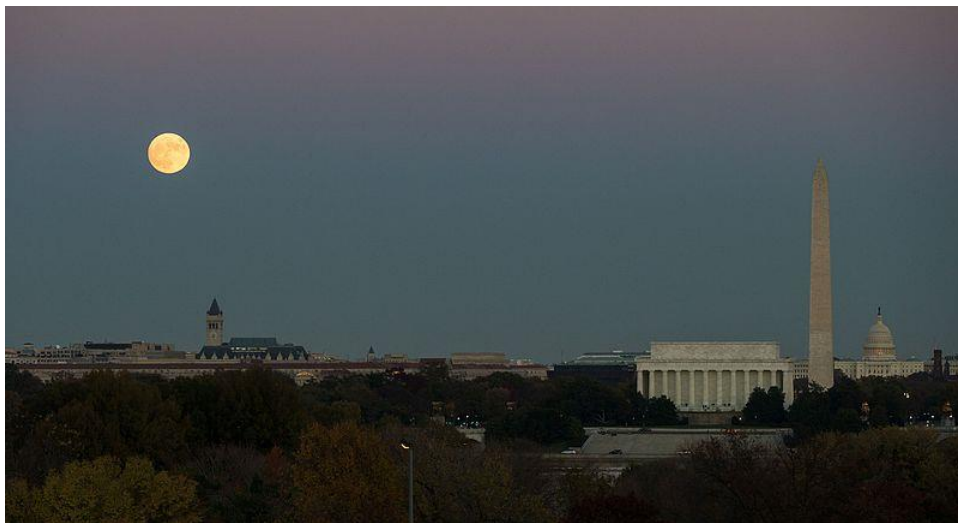
Cryptogram	K4 Key	Cryptogram Result	Plain Text
1	OBKRUOXO	UNYY CKBD	KRYP TOSI
2	GHULBSOL	WRLF PRRG	TEAM EAST
3	IFBBWFLR	FNWE GZJO	ROON DEAS
4	VQQPRNGK	BZEX PUXA	TNOR THEA
5	SSOTWTQS	KXPY AWLO	STBS TREE
6	JQSSEKZZ	HWOR ZSOS	TLEA DSTO
7	WATJKLUD	MWPN RHXU	ITAC RONO
8	IAWINFBN	PBAV BNTI	MITE ROFB
9	YPVTTMZF	CXUE ILNM	ERLI NCLO
10	PKWGDKZX	SKCU LIMN	CKSB RILL
11	TJCDIGKU	ELDV VOOS	IANE EMOR
12	HUAUEKCA	OQFM UTXQ	NING STAR

Looking at the composite message in more detail there are a number of features that allow us to make some further observations beyond this end result.

KRYPTO SITE AM EAST / ROOND EAST NORTH EAST B STREET LEADS TO IT / A
CRONOMITER OF BERLIN CLOCKS BRILLIANSE / MORNING STAR(R)

The first and most obvious feature is the correlation in the text between the opening statement and the end statement. Both would appear to be referencing the planet Venus in the morning or easterly phase of its synodic cycle. The second most obvious feature would appear to be something lunar in nature as the moon certainly rises for some of its cycle in an East North Easterly position on the horizon.

The reference to B STREET is compelling as further analysis shows that B Street is also the name for Constitution Avenue in Washington. If one thinks for a moment about the iconic shot of the Moon rising above the Capitol Building in Washington then those shots are typically taken whilst the camera is pointing in a North Easterly direction with the Moon quite literally hanging over B Street.



To see the Moon rise in a North Easterly direction over Constitution Avenue there are two locations of interest where you could be standing, both of which would qualify as a ‘crypto site’.

The first is the Pentagon, whose design will also be seen to link to the synodic cycle of Venus that we shall discuss shortly.



The second and perhaps most compelling location due to its civilian accessibility would be the Hirshhorn Museum that sits just off the National Mall. The building itself is famous for its round architectural design and as discussed at the head of the document also houses a copy of the Antipodes sculpture.



A solution beyond the solution

The reference to something astronomical in nature combined with the mention of a chronometer that is related to a clock or clocks in Berlin was a satisfying message to extract as the author's two previous 'Machine' papers published in 2017 have addressed these very issues of celestial mechanics and how they would appear to relate to the overall sculpture layout. For anyone that hasn't read those papers, they assert that all 4 panels of the sculpture participate in a schematic that can be overlayed on top of the sculpture text as a hidden steganographic layer.

These two previous papers covered in some detail how this hidden layer can be used to calculate the Metonic and Saros cycles, which are also the two main functions provided by the Antikythera Mechanism, which is acknowledged as being one of the worlds earliest analogue computers.



The solar/lunar cycles detailed in those papers stopped short of mentioning anything about the Antikythera Mechanism's planetary dial, but a cryptic analogue to this dial was found by the author in late 2017. The discovery of this dial is significant as it simultaneously clears up an additional outstanding question about the relevance of the K3 plain text. The evidence would suggest that this is in fact another cryptic clue related to geometry and implies that the Tutankhamun clue should be read as a homophone much like the CANDIL clue that was used in K4's second crib example. If we break up Tutankhamun phonetically then we can say it instead as TU (TAN) CAR (MOON). If we look at K4 again in light of this assertion then we can pick out the words CAR and TU.

E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R	*
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	*
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	*
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	*

The line that has been drawn from the top left of the U in TU to the bottom right corner of the CAR is expressed in angular terms as $\text{Tan}(\theta) = 3/31$.

Resolving this angle gives a figure of 5.5 degrees. This is a very interesting angle as it is a coarse grain figure for the offset of the Moon's orbit relative to Earths orbital plane. In other words this angle explains why we get eclipse cycles as they can only occur during a new moon that happens

whilst the Earth's orbital plane is being crossed. Many have commented in various forums about the presence of the KRYPTOS anagram that is tucked into K4's edges (shown here in green), but if we examine it in more detail then we can see that exhibits the exact same geometrical relationship as our previous example for TU and CAR having 3 letters in the vertical and 31 in the horizontal.

Drawing both of these lines simultaneously shows this remarkable set of relationships.

E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R	*
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	*
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	*
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	*

It will be seen instantly that the two lines intersect directly beneath the letters SOL. Delving slightly deeper with a more cryptic eye we can see that STTR is a Caesarean shift of 21 for the letters NOOM. Likewise a shift of 10 for the letters KLUDI gives UVENS and finally a shift of 16 for the letters OBKR gives ERAH. The author asserts that this is a cryptic model of the Sun, Earth, Moon and Venus.

The relationship above would imply that K4 does in fact implement the complete Antikythera scheme with the presence of this analogue to its planetary dial. This planetary dial may be cryptically hidden, but it is clearly showing rays hitting the Earth that are about to be blocked by the Moon. The presence of Venus along the path of the other ray would imply that the equivalent phenomena of Venus' solar conjunctions are also just as important as lunar eclipses.

The scheme and calculation of the Metonic and Saros cycles has already been covered in the previous papers, so beyond the discovery of the planetary dial the only new information to add that hasn't been shared previously is with regards to the synodic cycle of Venus.

Before discussing this in detail we will briefly need to revisit the general concept of the steganographic overlays. We are in fact given the chance to segue directly into this discussion by the first few lines of the previously derived K4 message itself.

A Kryptos.Groups.Io forum member with the handle LouiZSA made a very interesting observation in message #22071 to which I responded with this extension to the scheme. Despite only having two instances of EAST from the plain text clues given LoiZSA was somehow able to assert the presence of an interesting numerical pattern. Comparing this with the plain text results just derived for K4 it became apparent that they appeared to extend LouisZSA's observation further.

To understand its significance, let's tabulate the first 34 letters of our proposed plain text result by splitting it over a table with 9 columns and 3 rows.

If we remove the word KRYPTOS from the beginning and write out the remainder then the result fits rather neatly into this table.

1	2	3	4	5	6	7	8	9
I	T	E	A	M	E	A	S	T
R	O	O	N	D	E	A	S	T
N	O	R	T	H	E	A	S	T

This is significant as we have seen the exact same pattern before in the sculptures steganographic layer previously detailed in the 'Machine' papers. This layer not only holds the sculptures coordinates, but also a range of other numbers that when appraised as a complete scheme allow the sculpture layout to be used as a sophisticated analogue computer. Delimiting the cipher text panels in a 19/13 split and the tableau panels in an 18/13 split eventually leads to the following overlays.

Cipher text panel overlays

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	*	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	*	
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G	*	*	
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A	*	
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R	
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I	*	
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E	*	
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X	
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F	*	
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q	*	*	
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E	*	
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P	*	
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	S	W	J	L	L	A	E	T	G	*

E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A	
C	H	T	N	R	E	Y	U	L	D	S	L	L	S	L	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E	*	*	
T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E	*	
W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E	*	*	
T	F	O	L	S	E	D	T	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	C	R	
E	I	F	T	B	R	S	P	A	M	H	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	E	R	L	B	*	*	
T	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	R	T	N	R	T	I	
B	S	E	D	N	I	A	A	H	T	T	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B	*		
A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R	I	T
R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	P	N	E	O	H	M	G	F	M	F	E	U	H	E	*	*	*	
E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R	*	
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	*	
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	*	
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	*	

Tableau panel overlays

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D		
A	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P		
B	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T		
C	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O		
D	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S		
E	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A		
F	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B		
G	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C		
H	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D		
I	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E		
J	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F		
K	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G		
L	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H		
M	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I		
N	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	
O	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L		
P	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M		
Q	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N		
R	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q		
S	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U		
T	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V		
U	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W		
V	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X		
W	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z		
X	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K		
Y	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R		
Z	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y		
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D		

It will be seen in both instances that the large green block on the right hand side for both the cipher and tableau text is 364 letters, with the extraneous T and L in both cases bringing that figure to 365. In other words, both the cipher and tableau text have embedded 365 day calendars hidden by the stego scheme when appraised as a matrix that is 33 X 28 and 33 X 28 respectively.

For the case of the cipher text we additionally know that we have a missing X that can be added that will force one of the existing spaces to move into column 33 that will then be capable of doubling up as a leap day.

The fine structure contained within the overlay for the top cipher text panel is perhaps the most interesting part (to the authors mind) of the whole Kryptos piece. The steganographic overlay shows the presence of the complete set of coordinates for the sculpture embedded in the top left corner of the cipher text. These coordinates are initially defined by 3 repeating rows of 19 that give 19, 38 and 57 when summed. It is these initial 3 repeating rows of 19 that would appear to echo the 3 repeating rows that end in EAST shown in the previous 9 column table.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J		
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D			
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E			
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G				
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A			
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R		
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I			
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E			
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X		
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F			
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q				
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E			
D	N	Q	F	M	P	N	Z	G	L	F	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P		
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G			

To finish up the discussion on the embedded coordinates given by the stego system is can be seen that the seconds for the northerly coordinates are marked by 6 letters followed by a question mark and then an additional 5 letters to give us our 6.5. It can be seen that the end of the 6.5 region is abutted by the last row of the 77 letter section for the westerly coordinates. HHDDDDUVH marks the 8 minutes, which are then added to the 36 letters above to give the 44 seconds that complete the full set of sculpture coordinates.

Brilliant Chronometers

Much has been made of the various clocks of interest that reside in Berlin, but there are perhaps two that are of real interest for the purposes of this conclusion. The first of course is Dieter Binninger's 'Mengenlehreuhr' or set theory clock and the second is Bernard Gitton's 'Clock of Flowing Time' that extends over 3 floors of the Europa Center in Berlin.

The steganographic layout that we find within Kryptos would appear to be a hybrid of these two concepts, both of which are driven by the presence of an ingenious gearing system hidden within the actual coordinates themselves. The presence of this gear mechanism is not arbitrary and is instead found in a clue that can be extracted when looking at the plain text through the filter of the steganographic overlay. If we examine the boundaries of the 19/13 row scheme when viewing rows 14 and 15 then we see the following relationship of note.

T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	W	E	S	T	X	L	A	Y	E	R	T	W	O	
S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	E	S	T	H	L	A	Y	E	R	M	I	N	S

An anagrammatic rearrangement of the 13 letters for the row 15 line end gives.

T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	W	E	S	T	X	L	A	Y	E	R	T	W	O
S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	E	S	T	H	L	A	Y	E	R	M	I	N

That may seem like an unremarkable insight until we realise that the hours (H) and minutes (MIN) of the Kryptos coordinates as given in the K2 plain text sum exactly to 180.

$$38 + 57 + 77 + 8 = 180$$

This hour to minute relationship forms the backbone of a gearing system that we can drive by treating the sculpture as an analogue computer driven by cursor operations, whereby each letter can light up when selected as part of some predefined function. In the physical case of something like one of Bernard Gitton's water clocks this could also be thought of as filling a container with a set amount of water before filling up the next. At some logical point the sum (in this case 180) is reached and we would then drain the water away to start the process again of filling again.

It should be noted that the coordinate system used in the Kryptos stego overlay is schematic and not physical, by which it is meant that after counting out 38 blocks we then need to return to the start of the three initial rows in order to count out 57. Whilst this is not an issue when programming such a scheme it would be an issue if making a physical reproduction as another 38 blocks would be needed. Anyone reading the authors previous papers looking to make physical interpretations of the scheme should bear this in mind when sketching out their initial designs.

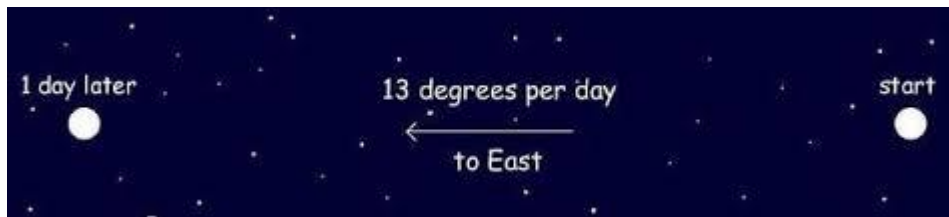
If any additional proof is needed that Kryptos can double up as an analogue computer then we need look no further than the computation of the K1 and K2 decrypts, which can both be performed by the sculpture by driving it as an analogue computation using these same cursor operations. If we in fact leave the letters that have participated in that computational process highlighted after decrypting K1 and K2 then we see that Kryptos has yet another additional secret to reveal.

EMUFPHZLR FAXYUS DJKZLDKRN SHGNFIVJ	ABCDEFGHIJKLMN O PQRST UVWXYZ ABCD
YQTQUXQBQVYU VLLTRE VJYQTMKYRDMFD*	AKRYPTOS ABCDEFGHIJ LMNQUVW XZKRYPT
VFPJQUDEEHZ WETZYVGVG HKKQETG FQJNCE*	B RYPTOS ABCDEFGHIJ LMNQUVW XZKRYPT
GGWHKK?DQMC PFQZDQMMIAGPF XHQRLG**	CYPTOS ABCDEFGHIJ LMNQUVW XZKRYPTO
TIMVMZJANQLVKQEDAGD VFRPJUNGEUNA*	DPTOS ABCDEFGHIJ LMNQUVW XZKRYPTOS
QZGZLEC GYUXUEENJTB JLBQCRTBJDFHRR	ETOS ABCDEFGHIJ LMNQUVW XZKRYPTOSA
YIZETKKEZEMV DUFK SJHKF WHKUWQLSZFTI*	FOS ABCDEFGHIJ LMNQUVW XZKRYPTOSAB
HHDDDUVH?D WKB FUF PWNTDFIYCUQZERE*	GS ABCDEFGHIJ LMNQUVW XZKRYPTOSABC
EVLDKFEZMOQQJLTTUGS YQPF EUNLAVIDX	H ABCDEFGHIJ LMNQUVW XZKRYPTOSABCD
FLGTEZ?FKZBSFQDQVGO GIPUFXHHDRKF*	IBCDEFGHIJ LMNQUVW XZKRYPTOSABCDE
FHQNTGCPUAECNUV PDJMQCLQUMUNEDFQ**	JCDEFGHIJ LMNQUVW XZKRYPTOSABCDEF
ELZQTVRRGKF FVVOEEEXBD MVPNFQXZLGR*	KDEFGHIJ LMNQUVW XZKRYPTOSABCDEF
DNQFMPN ZGLFLPM RJQYALMGNUPVDXVKP*	LDEFGHIJ LMNQUVW XZKRYPTOSABCDEF
DQUME BEBDMHDAFMJGZNU PLGEWJLLAETG*	MFGHIJ LMNQUVW XZKRYPTOSABCDEF
ENDY AHR OHNLSRHE OCPTEOIBIDYSHNAIA	NGHIJ LMNQUVW XZKRYPTOS ABCDEFGHIJL
CHTNREYULDSL SLLNOHSNOSMRWXMNE**	OHIJ LMNQUVW XZKRYPTOS ABCDEFGHIJL
TPRNGAT IHNRRARPE SLNNELEBLPIIACAE*	PIJ LMNQUVW XZKRYPTOS ABCDEFGHIJLM
WMTWN DITEENRAHCTEN EUDRETNHAEOE**	QJ LMNQUVW XZKRYPTOS ABCDEFGHIJLMN
T EOLSED TIWENHAE IOYTEYQH EENCATAYCR	RLMNQUVW XZKRYPTOS ABCDEFGHIJLMNQ
E IFTBRSPAMHHEWENATAMATEGYEERL B**	SMNQ UVW XZKRYPTOS ABCDEFGHIJLMNQ
T EEF OASFI OTUETUA EOT OARMAEERTNRTI	TNQ UVW XZKRYPTOS ABCDEFGHIJLMNQ
B SEDDNIAAHT TMSTEWPIEROAGRIEWFEB*	UQUVW XZKRYPTOS ABCDEFGHIJLMNQ
AEC TDDHIL CEIHSITEGOEASDDRDYDLORI T	VUVW XZKRYPTOS ABCDEFGHIJLMNQ
RKLMLEHAGT DHARDPNE OHMGFMFEUHE**	WVW XZKRYPTOS ABCDEFGHIJLMNQ
ECDMRIPFEIMEHNLSSTTRTVDOHW?OBKR*	XWXZKRYPTOS ABCDEFGHIJLMNQ
UOX OGHU LBSOLIFBBWFLRVQQPRNGKSSO*	YXZKRYPTOS ABCDEFGHIJLMNQ
TWTQS JQSSSEKZZWATJKLUDIAWINFBNYP*	ZZKRYPTOS ABCDEFGHIJLMNQ
VTTMZ FPKWGDKZX TJCDIGKUHU AU EKCAR*	ABCDEFGHIJKLMN O PQRST UVWXYZ ABCD

On the right hand side of the picture above we see all the tableau letters that participated in the decryption of K1 and K2. Taking those same letter positions as a mask and rotating them by 90 degrees before overlaying back on the cipher text gives a remarkable result. The portion of the mask that overlays K1 and K2 on the left hand side of the picture is exactly 97 letters. The portion that overlays K3 and K4 is 63 letters, which is the exact same length as the K1 message. It will also be noted that our rail position markers did not extend beyond row 26.

We saw in the 'Machine' papers that cycling through the 180 blocks of the stego gearing system allowed us to traverse the blocks for the 6.5 coordinate region every time we reached 180 before resetting to 0 again. These 11 blocks that were transited in the 6.5 coordinate block could be interpreted in a different way. If we thought of 6.5 degrees as an angle instead then it would mean for

every 360 blocks we traversed in the gearing scheme we would symbolically trace out 13 angular degrees. This is significant as every time the Earth turns once on its axis the moon appears to move 13 degrees in the sky relative to the previous night's position.



Revisiting the number of blocks transited before the gearing scheme resets we see that we have 180 blocks for the half turn followed by 11 blocks for the representation of 6.5 degrees, so in fact for every representation of a half day we traverse 191 blocks before resetting.

If we start this process by simultaneously running a cursor through the letter blocks of K3 and examine the letter underneath the cursor every time we reset the gearing to zero then we in fact inadvertently recreate the algorithm for the extraction of K3, which is effectively a 192 letter skip.

Making this connection that Kryptos really is capable of meaningful analogue computations by virtue of its stego layer is necessary to digest before we examine the synodic cycle of Venus in more details.

The Synodic Cycle of Venus

We have just seen previously that adding the northerly 6.5 seconds to the base 180 gearing allowed us to derive 13 degrees of lunar motion through the sky for every Earth day. By the exact same method, but using the 44 westerly seconds added to the base 180 scheme we can see that we are also given a coarse grain figure in days for the orbital period of Venus, which in this instance is given as 180 plus 44 or 224 days. Using the calendar that we are provided as part of the stego layer it is possible to track the course of its progress relative to the Earth year.

By combining the coordinate blocks used to describe the 44 westerly seconds and the blocks used to describe the 6.5 northerly seconds it is possible to completely capture the synodic cycle of Venus. The only slight adjustment we need to make is to lose three letters from the top and bottom left of the cipher text panels.

This gives an arrangement with similar aesthetics to the tableau panel notches in the top and bottom left hand corners, and could be considered as an operation undertaken under instruction by letting 'the HOT AIR escape' from the top and bottom panels respectively.

Reviewing our 19/13 steganographic division under this condition gives the following layout.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E		
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G			
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A		
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R	
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I		
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E		
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X	
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F		
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q			
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E		
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P		
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G		
E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A	
C	H	T	N	R	E	Y	U	L	D	S	L	L	S	L	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E			
T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E		
W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E			
T	F	O	L	S	E	D	T	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	C	R	
E	I	F	T	B	R	S	P	A	M	H	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	E	R	L	B			
T	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	R	T	N	R	T	I	
B	S	E	D	D	N	I	A	A	H	T	T	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B		
A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R	I	T
R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	P	N	E	O	H	M	G	F	M	F	E	U	H	E				
E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R		
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O		
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P		
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R		

Combining the stego coordinate scheme for the northerly and westerly seconds highlights the following blocks of interest.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E		
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G			
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A		
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R	
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I		
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E		
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X	
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F		
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q			
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E		
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P		
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G		
E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A	
C	H	T	N	R	E	Y	U	L	D	S	L	L	S	L	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E			
T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E		
W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E			
T	F	O	L	S	E	D	T	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	C	R	
E	I	F	T	B	R	S	P	A	M	H	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	E	R	L	B			
T	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	R	T	N	R	T	I	
B	S	E	D	D	N	I	A	A	H	T	T	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B		
A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R	I	T
R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	P	N	E	O	H	M	G	F	M	F	E	U	H	E				
E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R		
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O		
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P		
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R		

If we use the highlighted blocks for the westerly and northerly seconds as positions to mark our days then we can now chart the synodic cycle for Venus using the following steps.

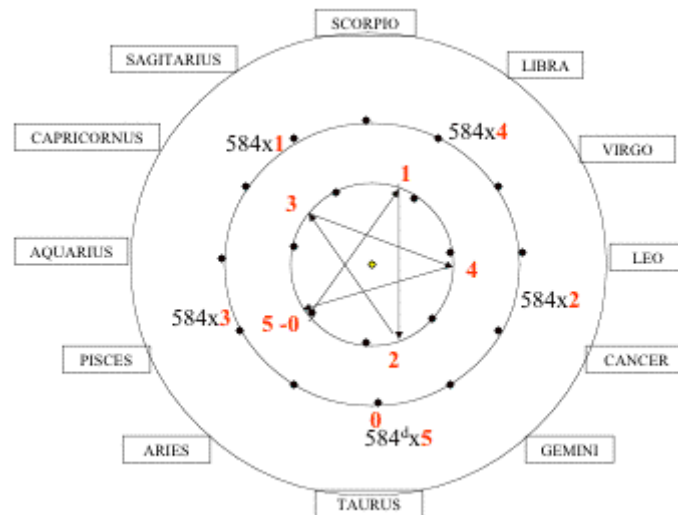
Venus appears as a morning star for 263 days, which we can track in the light blue area in the top panel.

Venus then disappears for 50 days behind the Sun, which can be tracked in the gold block given by the 44 plus 6.

Venus is then an evening star for another 263 days which we can mark in the light blue area in the bottom panel.

Finally, Venus transits in front of the Sun and is not visible for 8 days, which we can record in the 8 minute block of our coordinate scheme highlighted above in orange.

The cycle then repeats. As we can simultaneously set off a cursor in the calendar portion of the panels it will of course be possible to determine the dates on which superior and inferior conjunctions will occur. As we saw with the Metonic and Saros cycles, the Synodic Cycle of Venus also gives rise to regularly defined intervals of time, but in this instance there is something of an added bonus as the position in the sky at which these conjunctions occur give a well-defined and recurring geometric pattern, which is that of the pentagram. The following diagram of this phenomenon has been taken from the European Southern Observatory public outreach site, and shows how this geometric relationship plays out over time with respect to the various constellations of the zodiac.



To conclude, it can be seen that all the astronomical phenomenon that have been highlighted by the sculptures numerical layer are all about repeating cycles that continue with clockwork regularity despite the chaotic nature of the overall universe through which these kinetic systems are travelling. This paper has outlined the principle of invariance under transformation as it applies to the K4 cipher, but it is not a great leap of the imagination to see that these astronomical cycles that are encoded within the sculptures numerical overlay are also telling us something about their invariance when transformed over time itself. It is hoped that this paper has been useful and that as a result you will finally be able to see Kryptos in a new light.

Matt Reid 12/3/21

Further Reading

For anyone wanting to read more about the various subjects that have been touched upon in this paper I would recommend the following books.

‘A Portable Cosmos’ by Alexander Jones is highly recommended. This is an excellent work on both the history and reconstructed mechanics of the Antikythera Mechanism.

ISBN 978-0-19-973934-9

I would also recommend Carl Sagan’s ‘Contact’ novel to anyone that has an active interest in Kryptos. Whether Carl Sagan was a consultee to the Kryptos project or not will no doubt be known in time, but I can’t recommend this book highly enough. For anyone that has seen the film, but not read the book, the endings are different with one being more profound than the other.

ISBN 0-09-946950-2

Simon Singh’s ‘The Code Book’ is still a classic introduction to Cryptography for the modern reader and I recently had chills upon noticing that the books acknowledgements include thanks for the brilliant Sonya Holbraad who was once also my maths tutor at a pivotal point in my academic career.

ISBN 185-70-2889-9

Codebreaking A Practical Guide, by Elonka Dunin and Klaus Schmeh is probably one of the most recent introductory cryptography books to hit the shelves and has a great level of depth for anyone wanting to take their interest further.

ISBN 978-1472144218

Cubed: The Puzzle of Us All is another new comer to the book shelves and is written by Erno Rubik the inventor of the cube that bears his name. This short, but fun book is not just about puzzles, but more importantly about the role of the creative process, so educators may also want to pick up a copy of this one and surreptitiously leave it lying in view of any students with curious minds.

ISBN 978-1474613118

Other than Rubik’s work on ‘invariance under transformation’ there appears to be no other record of a similar puzzle type cipher that I can find a reference to other than an academic paper by Douglas W Mitchell from 1992 entitled “Rubik's Cube” as a transposition device. This paper doesn’t appear to point back to anything relating to the method discussed in this paper, but I would be interested to hear from anyone that can find precedence for this scheme.

Afterward

In compressing these findings into a paper that is almost academic in nature it is appreciated that a large part of any potential audience may have been lost, along also with any sense of the narrative and the sense of adventure that ran through each of these discoveries. Please bear in mind that each one represented an individual moment of wonder, intrigue or elation rather than the clinical end results discussed in the previous sections. This is a long paper and there will no doubt be the need for many corrections and clarifications as it has been written in isolation, but it is hoped that professionals and amateurs alike in the crypto community will kick the tyres on both the IUT scheme as well as the assertion that Kryptos when viewed as a holistic piece is also an analogue calculator capable of a wide range of astronomical predictions.

About the author

Matt is a Software Architect based in the UK with undergraduate and post graduate degrees in Physics and Distributed Computing. For questions relating to this paper or previous Kryptos work he can be contacted at matzreid888@yahoo.co.uk.